IDAT7215
Computer Programming for Product Development and Applications

Lecture 5-1: Unsupervised Learning

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Outline

- Introduction
- Types of Unsupervised Learning
- Clustering
- K-Means Clustering Algorithm
- Hierarchical Algorithm
- Association Learning
- Apriori Algorithm
- Summary

Unsupervised Machine Learning: An overview

Supervised learning:

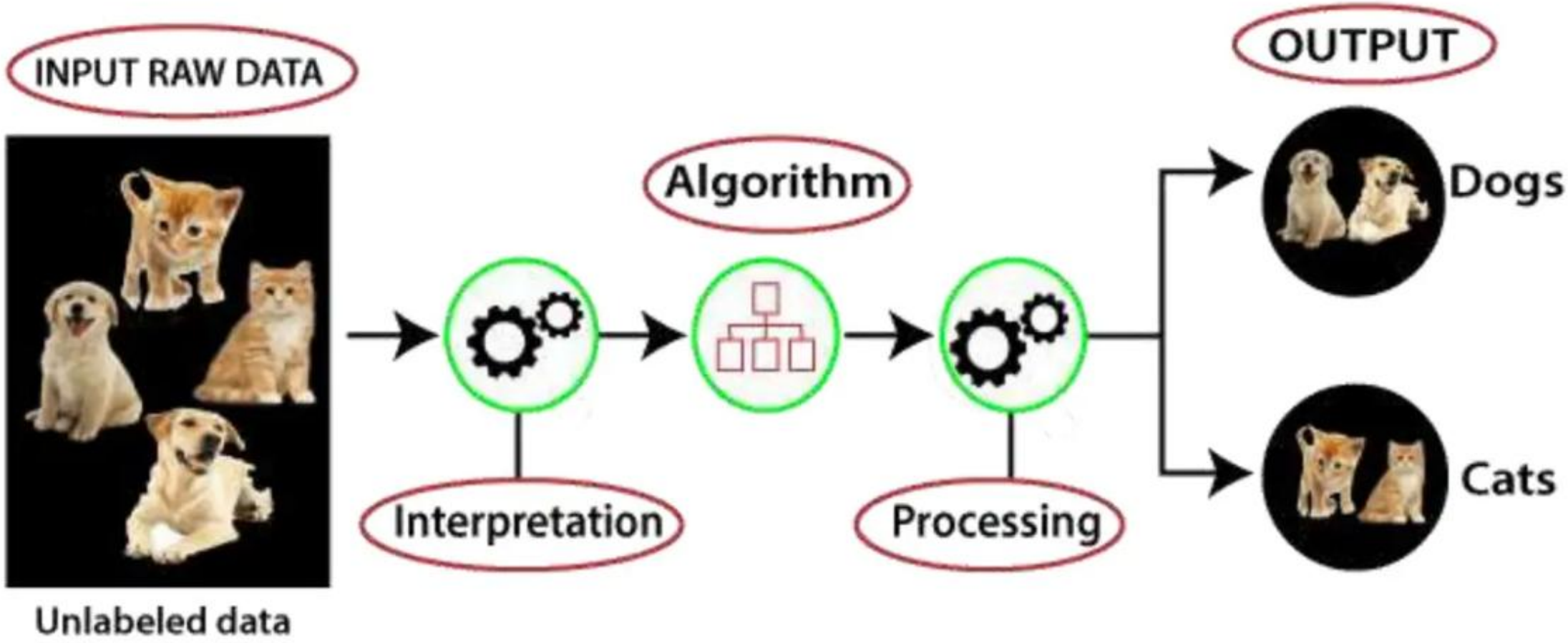
- Trained label data

Unsupervised learning:

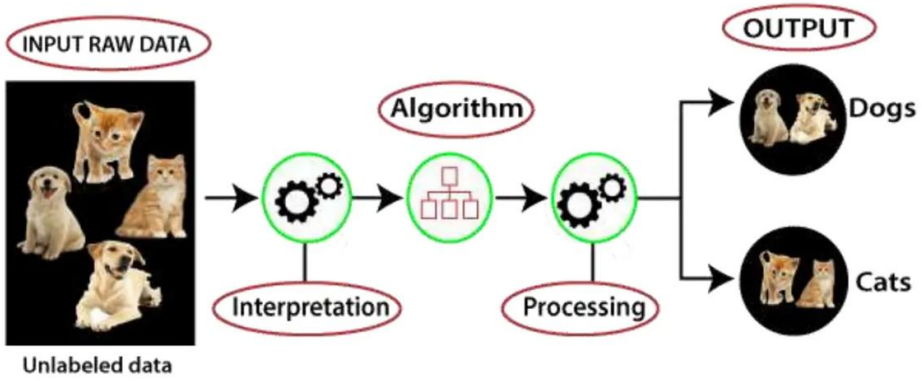
- Do not have labelled data.
- Need to find hidden patterns.
- Models are not supervised using the training dataset.
- Goal is to find the underlying structure of the dataset, group the dataset according to similarities.

Unsupervised Machine Learning: An overview

Example of Unsupervised Learning:



Unsupervised Machine Learning: An overview



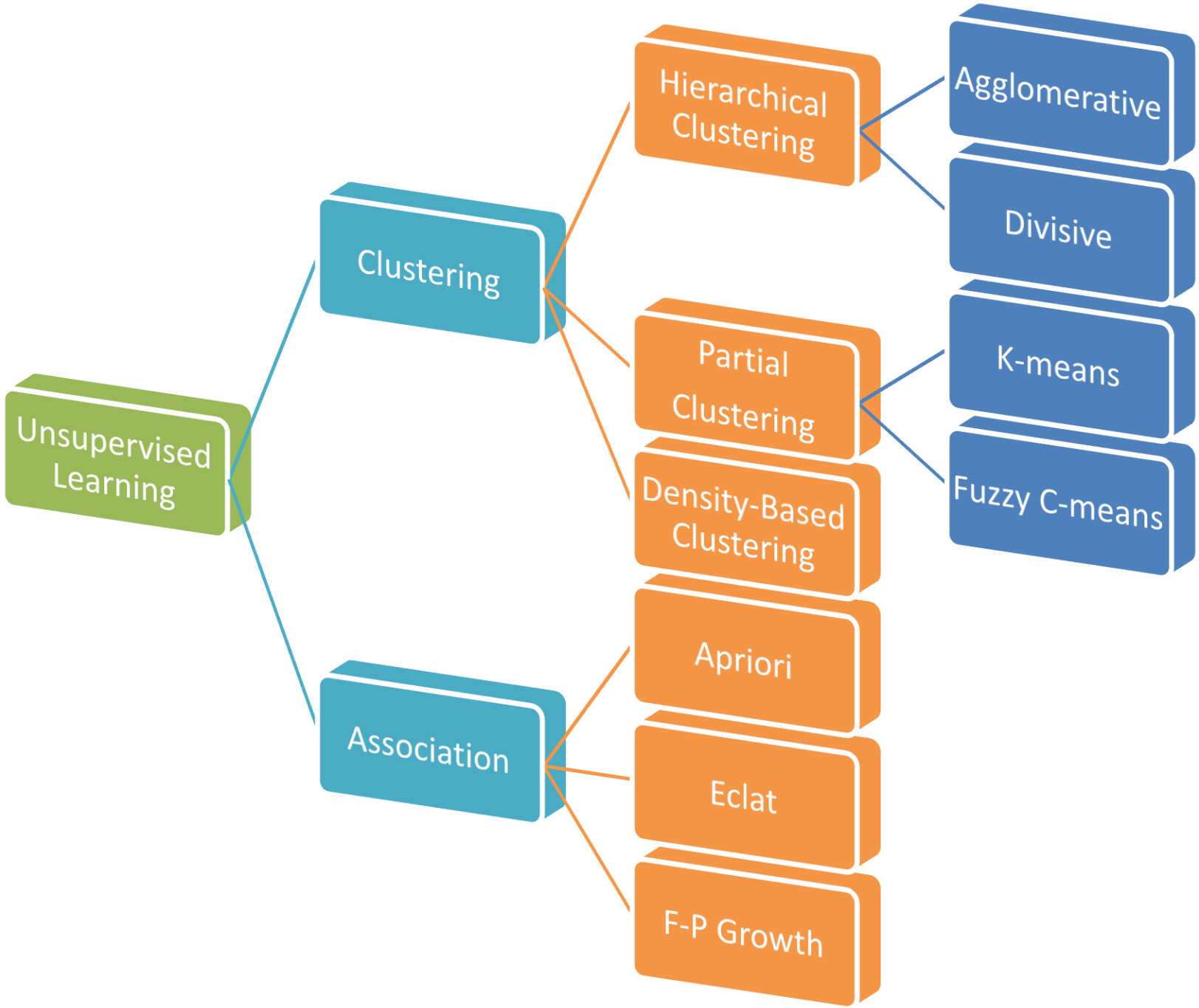
Example of Unsupervised Learning:

- The algorithm is never trained on the given dataset.
- The task of an unsupervised learning algorithm is to identify the image features on its own.
- An unsupervised learning algorithm will perform this task by clustering the image dataset into groups according to similarities between images.

Why Use Unsupervised Learning

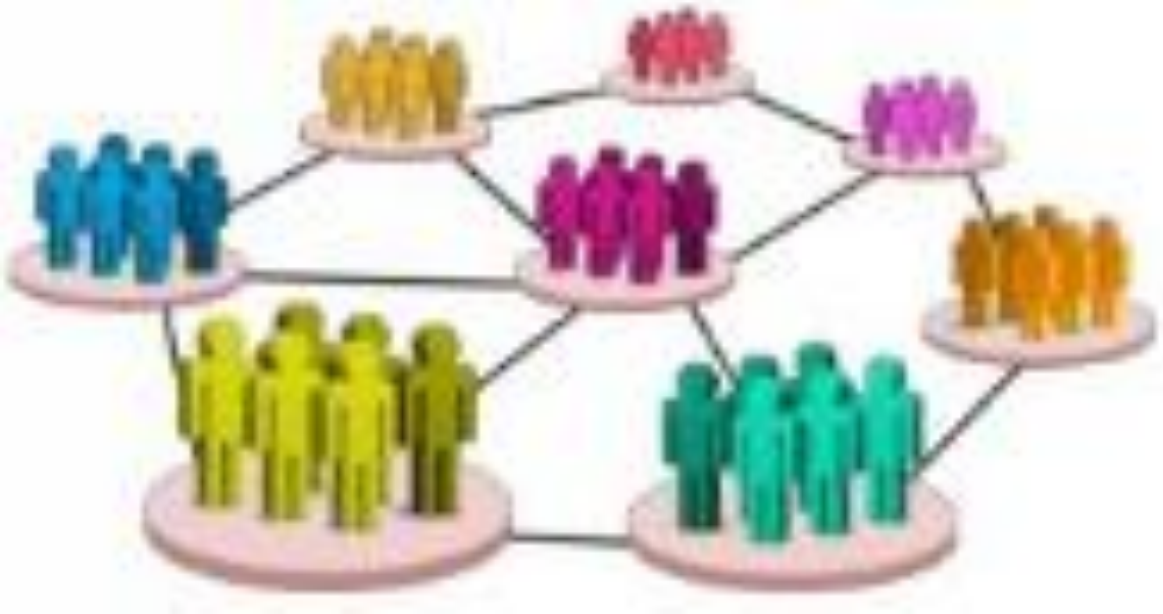
Importance of unsupervised learning:

- Unsupervised learning helps find useful insights from the data.
- It is much similar to a human learns to think by their own experiences, which makes it closer to the real AI.
- It works on unlabeled and uncategorized data, which make unsupervised learning more important.


Unsupervised Learning
Clustering
Association
Hierarchical Clustering
Partial Clustering
Density-Based Clustering
Apriori
Eclat
F-P Growth
Agglomerative
Divisive
K-means
Fuzzy C-means

What is Clustering

- Clustering means grouping objects based on the information found in the data describing the objects or their relationship.
- The goal is that the objects in one group (called a cluster) should be similar but different from objects in another group.
- It deals with finding structure in a collection of unlabeled data.



Clustering Use Cases

- **Marketing**

Discovering district groups in customer databases, such as customers who make a lot of long-distance calls.

- **Insurance**

Identifying groups of crop insurance policyholders with a high average claim rate. Farmers crash crops when it is “profitable”.

- **Land Use**

Identification of areas of similar use in the GIS database.

- **Seismic Studies**

Identifying probable areas for oil/gas exploration based on seismic data.



What is good clustering

1. A good clustering method will produce high-quality clusters with

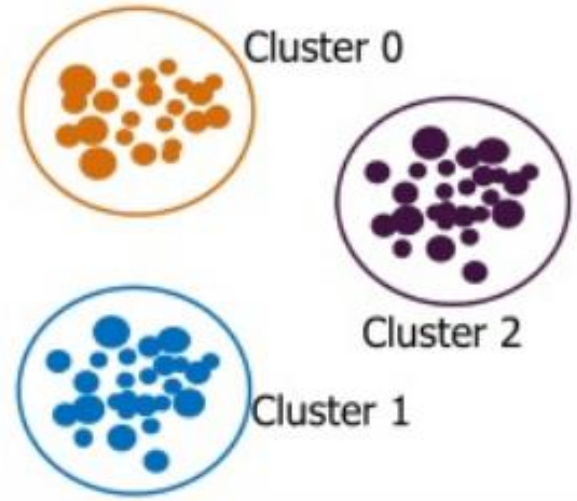
- high intra-class similarity.
- low inter-class similarity.

2. The quality of a clustering result depends on both the similarity measure used by the method and its implementation**.**

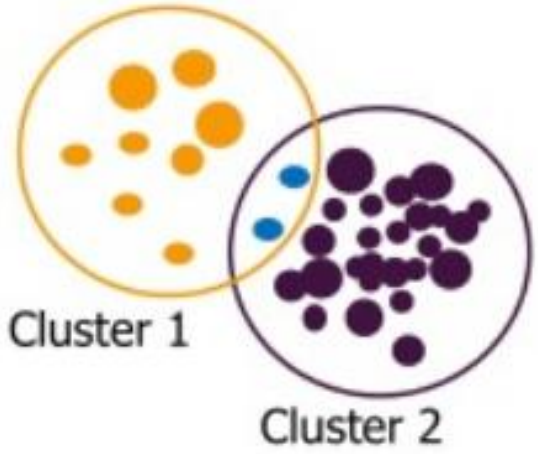
3. The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns**.**

Types of Clustering

1. **Exclusive Clustering**
   - An item belongs exclusively to one cluster, not several.
   - K-means does this sort of exclusive clustering.



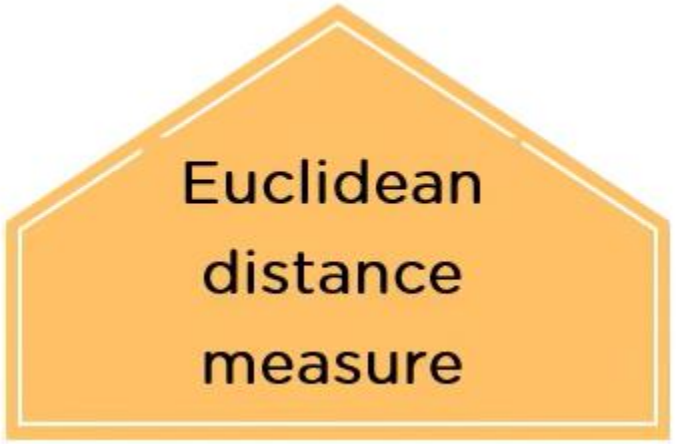
2. **Overlapping Clustering**
   - An item can belong to multiple clusters.
   - Its degree of association with each cluster is known.
   - Fuzzy/C-means does this sort of exclusive clustering.

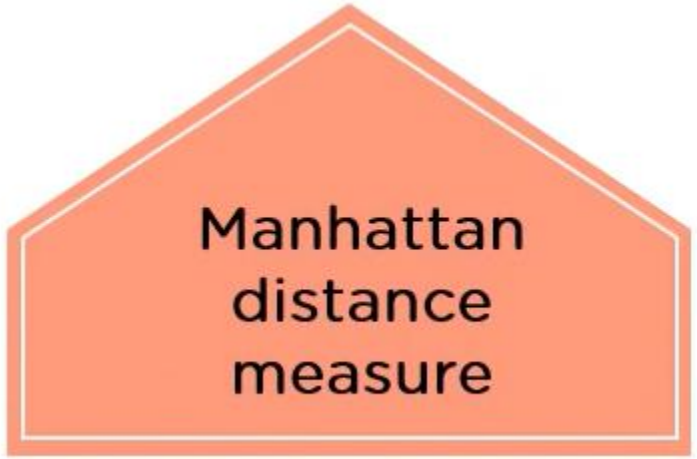


3. **Hierarchical Clustering**
   - When two clusters have a parent/child relationship or a tree-like structure then it is Hierarchical clustering.

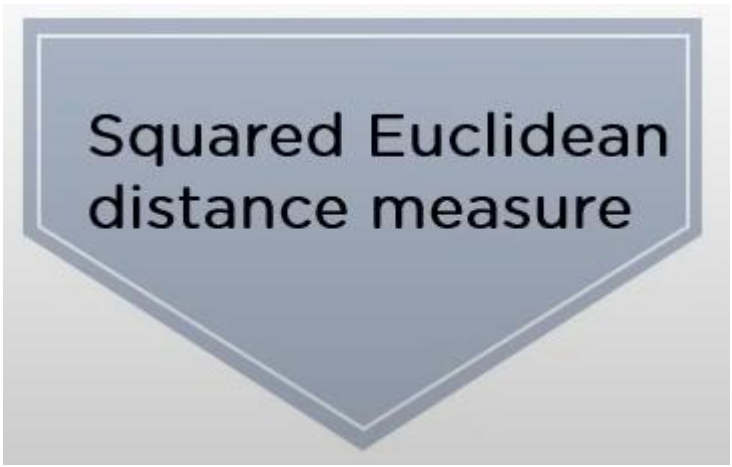


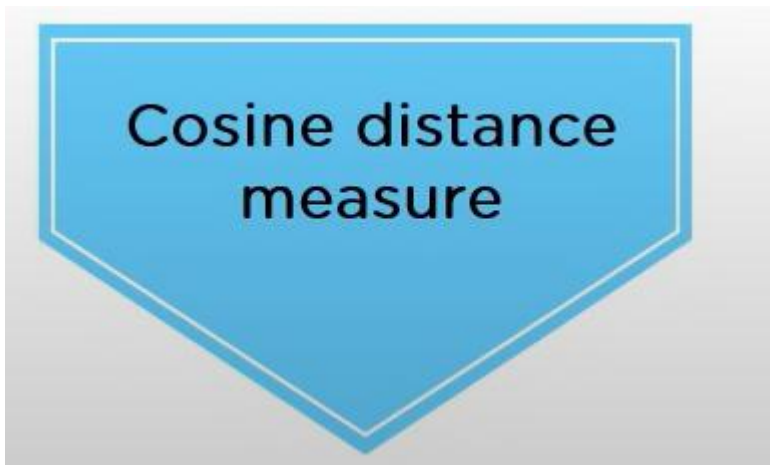
Distance Measure





The distance measure will determine the similarity between two elements, and it will influence the shape of the clusters.

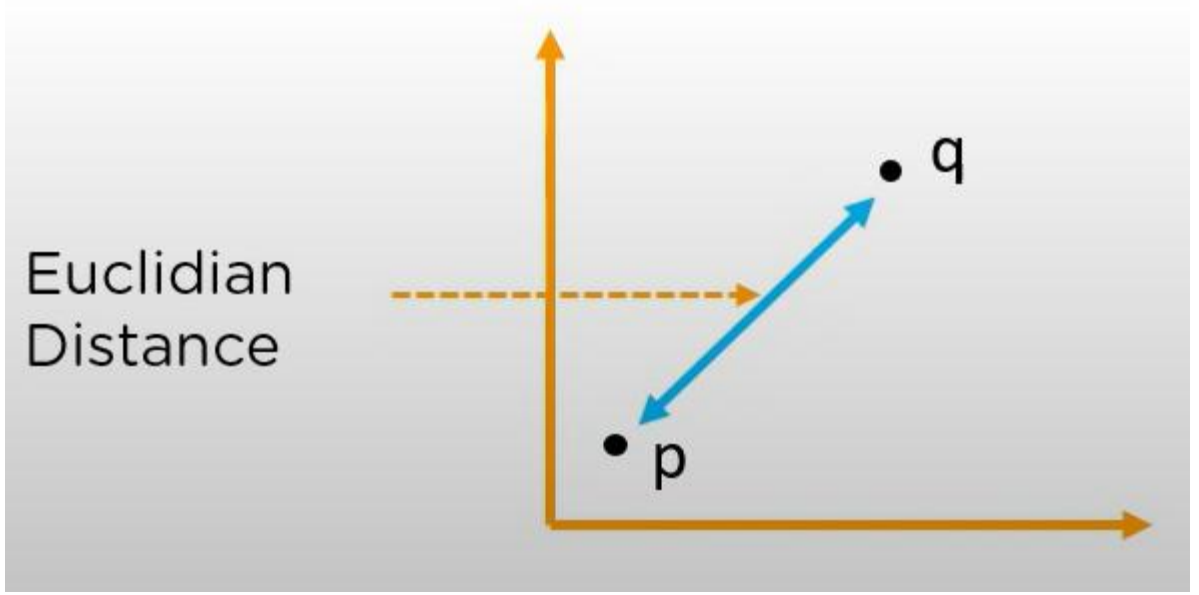




Distance Measure

1. **Euclidean distance measure**
   - The Euclidean distance is the "ordinary" straight line.
   - It is the distance between two points in Euclidean space.

$$d = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$



Distance Measure

2. Squared Euclidean distance measure

- The Euclidean squared distance metric uses the same equation as the Euclidean distance metric but does not take the square root.

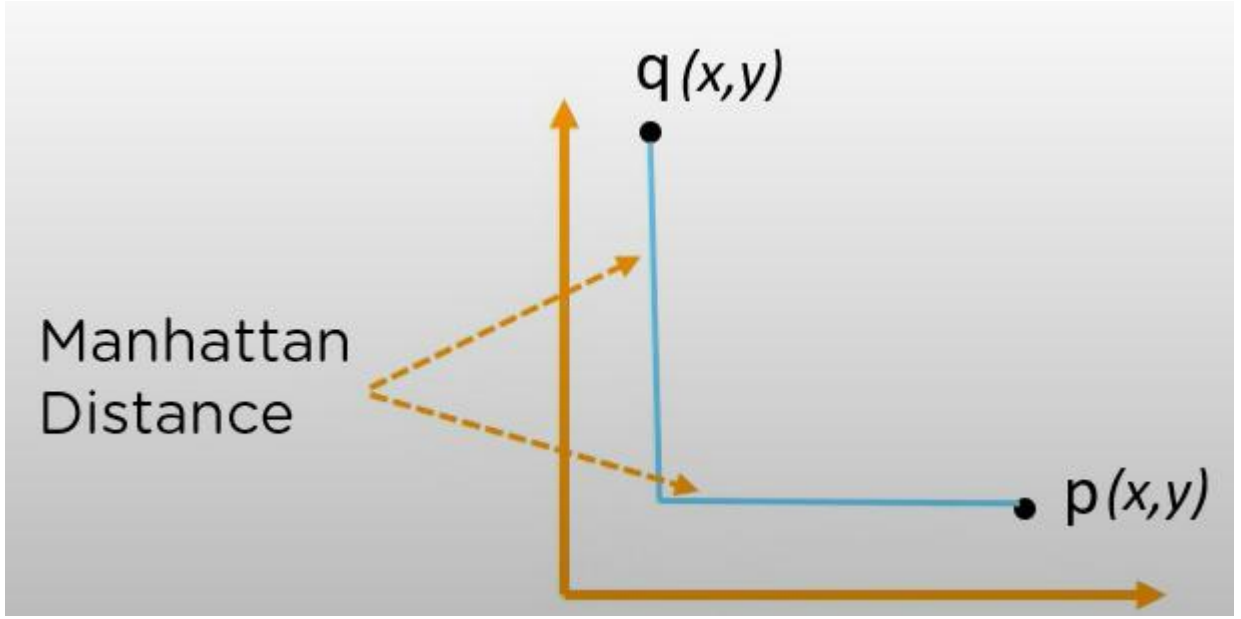
$$d = \sum_{i=1}^{n} (q_i - p_i)^2$$

Distance Measure

3. Manhattan distance measure

- The Manhattan distance is the simple sum of the horizontal and vertical components or the distance between two points measured along axes at right angles.

$$d = \sum_{i=1}^{n} |q_x - p_x| + |q_y - p_y|$$



Distance Measure

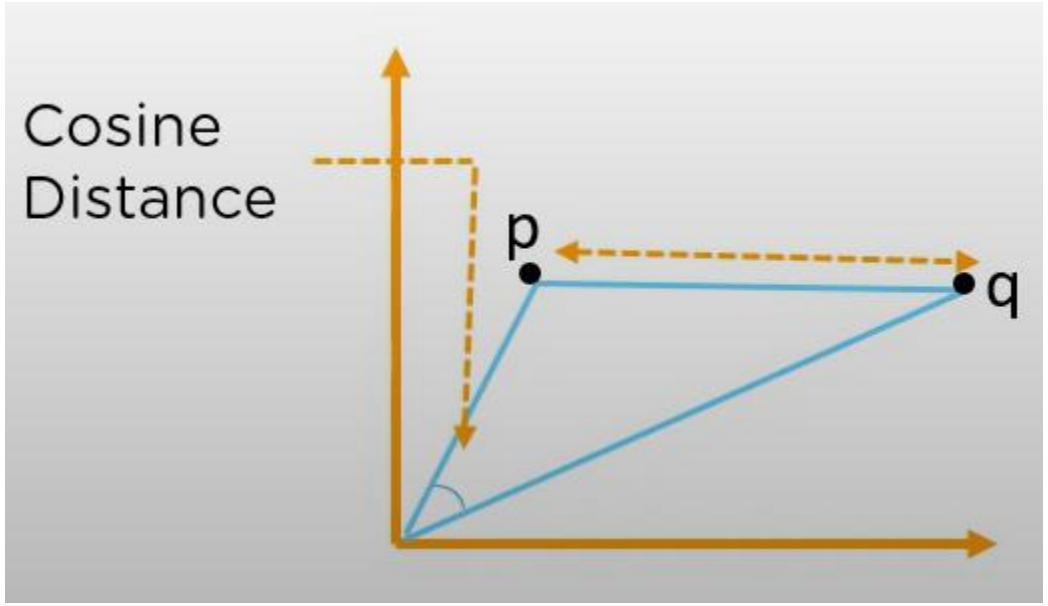
4. Cosine distance measure

- The cosine distance is a measure of similarity between two non-zero vecotrs in an inner product space.

$$Cosine\ similarity = \frac{\boldsymbol{p}.\boldsymbol{q}}{|\boldsymbol{p}||\boldsymbol{q}|}$$

Where

- **p.q** is the dot product of two vectors.
- $|\boldsymbol{p}|$ is the magnitude (or norm) of vector **p**.
- $|\boldsymbol{q}|$ is the magnitude (or norm) of vector **q**.



$$Cosine\ distance = 1 - cosine\ similarity$$

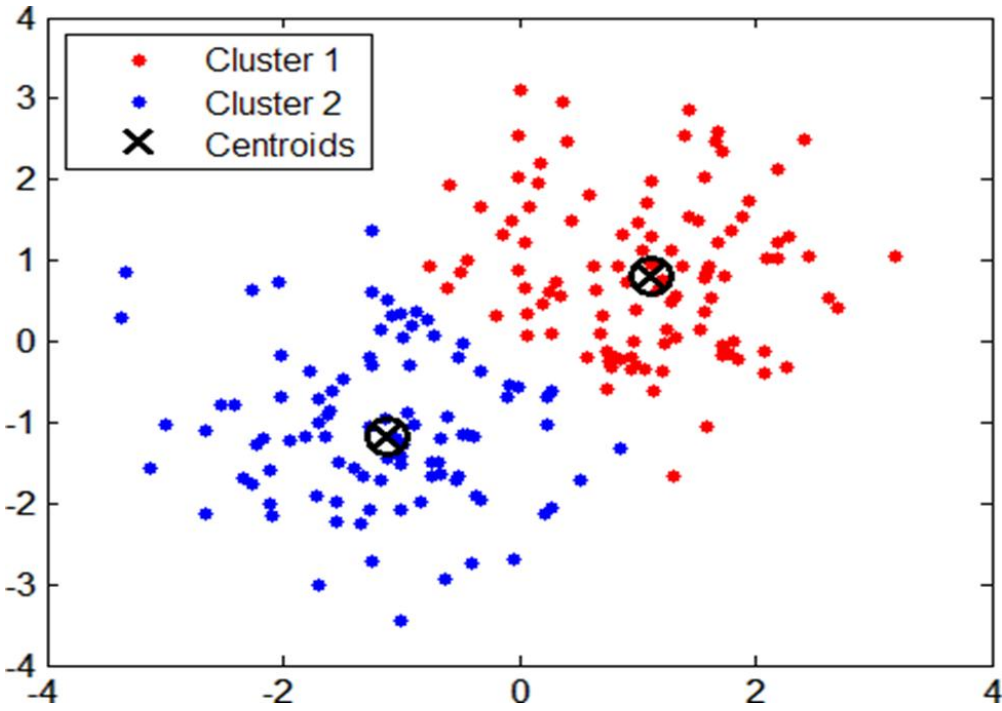
K-means Clustering

- The **k-means algorithm** is an algorithm to cluster n objects based on attributes into k partitions, where $k < n$.

- K-means algorithm is the simplest partitioning method for clustering analysis and widely used in data mining applications.

- Each cluster is represented by the **centre of the cluster** and the algorithm converges to stable centriods of clusters.

- It is similar to the expectation-maximization algorithm for mixtures of Gaussians in that they both attempt to find the centers of natural clusters in the data.



K-means Algorithm

Basic idea: randomly initialize the *k* cluster centers and iterate between the two steps:

1. Randomly initialize the cluster centers, c_1, ..., c_K

2. Given cluster centers, determine points in each cluster
   - For each point p, find the closest c_i. Put p into cluster I

3. Given points in each cluster, solve for c_i
   - Set c_i to be the mean of points in cluster I

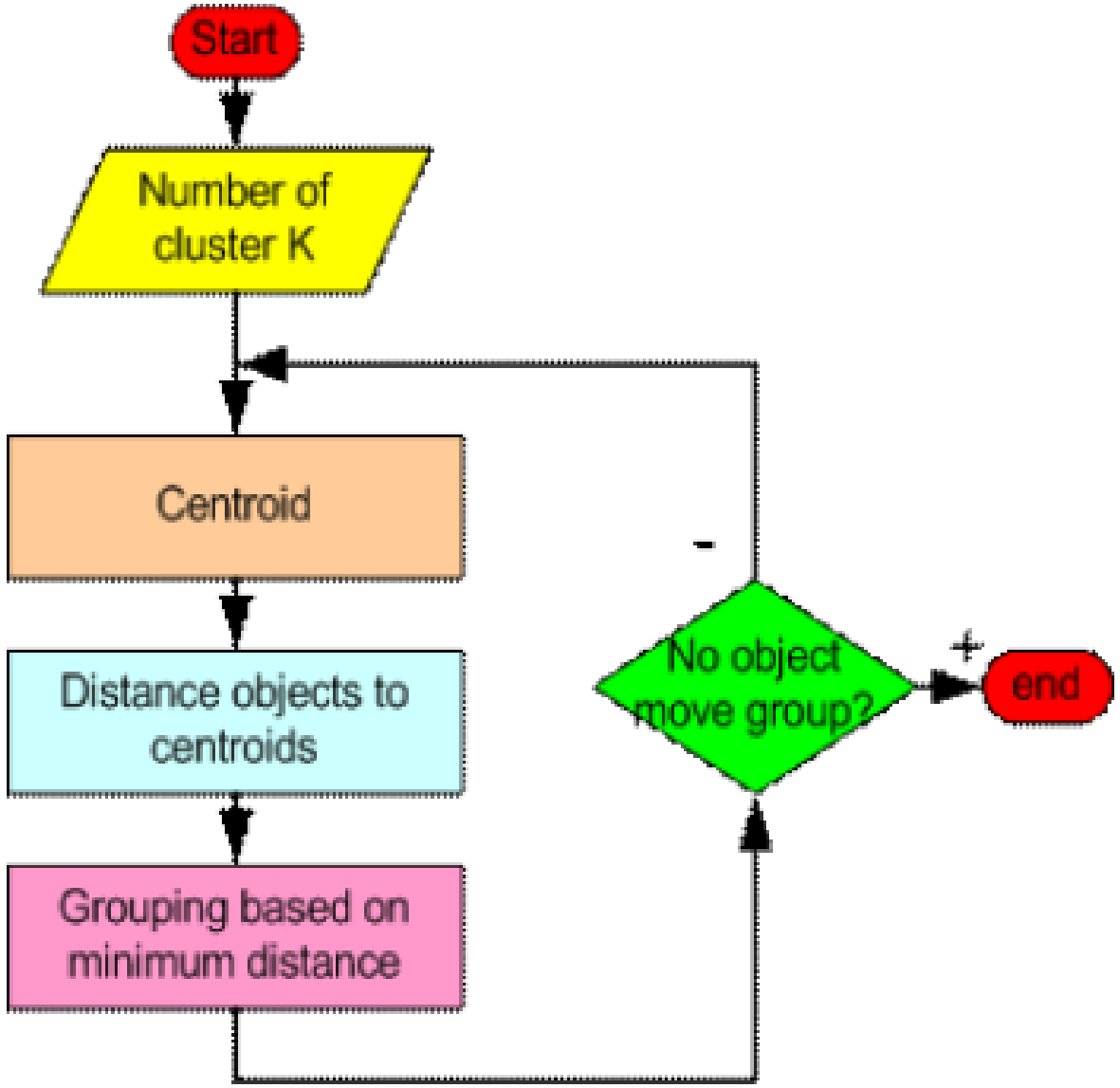
4. If c_i has changed, repeat Step 2

Properties

- Will always converge to *some* solution
- Can be a "local minimum"
  - does not always find the global minimum of objective function:

$$\sum_{\text{clusters } i} \sum_{\text{points p in cluster } i} \|p - c_i\|^2$$

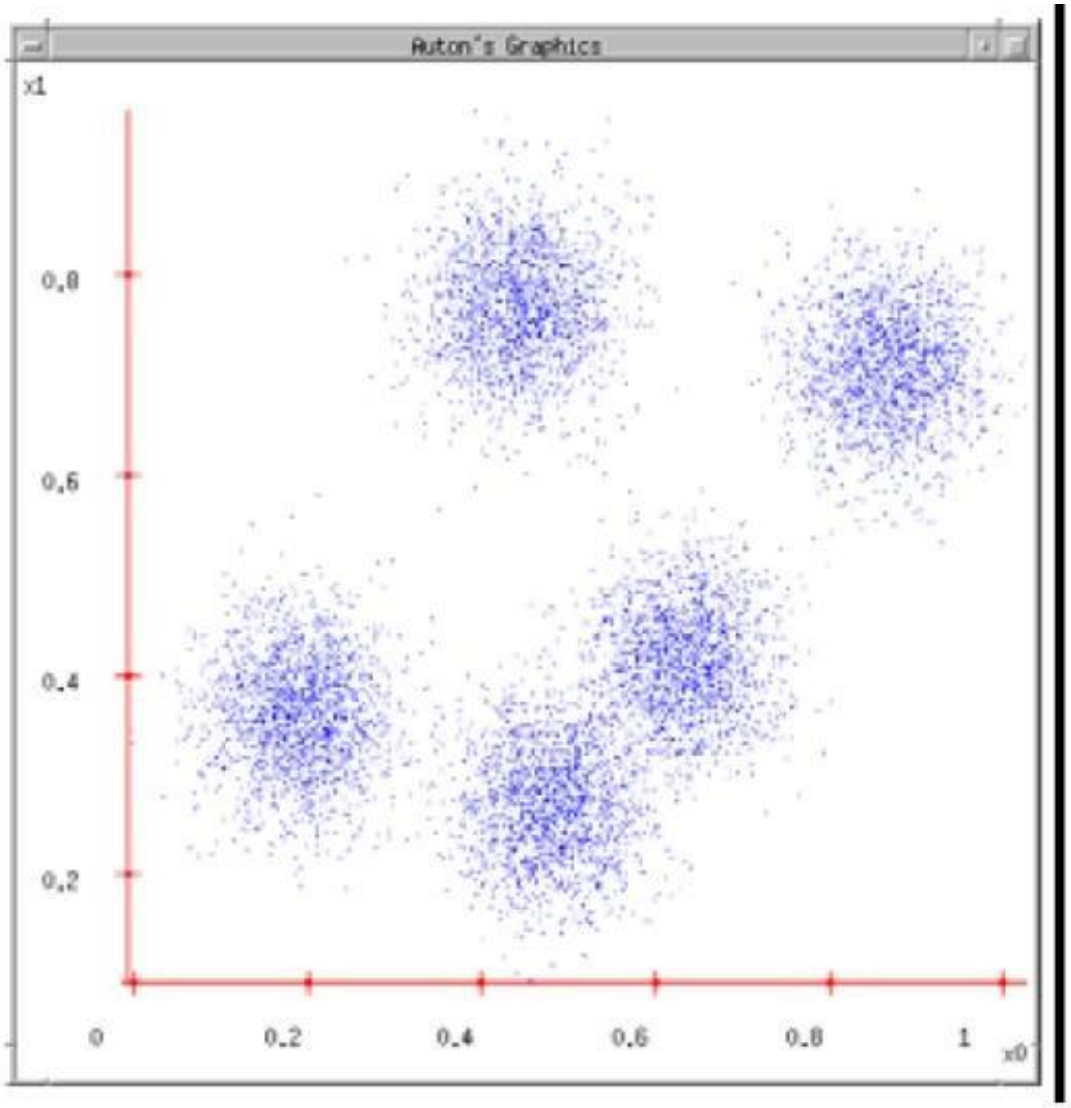
K-means Algorithm



K-means Example

K-means

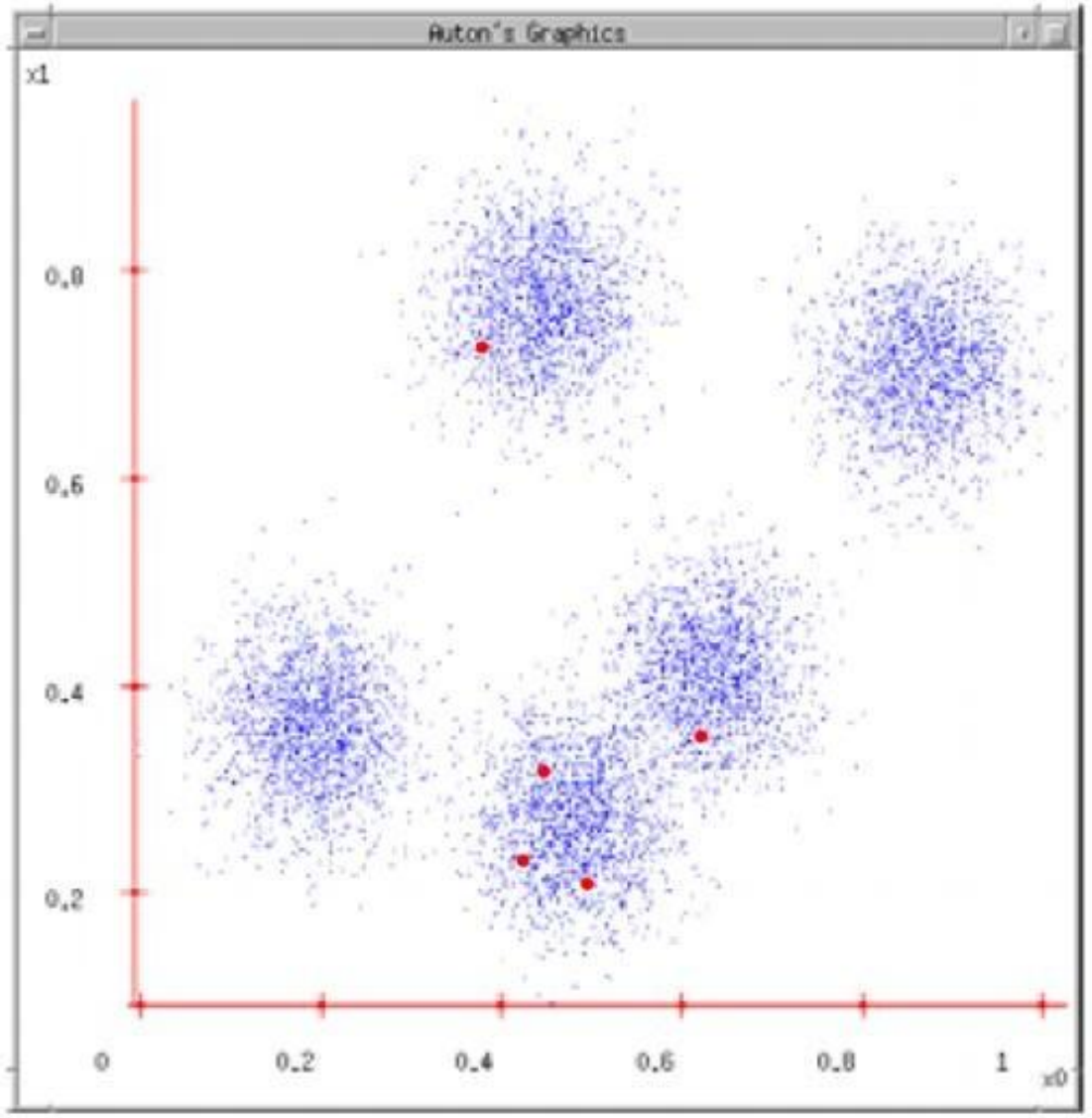
1. Ask user how many clusters they'd like. *(e.g. k=5)*



K-means Example

K-means

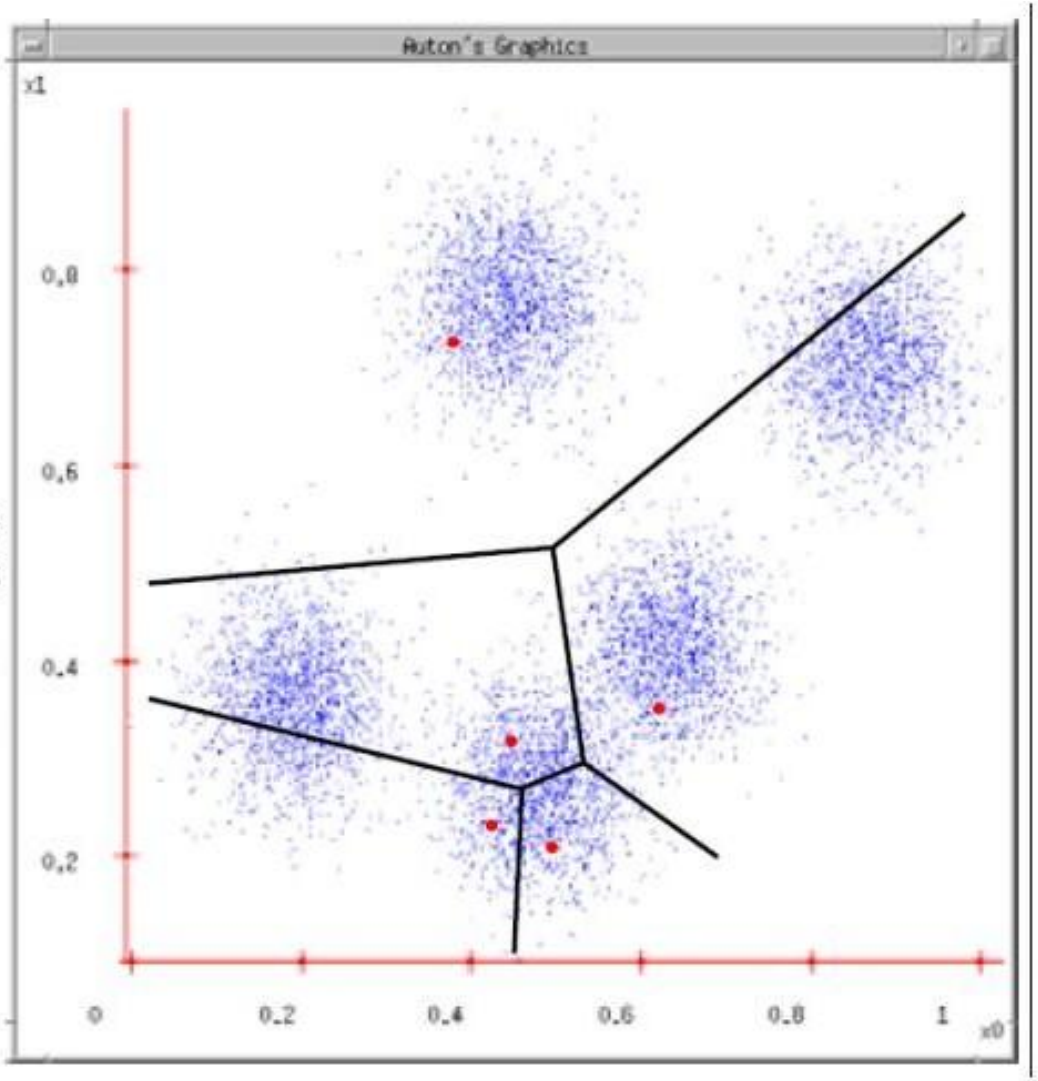
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations



K-means Example

K-means

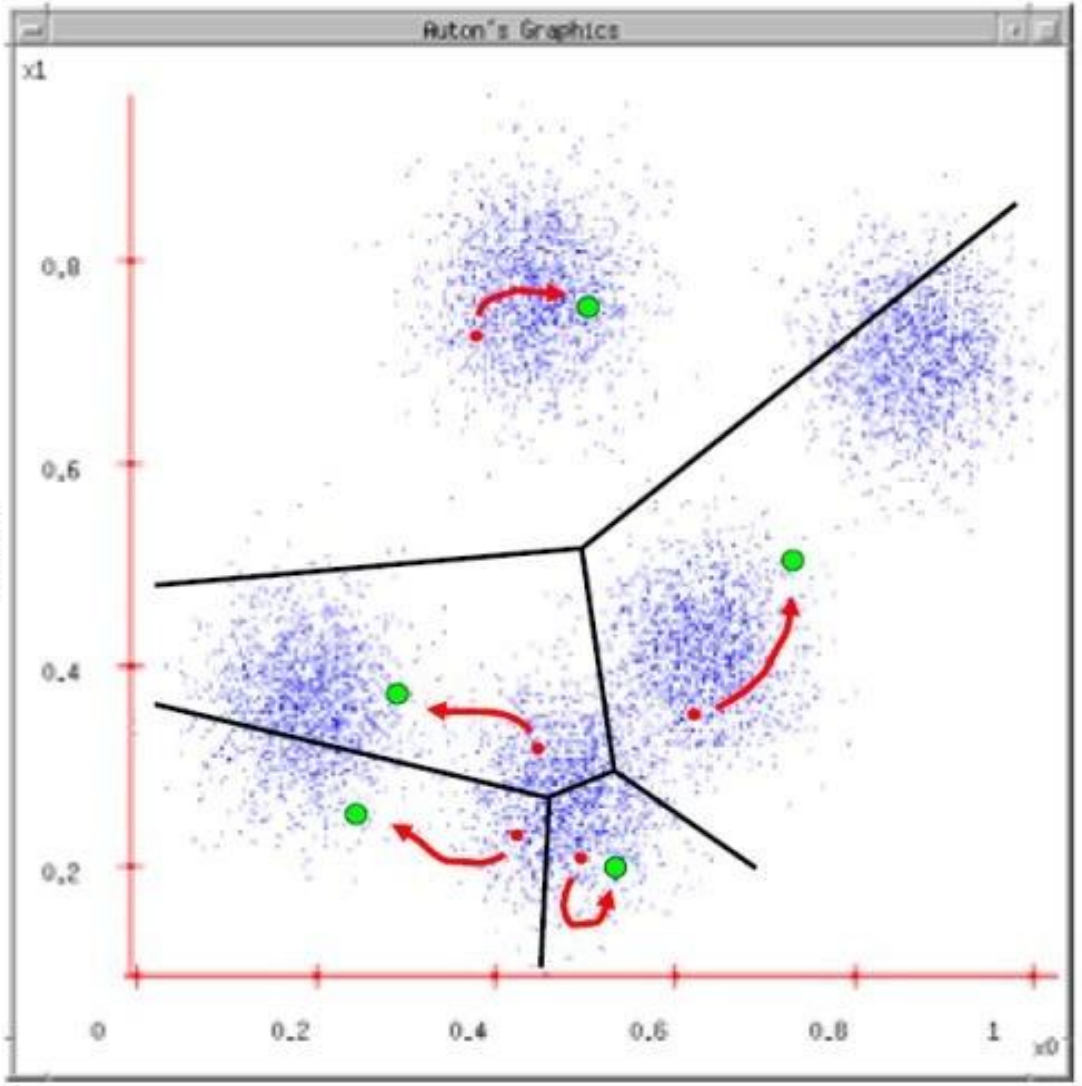
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to. (Thus each Center "owns" a set of datapoints)



K-means Example

K-means

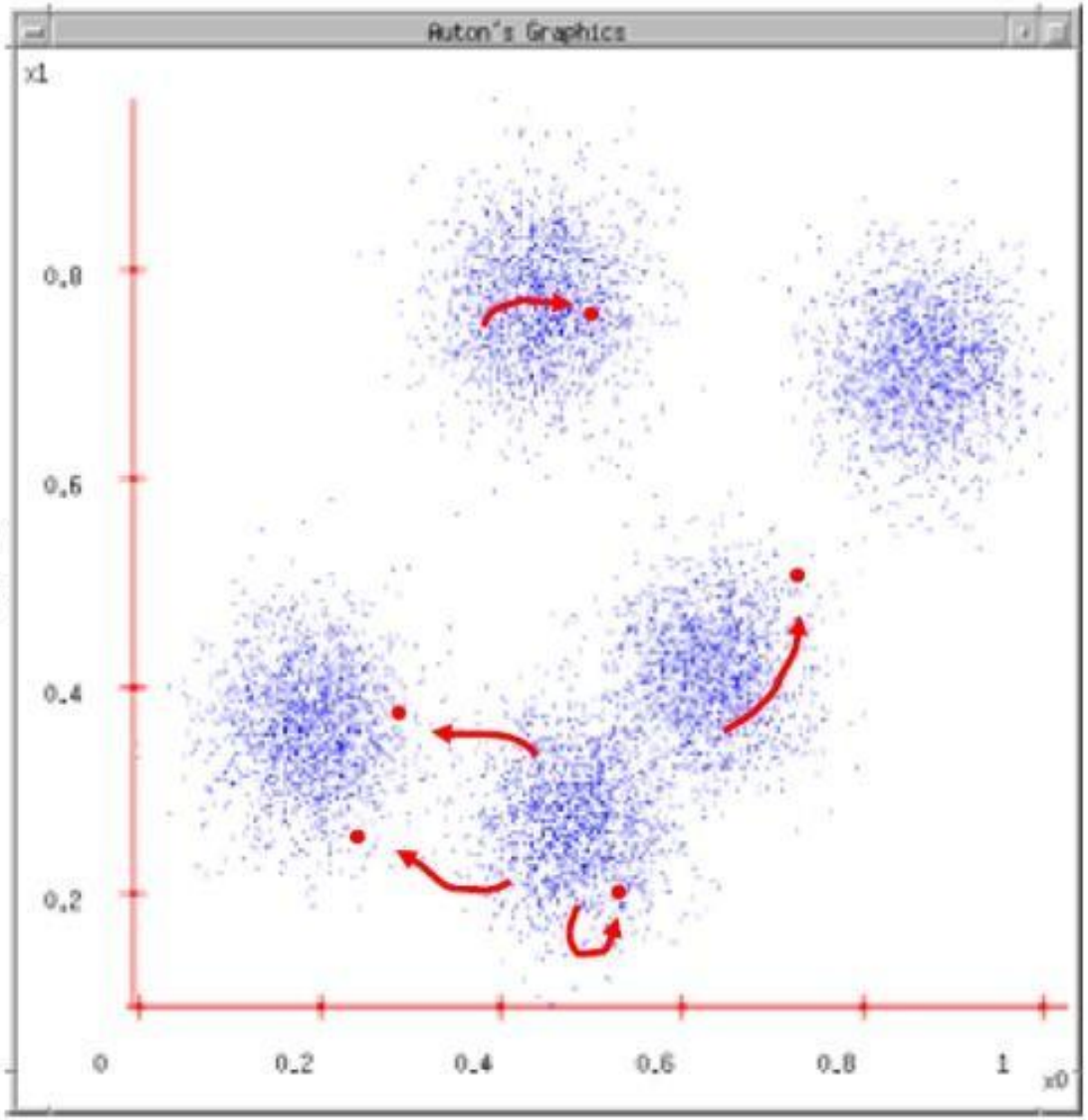
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns



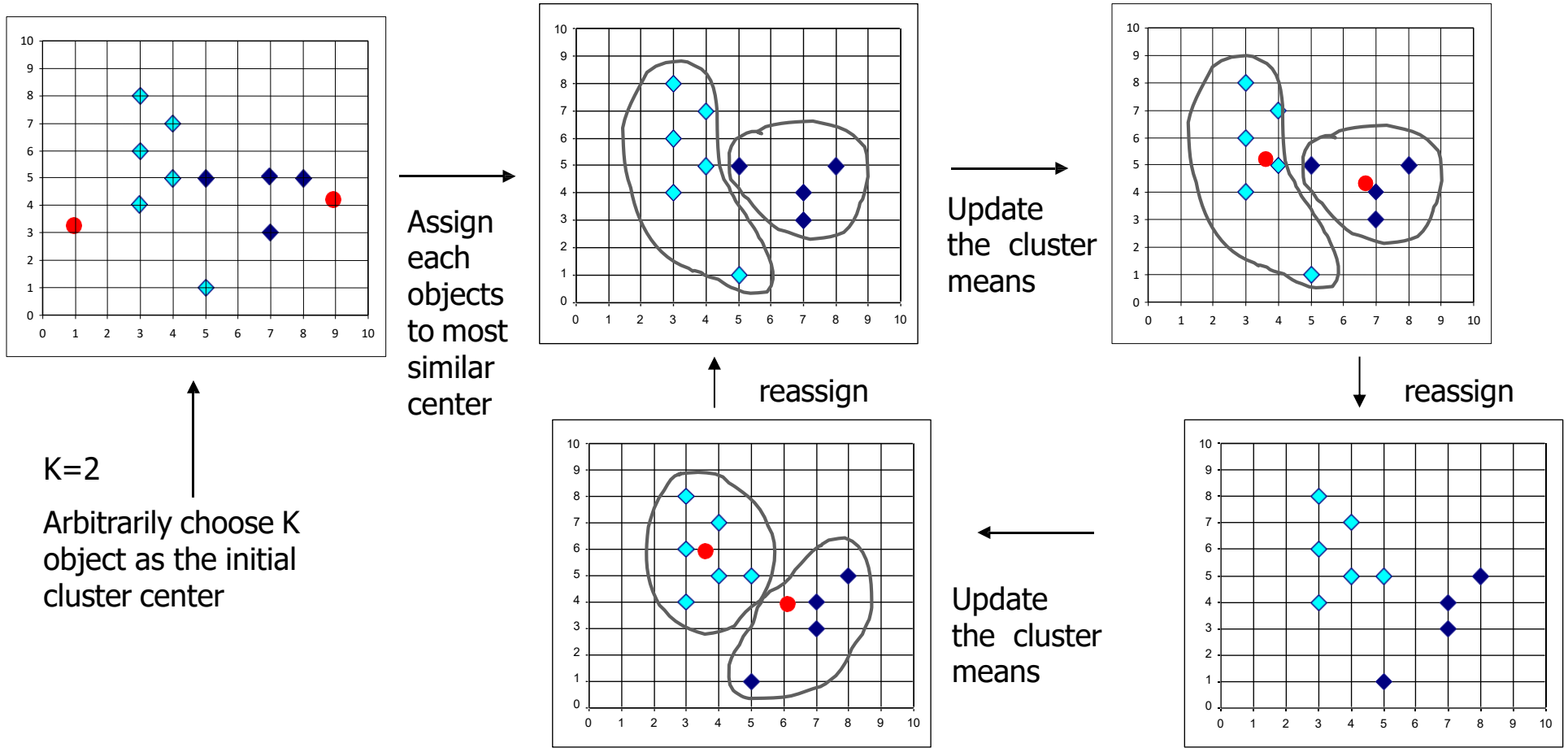
K-means Example

K-means

1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns...
5. ...and jumps there
6. ...Repeat until terminated!



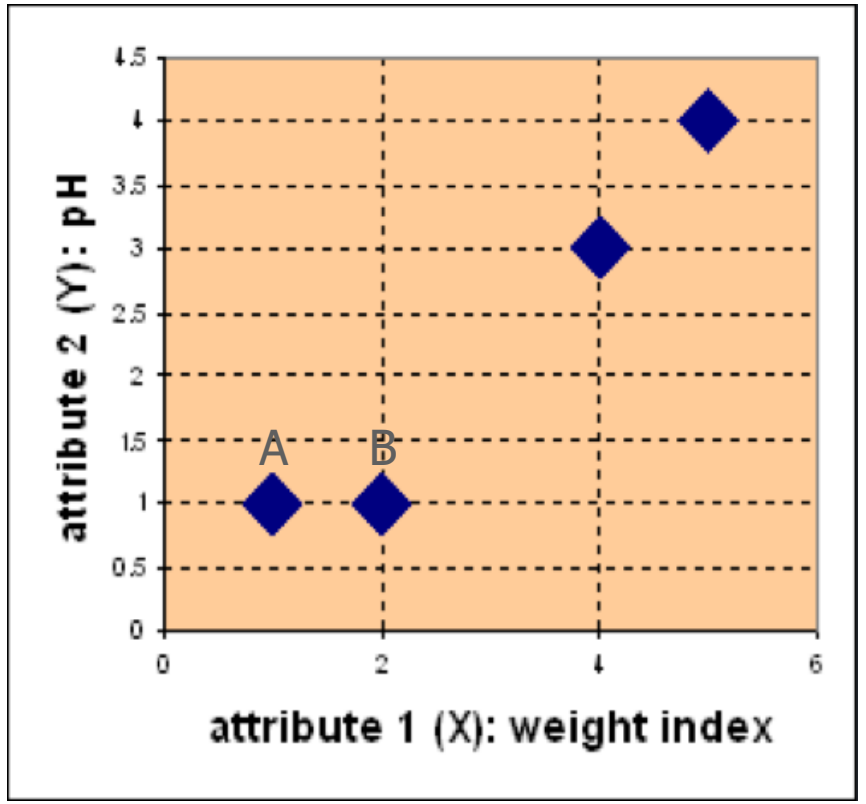
The K-Means Clustering Method



K-means Example-2

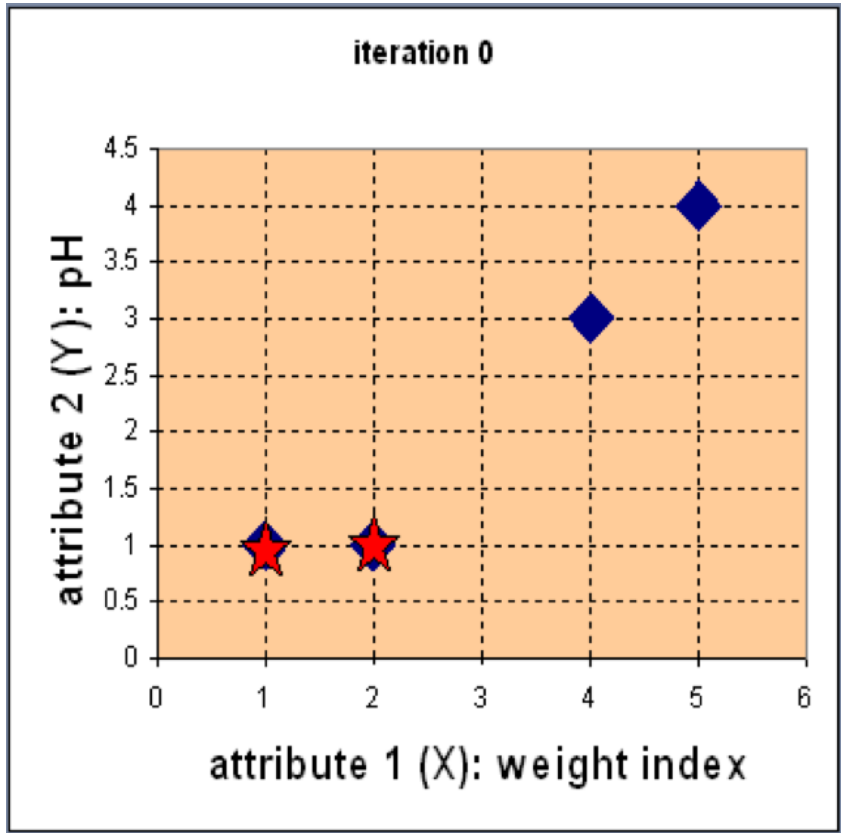
Suppose we have 4 types of medicines and each has two attributes (pH and weight index). Our goal is to group these objects into the *K=2* group of medicine.

Medicine	Weight	pH-Index
A	1	1
B	2	1
C	4	3
D	5	4



K-means Example-2

Step 1: Use initial centre points for partitioning



$$c_1 = A, c_2 = B$$

$$D^0 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 1 & 0 & 2.83 & 4.24 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) & group-1 \\ c_2 = (2,1) & group-2 \end{matrix}$$

$$\begin{matrix} A & B & C & D \\ \begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} & & & \begin{matrix} X \\ Y \end{matrix} \end{matrix}$$

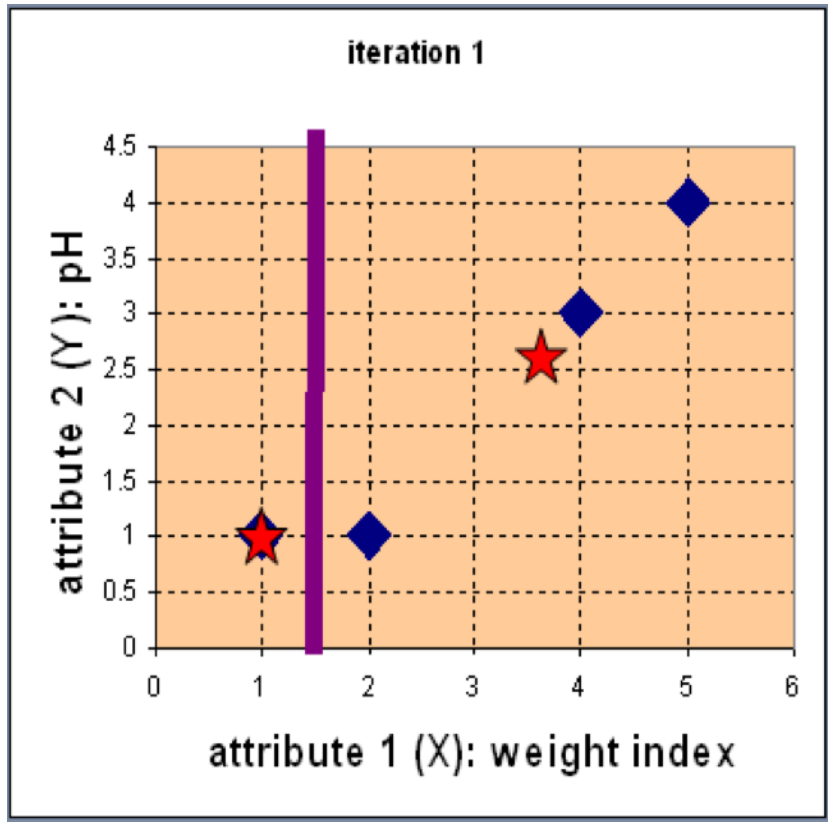
$$d(D,c_1) = \sqrt{(5-1)^2 + (4-1)^2} = 5$$

$$d(D,c_2) = \sqrt{(5-2)_2 + (4-1)_2} = 4.24$$

Assign each object to the cluster with the nearest seed point

K-means Example-2

Step 2: Compute new centroids of the current partition



Knowing the members of each cluster, now we compute the new centroid of each group based on these new memberships.

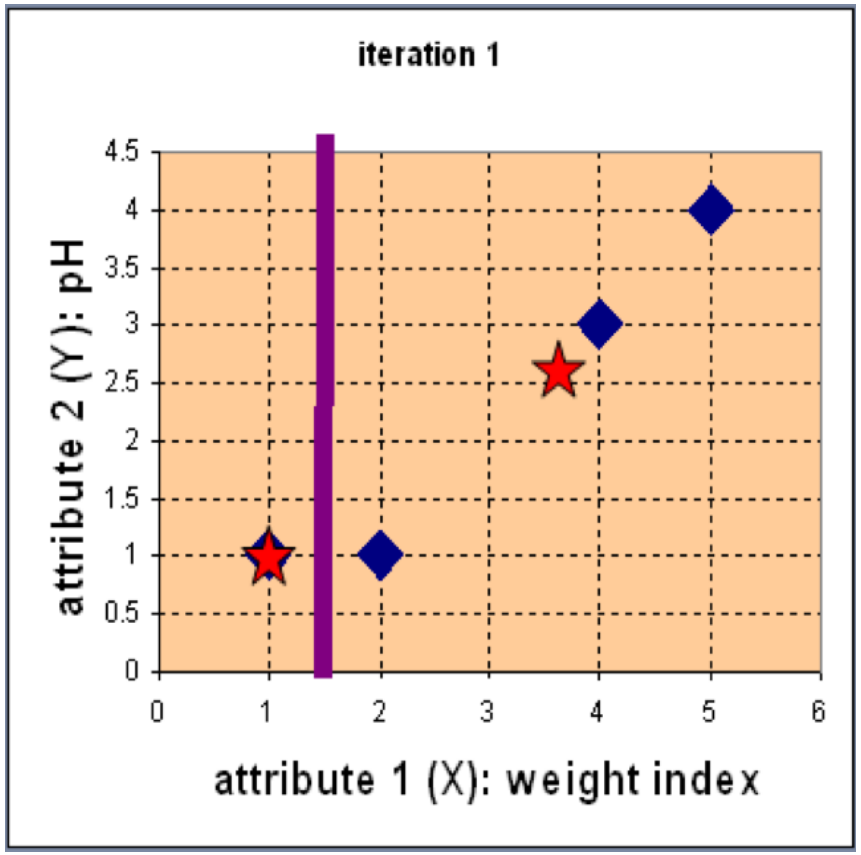
$$c_1 = (1, 1)$$

$$c_2 = \left(\frac{2+4+5}{3}, \frac{1+3+4}{3} \right)$$

$$= (\frac{11}{3}, \frac{8}{3})$$

K-means Example-2

Step 2: Renew membership based on new centroids



Compute the distance of all objects to the new centroids

$$D^1 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 3.14 & 2.36 & 0.47 & 1.89 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) & group-1 \\ c_2 = (\frac{11}{3}, \frac{8}{3}) & group-2 \end{matrix}$$

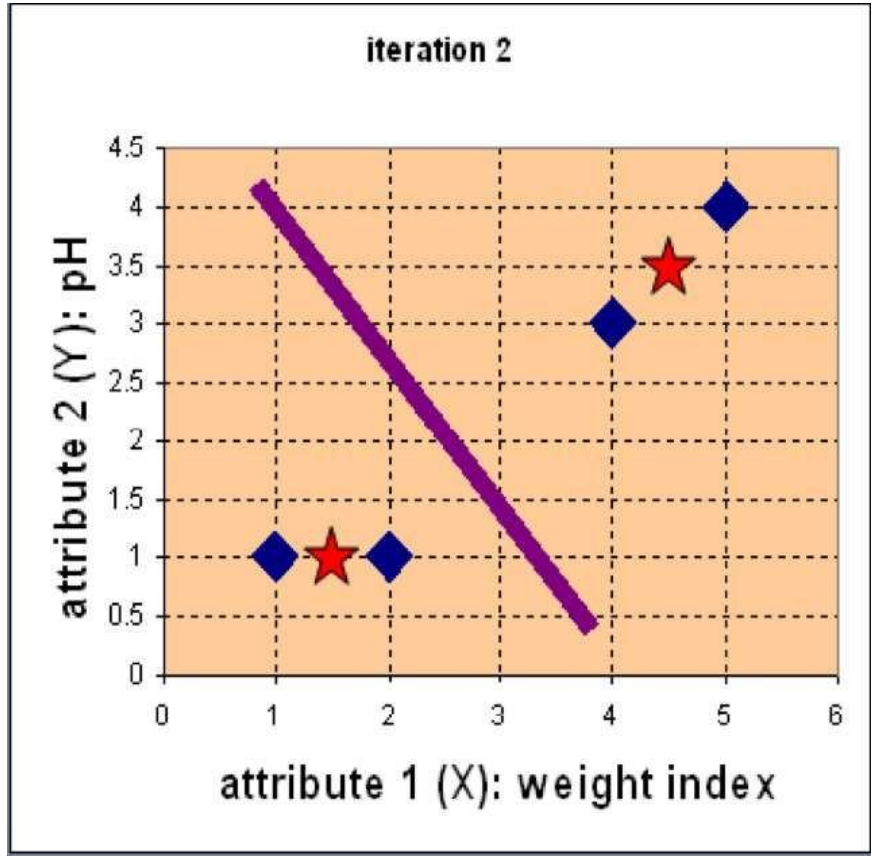
$$\begin{matrix} A & B & C & D \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \quad \begin{matrix} X \\ Y \end{matrix}$$

Assign the membership to objects

K-means Example-2

Step 3: Repeat the first two steps until its convergence



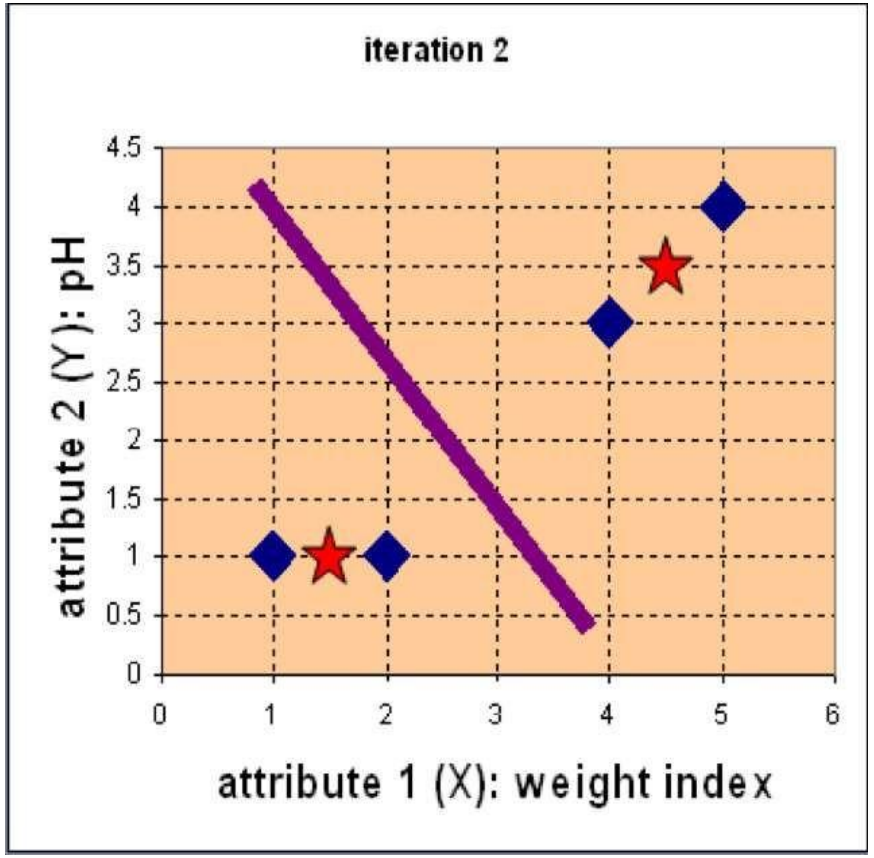
Knowing the members of each cluster, now we compute the new centroid of each group based on these new memberships.

$$c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1\frac{1}{2}, 1)$$

$$c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4\frac{1}{2}, 3\frac{1}{2})$$

K-means Example-2

Step 3: Repeat the first two steps until its convergence



Compute the distance of all objects to the new centroids

$$D^2 = \begin{bmatrix} 0.5 & 0.5 & 3.20 & 4.61 \\ 4.30 & 3.54 & 0.71 & 0.71 \end{bmatrix} \quad \begin{matrix} c_1 = (1\frac{1}{2}, 1) & group-1 \\ c_2 = (4\frac{1}{2}, 3\frac{1}{2}) & group-2 \end{matrix}$$

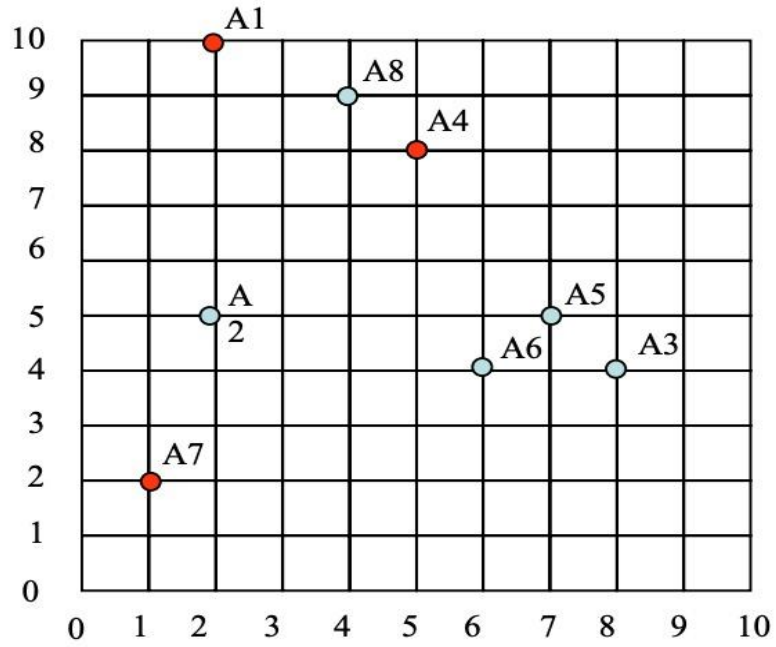
$$\begin{matrix} A & B & C & D \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \begin{matrix} X \\ Y \end{matrix}$$

Stop due to no new assignment
Membership in each cluster no longer change

Do it yourself

- Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters: A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9).



The distance matrix based on the Euclidean distance is given below:

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	$\sqrt{25}$	$\sqrt{72}$	$\sqrt{13}$	$\sqrt{50}$	$\sqrt{52}$	$\sqrt{65}$	$\sqrt{5}$
A2		0	$\sqrt{37}$	$\sqrt{18}$	$\sqrt{25}$	$\sqrt{17}$	$\sqrt{10}$	$\sqrt{20}$
A3			0	$\sqrt{25}$	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{53}$	$\sqrt{41}$
A4				0	$\sqrt{13}$	$\sqrt{17}$	$\sqrt{52}$	$\sqrt{2}$
A5					0	$\sqrt{2}$	$\sqrt{45}$	$\sqrt{25}$
A6						0	$\sqrt{29}$	$\sqrt{29}$
A7							0	$\sqrt{58}$
A8								0

Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7. Run the k-means algorithm for 1 epoch only.

At the end of this epoch show:

a) The new clusters (i.e. the examples belonging to each cluster)

b) The centers of the new clusters

How can we choose a "good" K for K-means clustering?

- Choose the number of clusters by visually inspecting your data points

- **Eblow Method:**
  - First of all, compute the sum of squared error (SSE) for some values of k (for example 2, 4, 6, 8, etc.). The SSE is defined as the sum of the squared distance between each member of the cluster and its centroid.

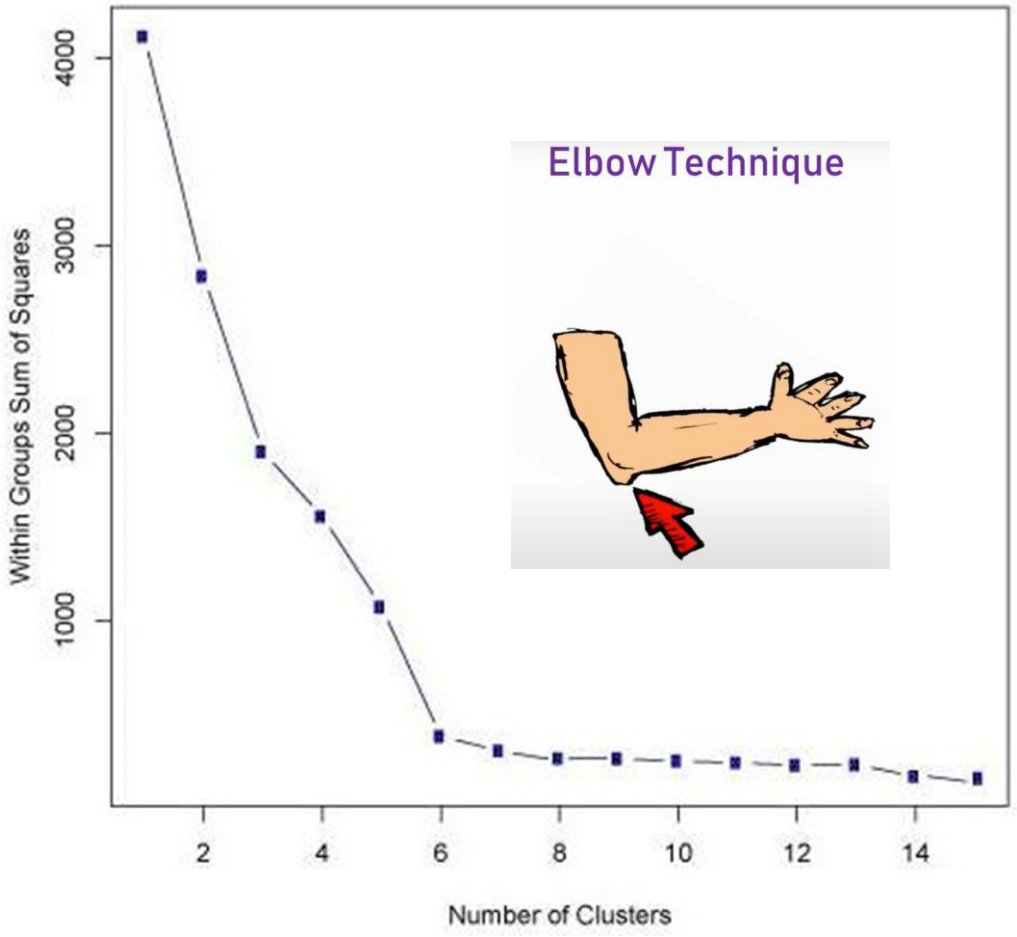
  - Mathematically:

$$SSE = \sum_{i=1}^{K} \sum_{x \epsilon c_i} dist(x_i, c_i)^2$$

  - If you plot k against the SSE, you will see that *the error decreases as k gets larger*; this is because when the number of clusters increases, they should be smaller, so distortion is also smaller. The idea of the elbow method is to choose the k at which the SSE decreases abruptly.

Elbow Effect

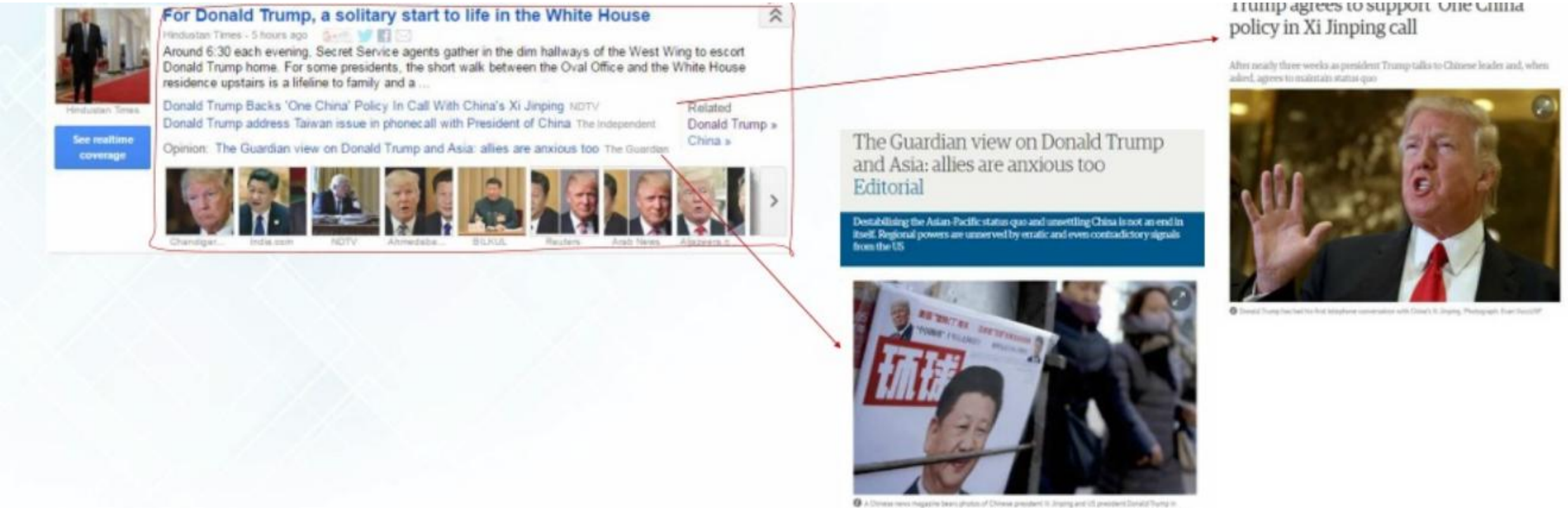
- In this case, k=6 is the value that the Elbow method has selected. Take into account that the Elbow method is a heuristic and, as such, it may or may not work well in your particular case.
- Sometimes there is more than one elbow or no elbow at all. In those situations, you usually end up calculating the best k by evaluating how well k-means performs in the context of the particular clustering problem you are trying to solve.



Application of K-Means

- Various news URLs related to Trump and Modi are grouped under a section.

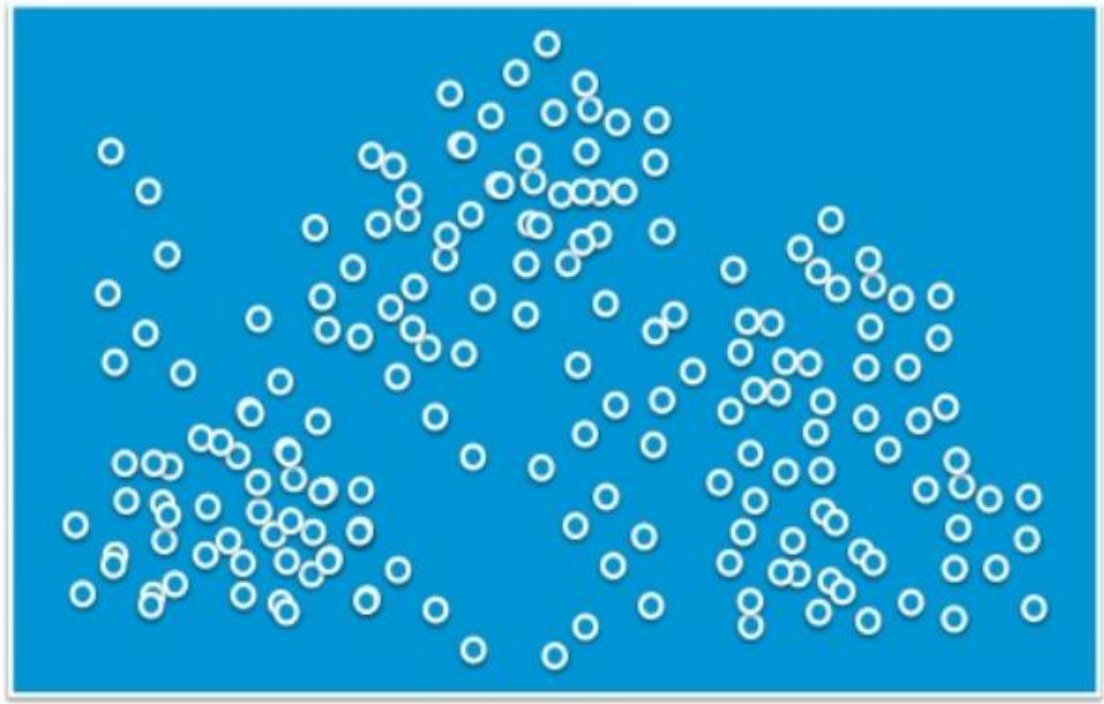
- K-means clustering automatically clusters new stories about the same topic into pre-defined clusters.



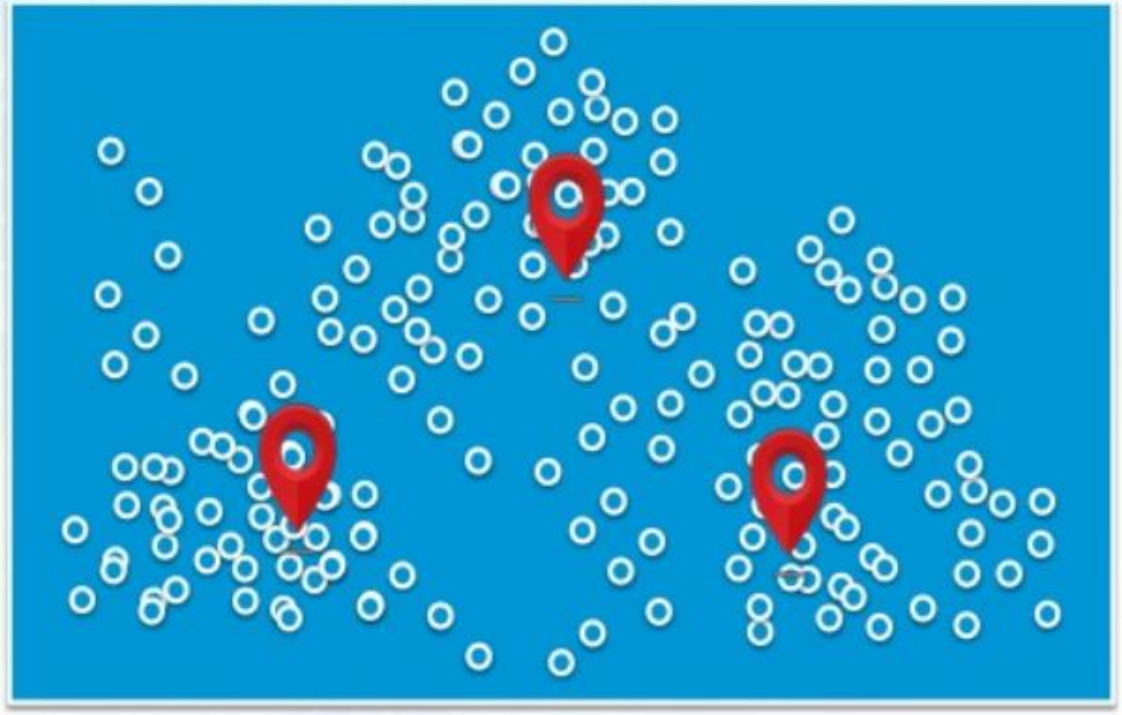
Application of K-Means

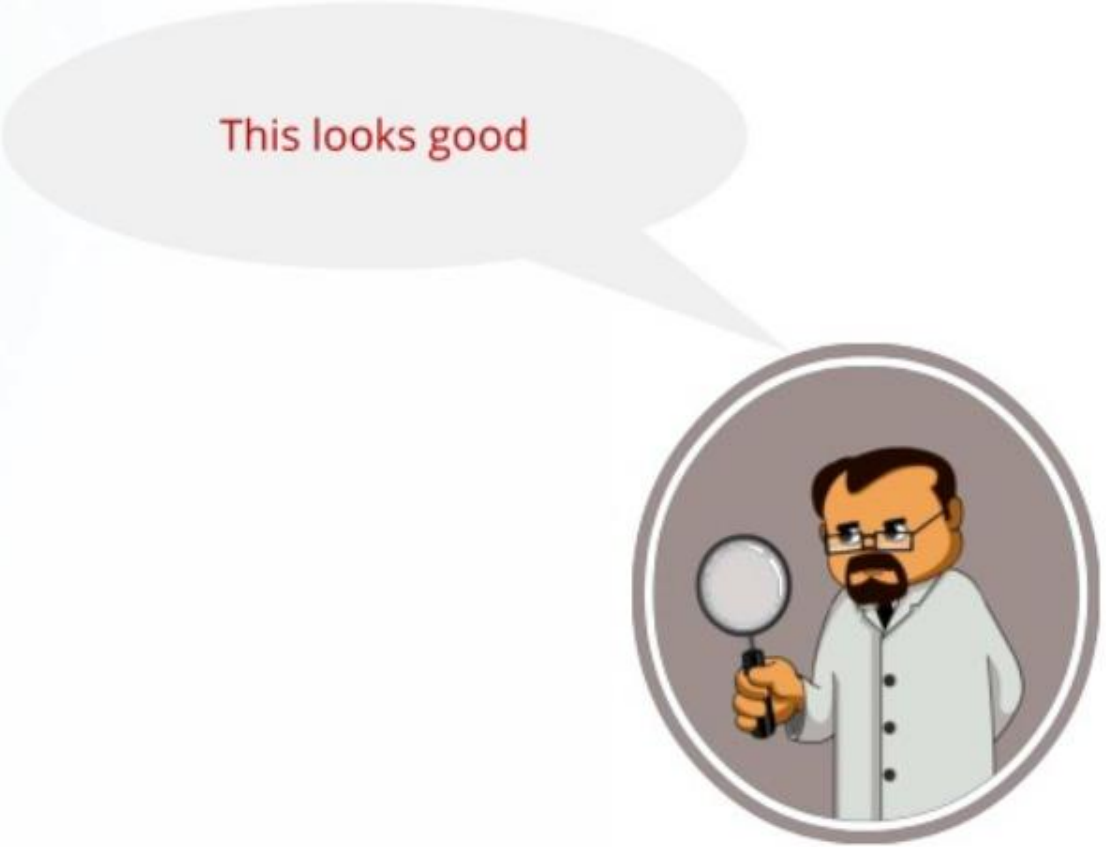
- The plot of students in an area is as given below.





Example-Solution





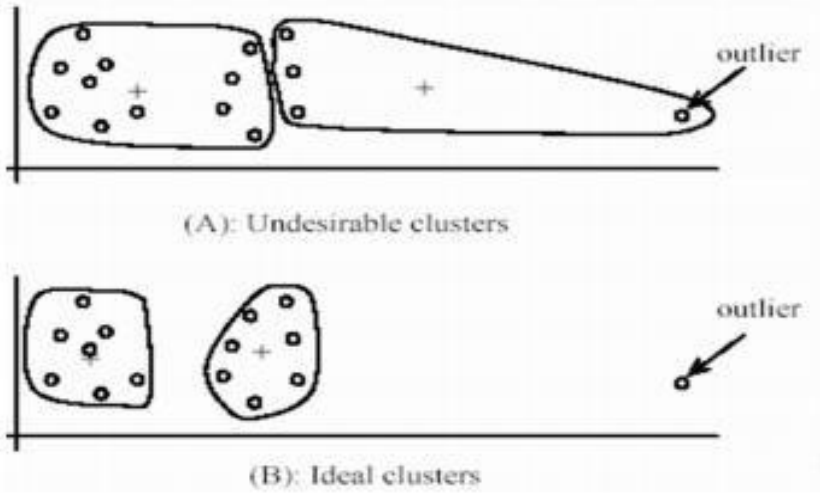
Pros and Cons of K-means method

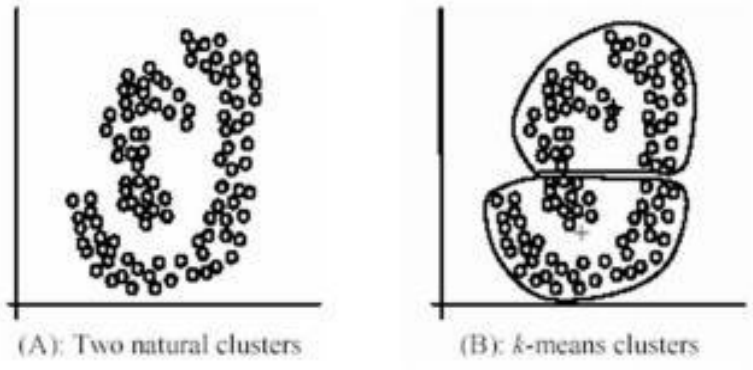
Pros

- Simple, fast to compute
- Converges to a local minimum of within-cluster squared error

Cons

- Setting k?
- Sensitive to initial centers
- Sensitive to outliers
- Detects spherical clusters





Implementation of K-means

IDAT7215 Lab 5 Tutorial- Unsupervised Machine Learning Algorithms

K-Means Algorithm Tutorial

[1]:
```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import MinMaxScaler
from matplotlib import pyplot as plt
%matplotlib inline
```

[2]:
```
# Load the dataset
df=pd.read_csv("C:/Users/itsme/Downloads/Lab 05-UnSupervised Learning/income.csv")

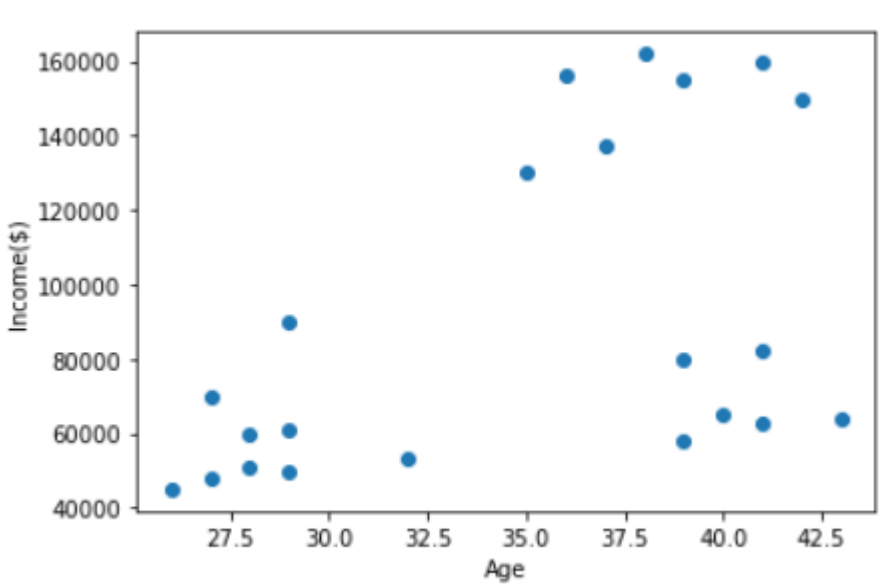
df.head()
```

[2]:

	Name	Age	Income($)
0	Rob	27	70000
1	Michael	29	90000
2	Mohan	29	61000
3	Ismail	28	60000
4	Kory	42	150000

Implementation of K-means

```
In [3]: # Plot the Age against income
        plt.scatter(df.Age,df['Income($)'])
        plt.xlabel('Age')
        plt.ylabel('Income($)')
```




```
In [4]: # Fit the K-means algorithm on the dataset considering age and income
        km = KMeans(n_clusters=3)
        y_predicted = km.fit_predict(df[['Age','Income($)']])
        y_predicted
```

```
Out[4]: array([0, 0, 2, 2, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0, 2])
```

```
In [5]: # See the cluster number against the datapoint
        df['cluster']=y_predicted
        df.head()
```

Out[5]:

	Name	Age	Income($)	cluster
0	Rob	27	70000	0
1	Michael	29	90000	0
2	Mohan	29	61000	2
3	Ismail	28	60000	2
4	Kory	42	150000	1

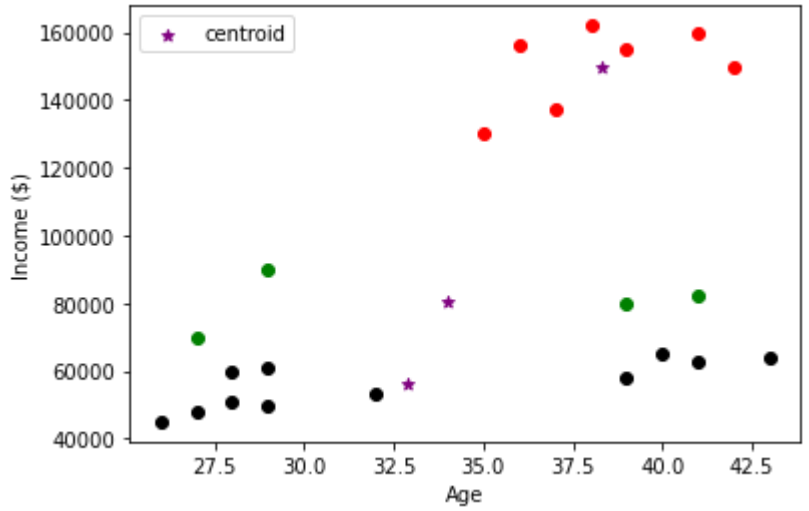
```
In [6]: # What are the cluster centre points
        km.cluster_centers_
```

```
Out[6]: array([[3.40000000e+01, 8.05000000e+04],
               [3.82857143e+01, 1.50000000e+05],
               [3.29090909e+01, 5.61363636e+04]])
```

Implementation of K-means

In [7]:
```
# Visualize the cluster with centre point against the datapoints with different colors

df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:,1],color='purple',marker='*',label='centroid')
plt.xlabel('Age')
plt.ylabel('Income ($)')
plt.legend()
```



Implementation of K-means

Preprocessing using min max scale

In [8]:
```
scaler = MinMaxScaler()

scaler.fit(df[['Income($)']])
df['Income($)'] = scaler.transform(df[['Income($)']])   # Transform the Income feature between 0 and 1.

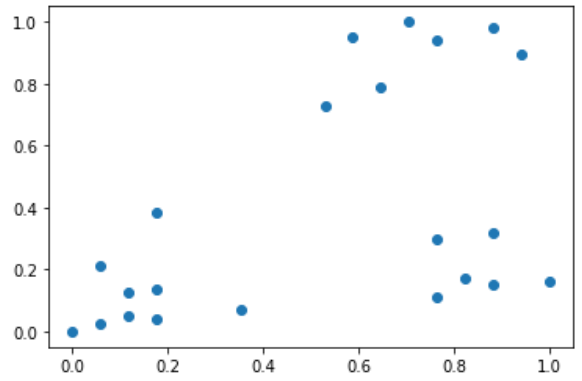
scaler.fit(df[['Age']])
df['Age'] = scaler.transform(df[['Age']])               # Transform the Age feature between 0 and 1.
```

In [9]:
```
df.head()
```

Out[9]:

	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	0
1	Michael	0.176471	0.384615	0
2	Mohan	0.176471	0.136752	2
3	Ismail	0.117647	0.128205	2
4	Kory	0.941176	0.897436	1

In [10]:
```
plt.scatter(df.Age,df['Income($)']);
```



Implementation of K-means

In [11]:
```
km = KMeans(n_clusters=3)
y_predicted = km.fit_predict(df[['Age','Income($)']])
y_predicted
```

Out[11]:
```
array([1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2])
```

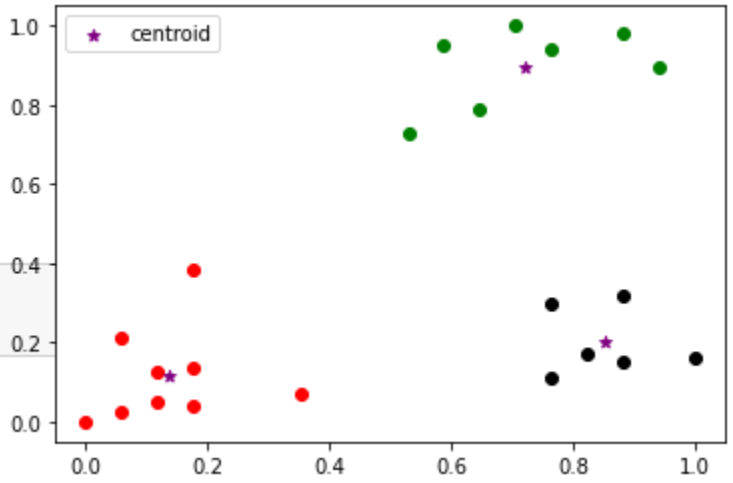
In [12]:
```
df['cluster']=y_predicted
df.head()
```

Out[12]:

	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	1
1	Michael	0.176471	0.384615	1
2	Mohan	0.176471	0.136752	1
3	Ismail	0.117647	0.128205	1
4	Kory	0.941176	0.897436	0

In [13]:
```
km.cluster_centers_
```

Out[13]:
```
array([[0.72268908, 0.8974359 ],
       [0.1372549 , 0.11633428],
       [0.85294118, 0.2022792 ]])
```



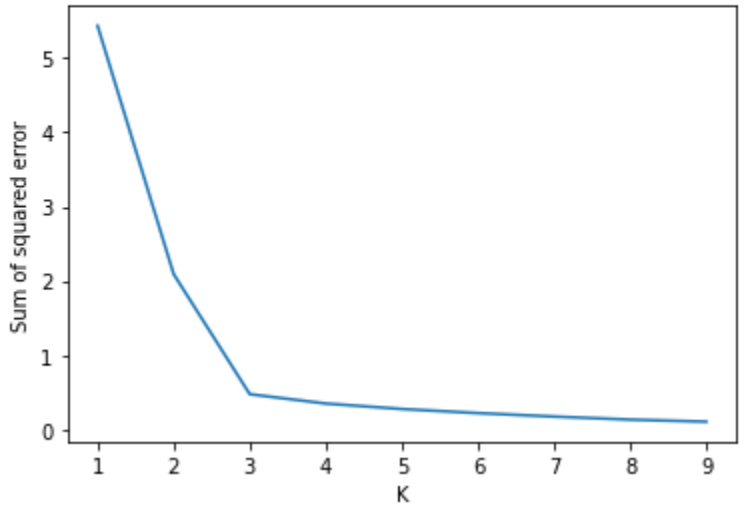
In [14]:
```
df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:,1],color='purple',marker='*',label='centroid')
```

Implementation of K-means

Elbow Plot

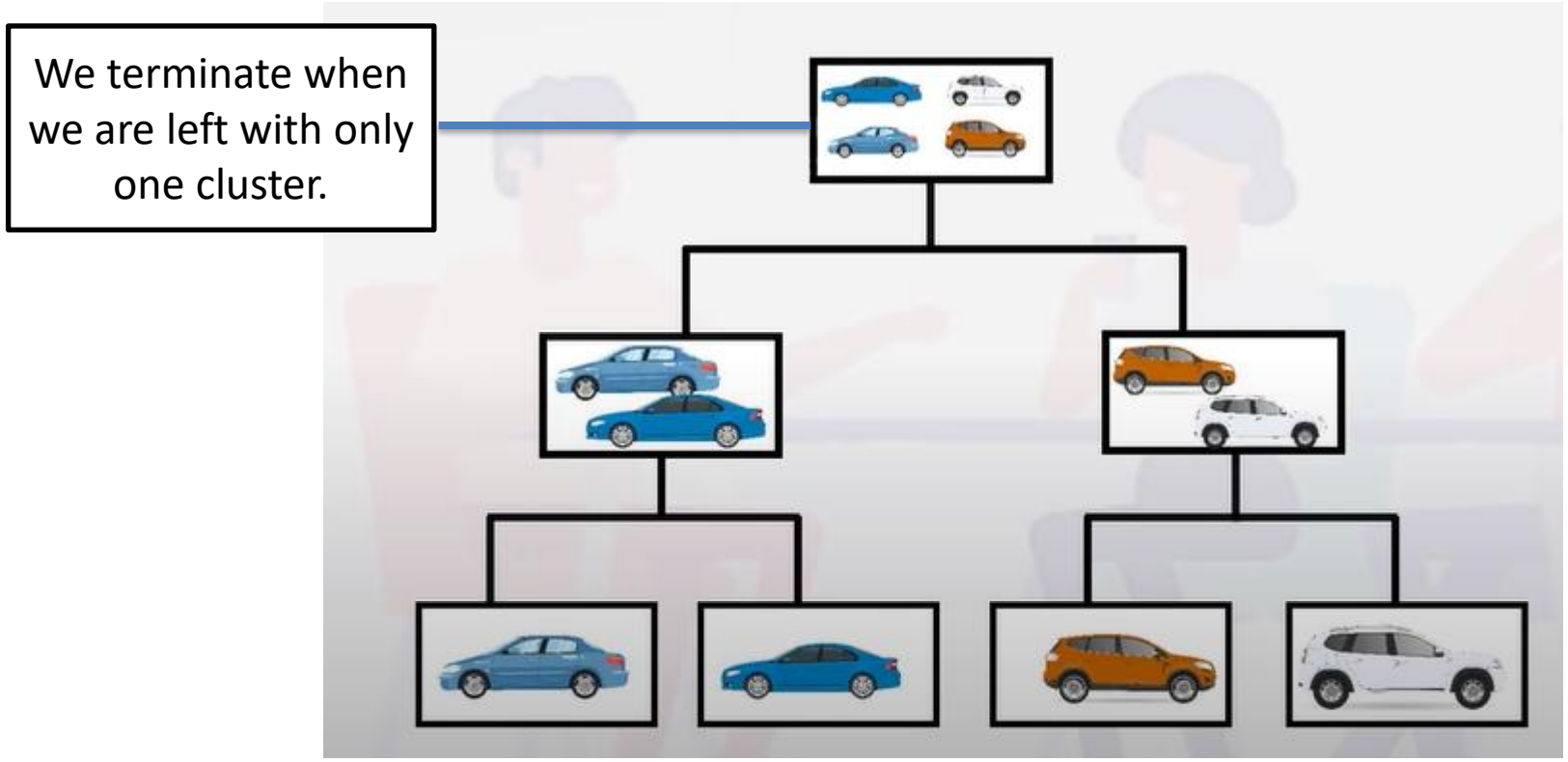
In [15]:
```
sse = []
k_rng = range(1,10)
for k in k_rng:
    km = KMeans(n_clusters=k)
    km.fit(df[['Age','Income($)']])
    sse.append(km.inertia_)
```

In [16]:
```
plt.xlabel('K')
plt.ylabel('Sum of squared error')
plt.plot(k_rng,sse);
```



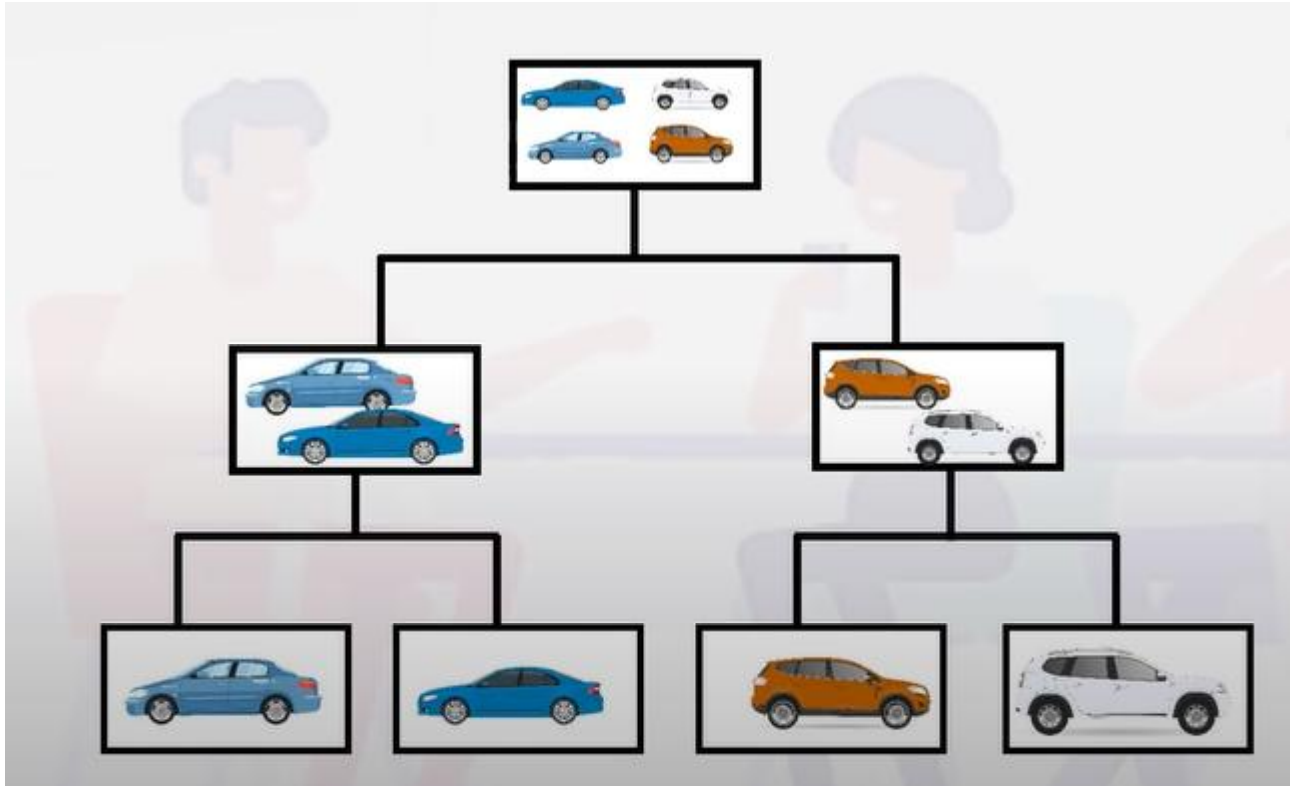
What is Hierarchical Clustering

1. Let's Consider that we have a set of cars and we have to group similar ones together
2. We get two clusters of car types, Sedan and SUV
3. At last we group everything into one cluster



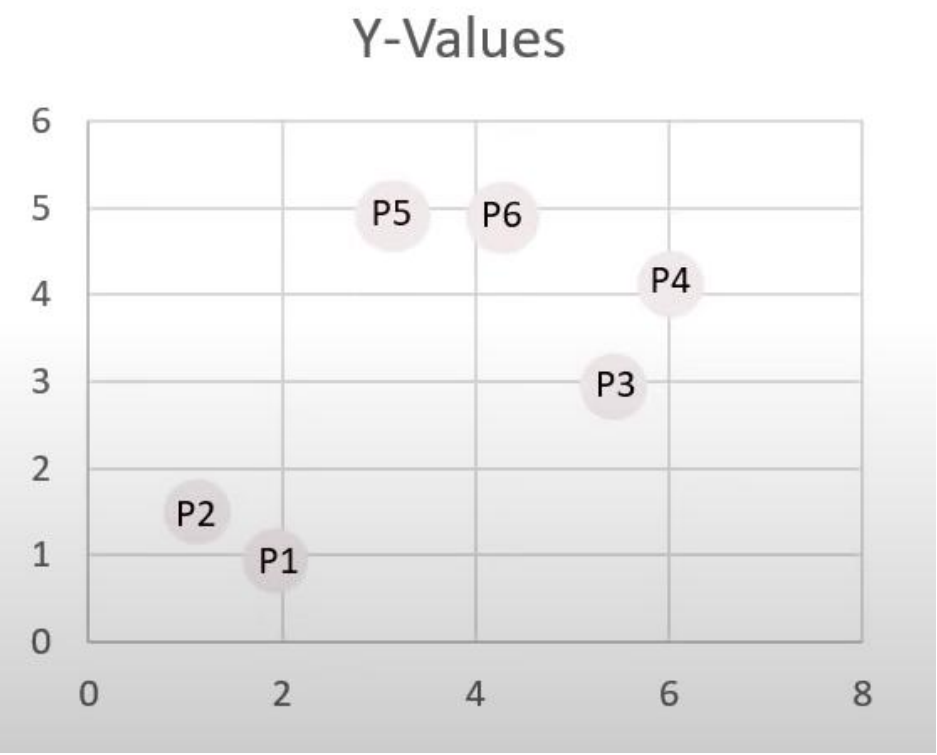
What is Hierarchical Clustering

- Hierarchical clustering separates data into different groups based on some similarity measure.
- Hierarchical clustering (also called hierarchical cluster analysis or HCA) is a method of cluster analysis that seeks to build a hierarchy of clusters.



How Hierarchical Clustering Works

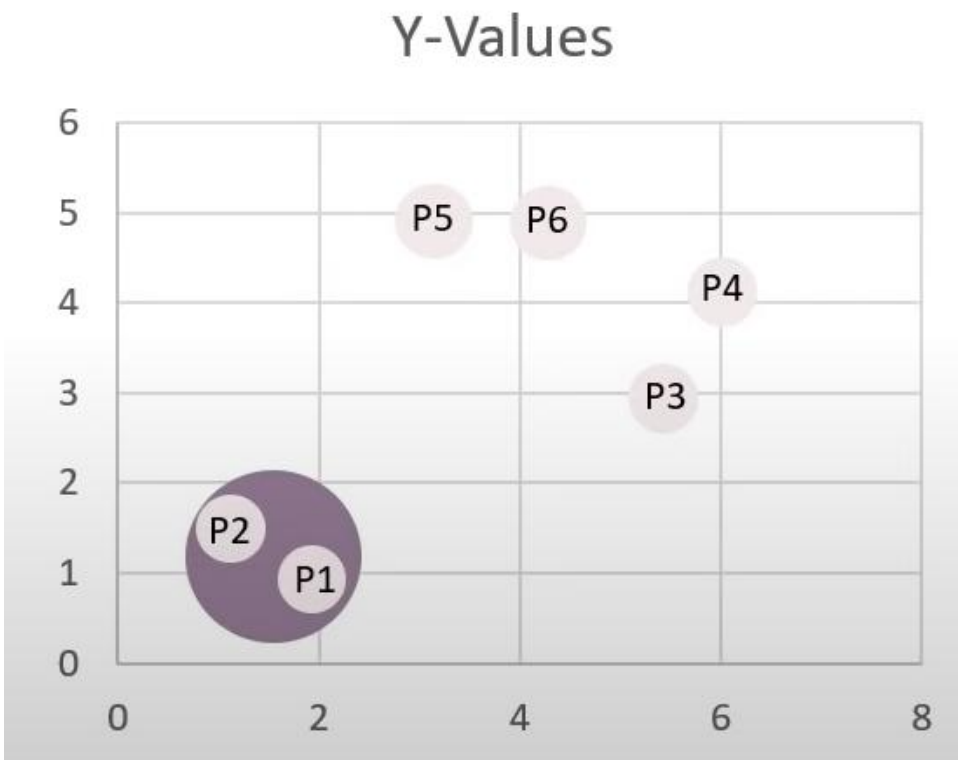
1. **Measure the distance**
   - Each data point is a cluster of its own.
   - We try to find the least distance between two data points/clusters.

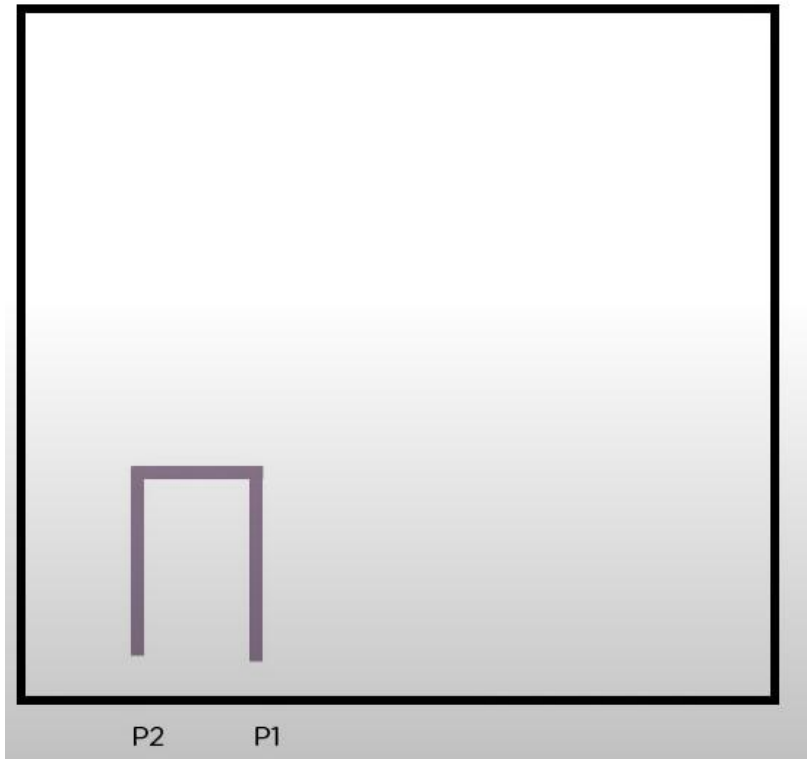


How Hierarchical Clustering Works

2. Grouping

- The two nearest clusters/data points are merged together.
- This is represented in a tree-like structure called Dendrogram.

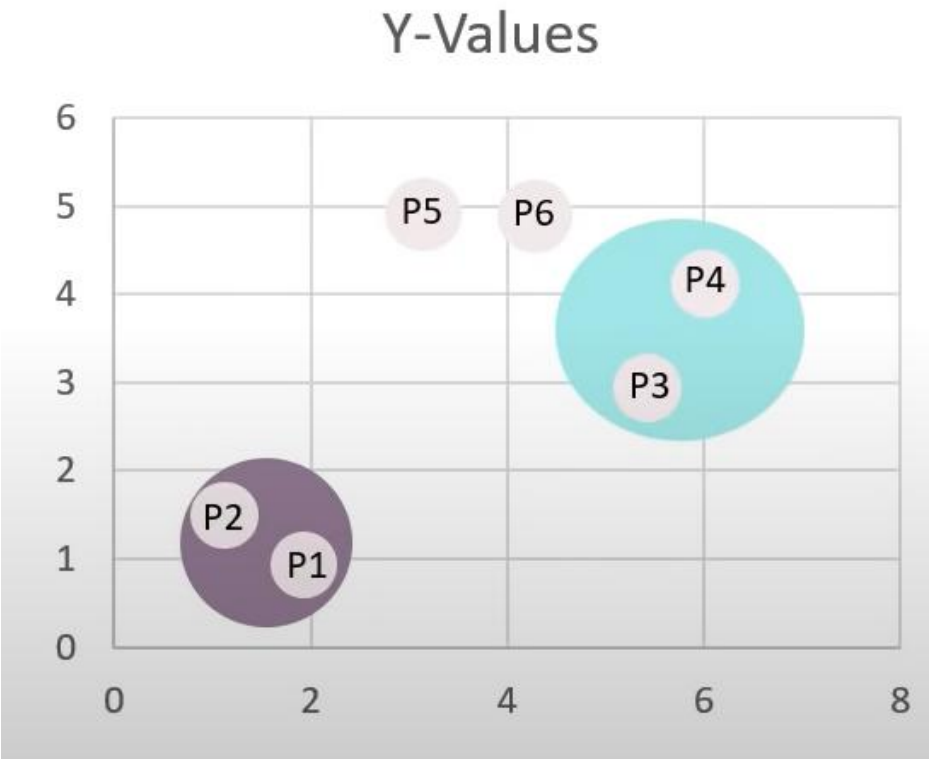


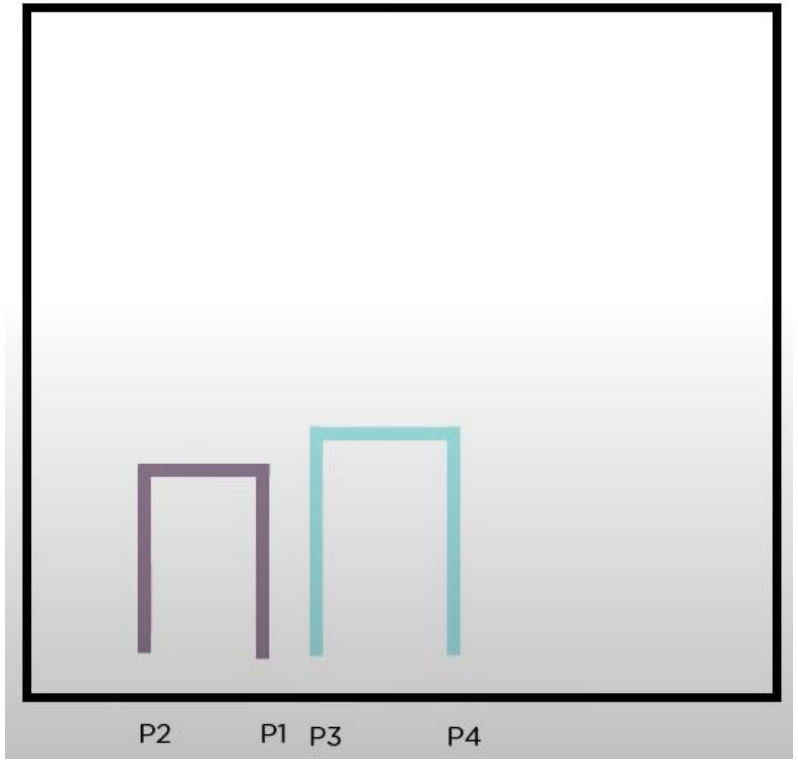


How Hierarchical Clustering Works

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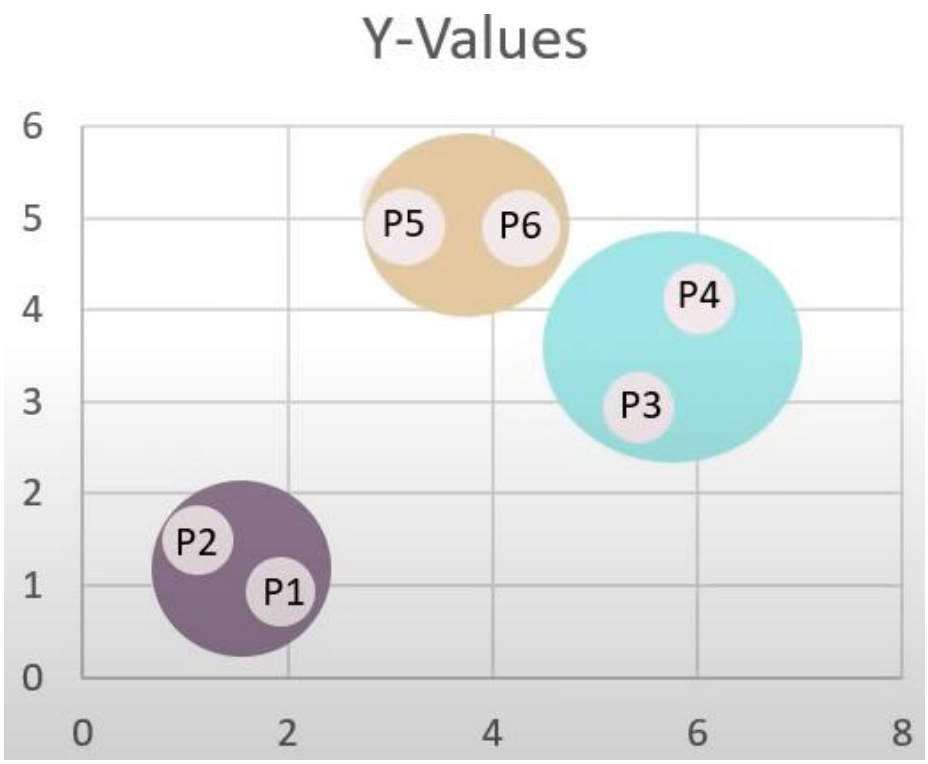


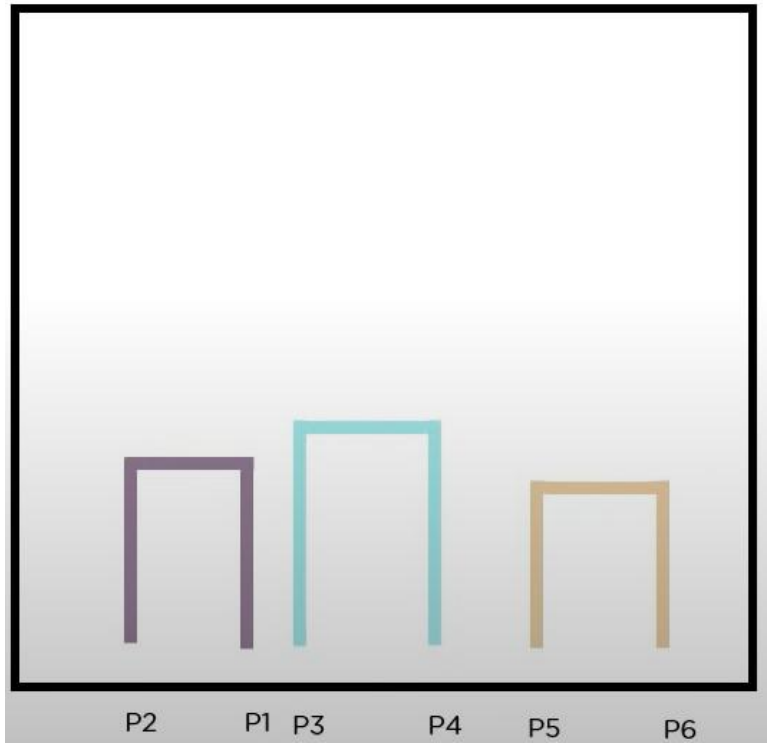


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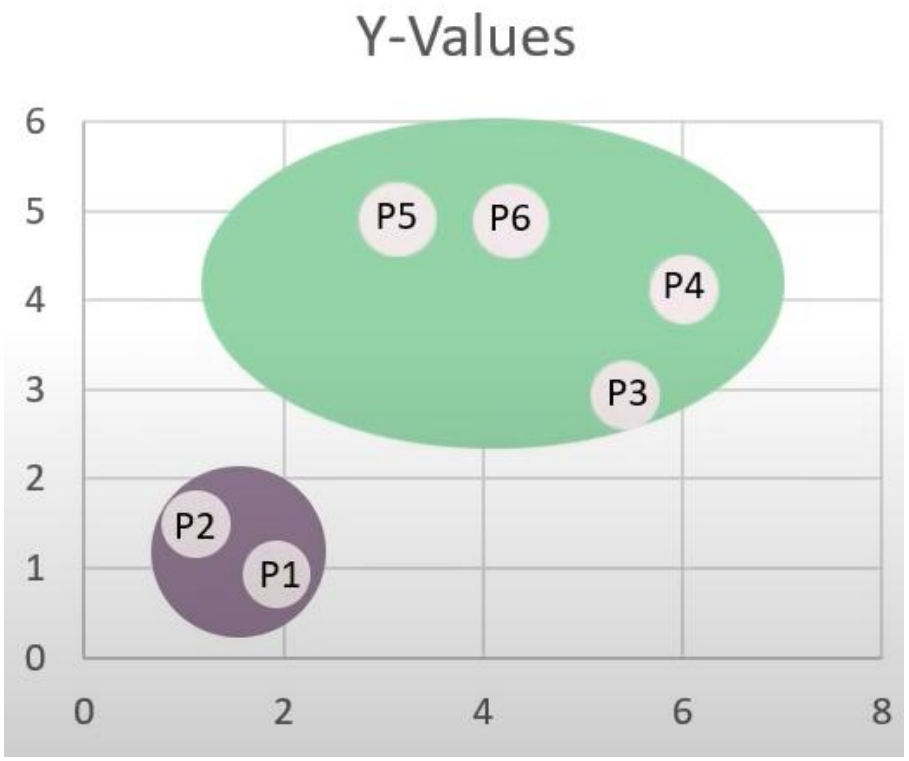


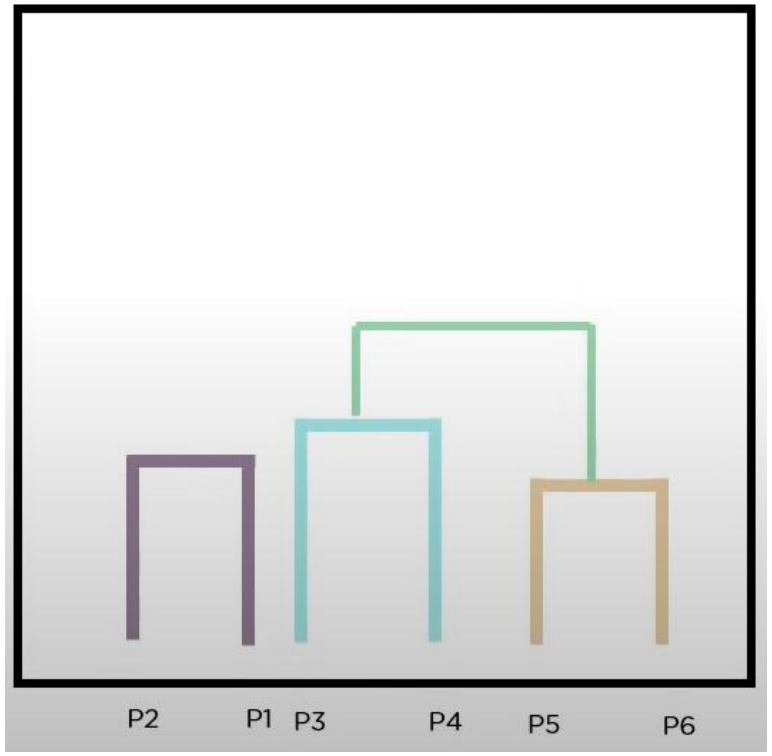


How Hierarchical Clustering Works

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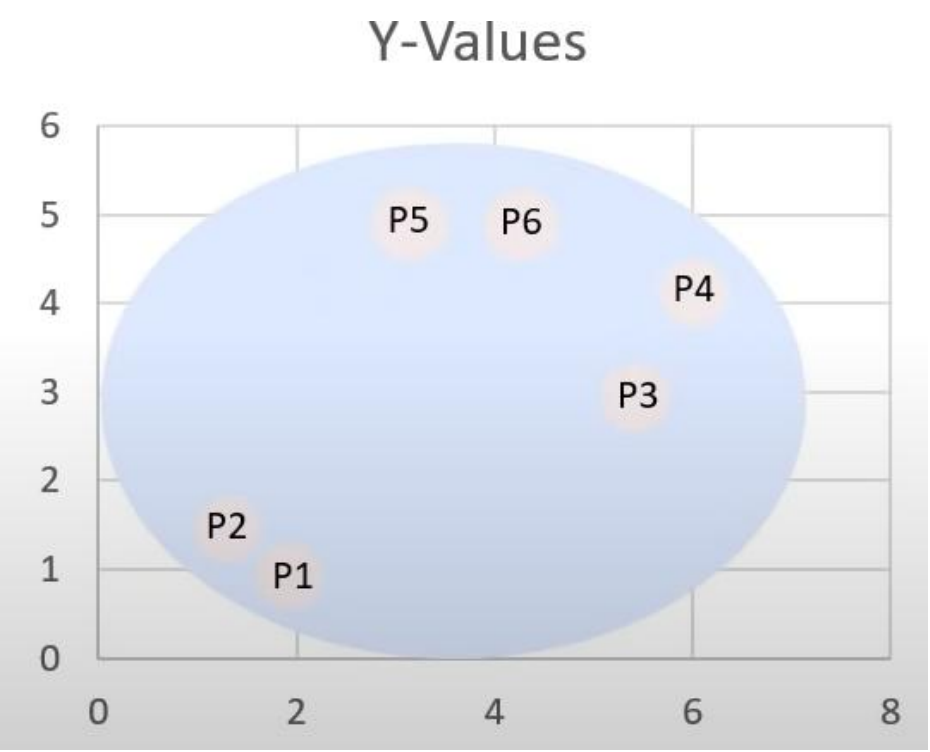


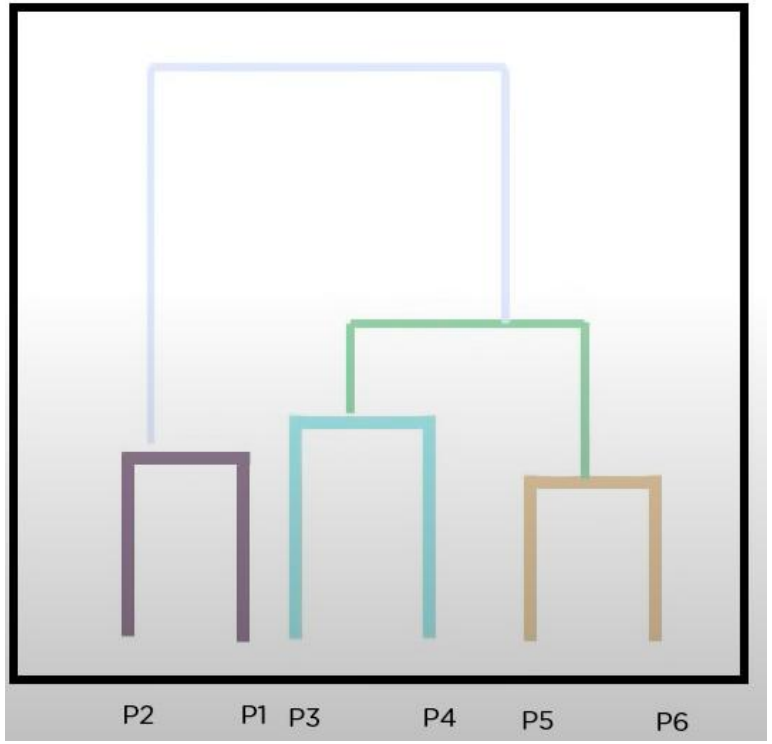


How Hierarchical Clustering Works

2. Grouping

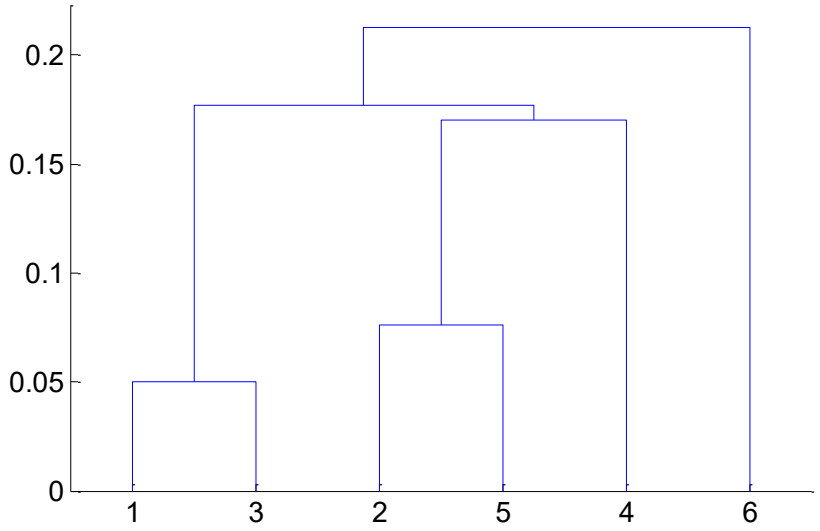
- We terminate when we are left with one cluster.

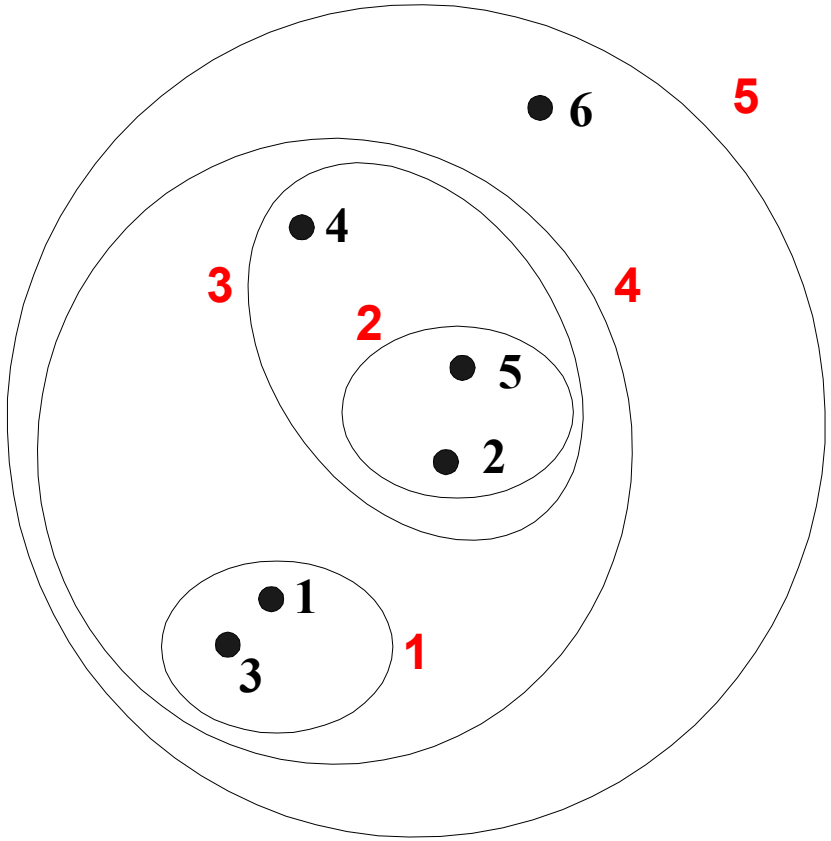




Hierarchical algorithms:

- Hierarchical clustering (also called hierarchical cluster analysis or HCA) is a method of cluster analysis that seeks to build a hierarchy of clusters.





- Produce a nested sequence of clusters, a tree, also called Dendrogram.

Types of hierarchical clustering

Agglomerative (bottom-up) clustering:

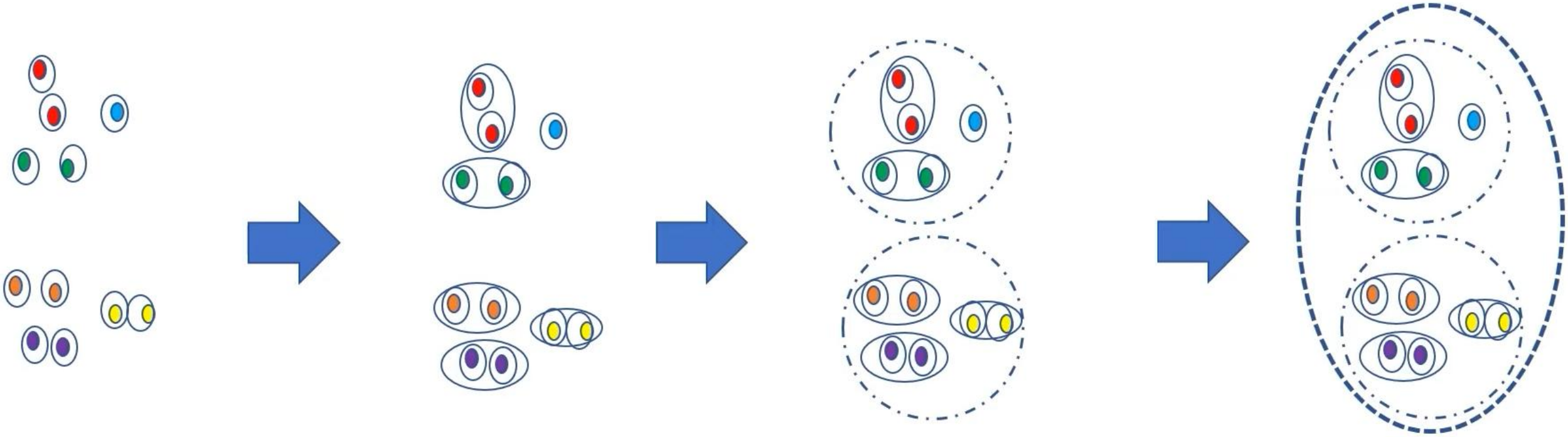
- It builds the dendrogram (tree) from the bottom level, and merges the most similar (or nearest) pair of clusters stops when all the data points are merged into a single cluster (i.e., the root cluster).

Divisive (top-down) clustering:

- It starts with all data points in one cluster, the root.
- Splits the root into a set of child clusters.
- Each child cluster is recursively divided further stops when only singleton clusters of individual data points remain, i.e., each cluster with only a single point

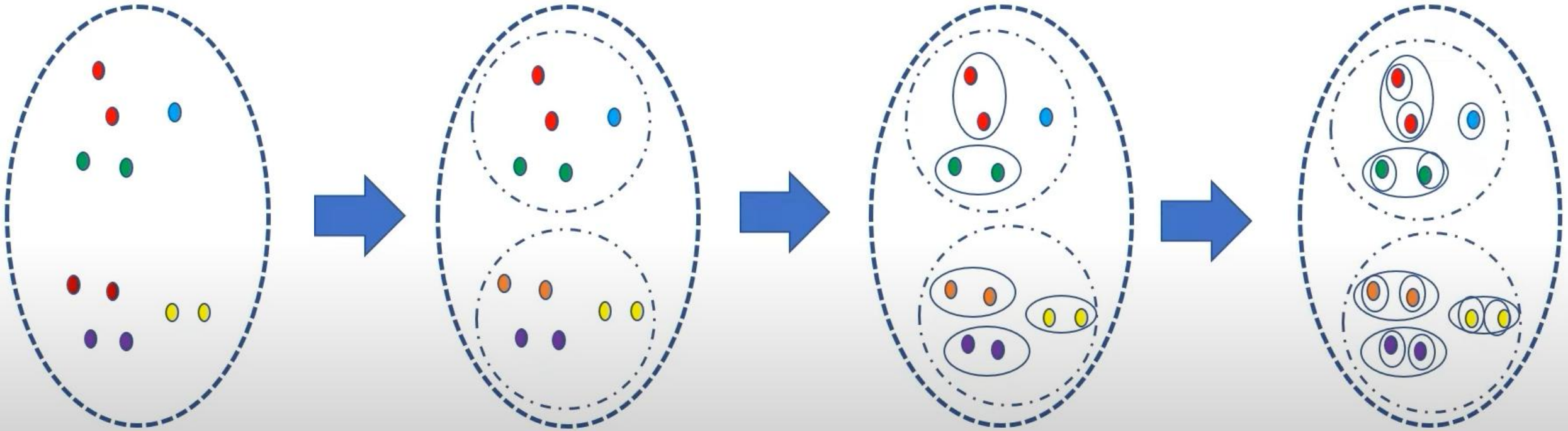
How Agglomerative Clustering Works

Each observation is treated as a separate cluster.

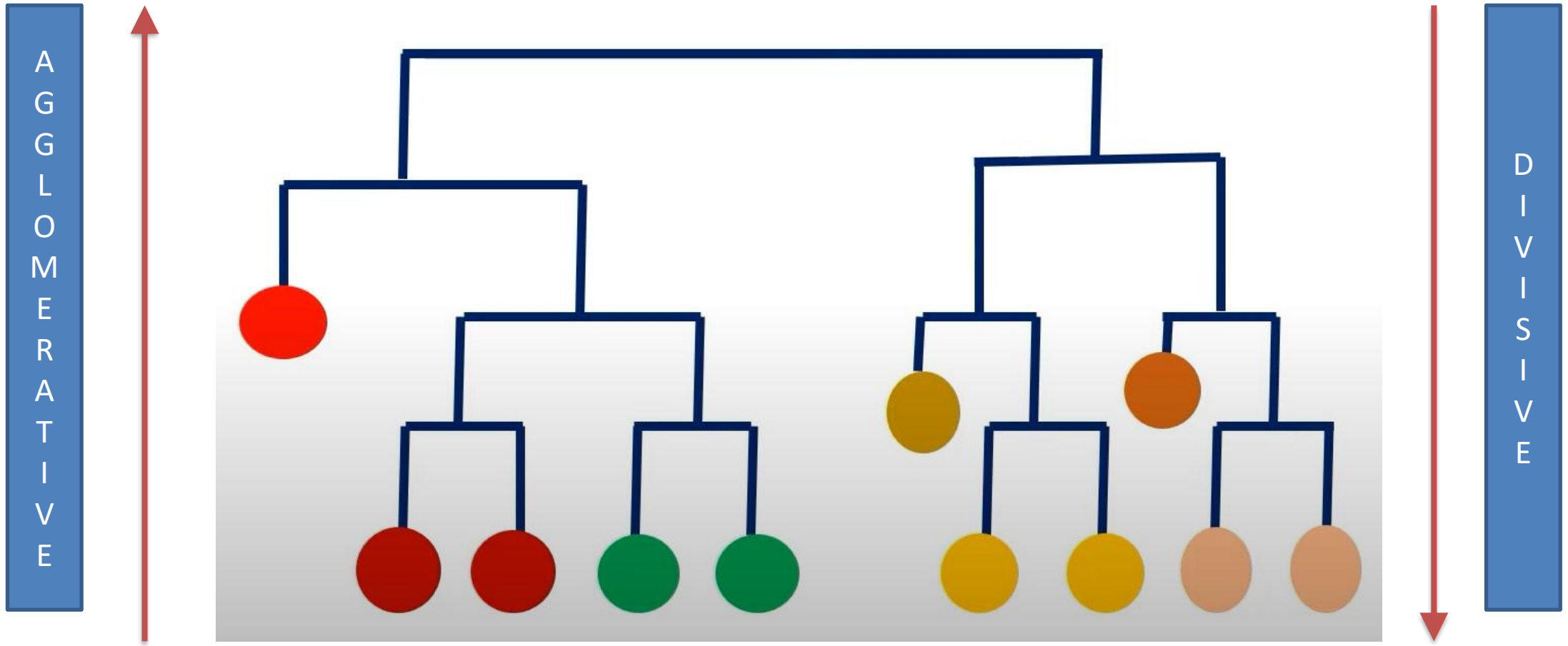


How Divisive Clustering Works

All data points are treated as a single cluster.

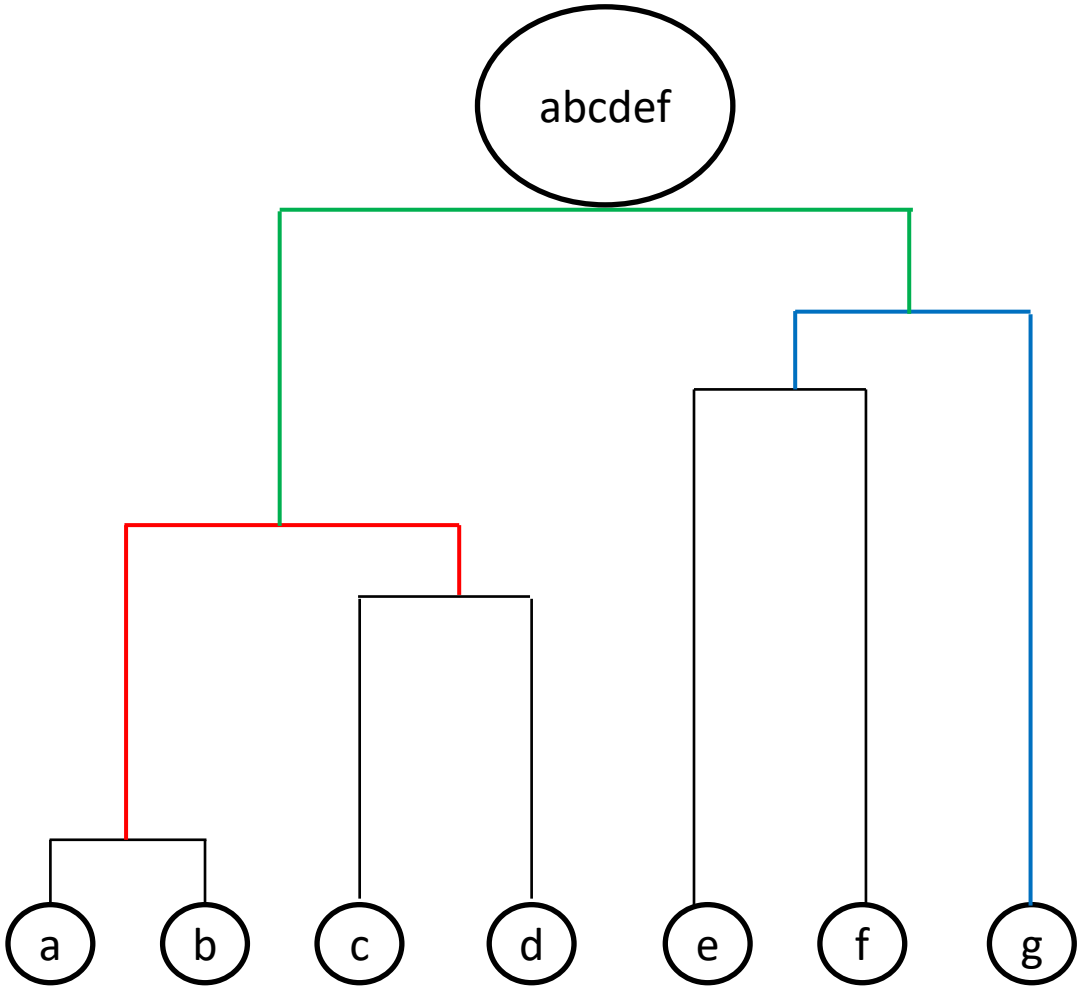


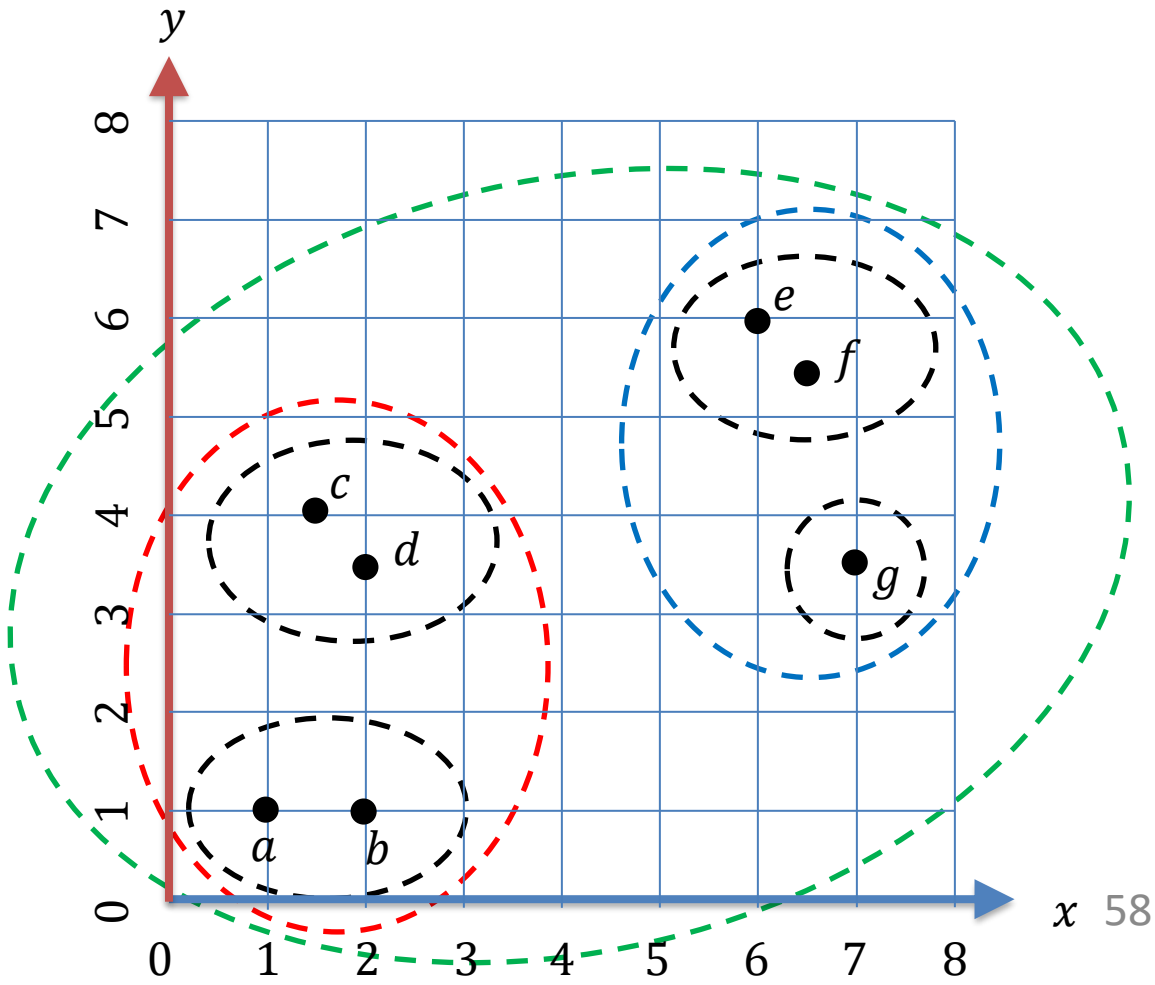
What is Agglomerative and Divisive Clustering



Example

Draw the Dendrogram using Agglomerative method.





Example of Agglomerative Clustering

Let's assume that we have 6 data points in Euclidean space



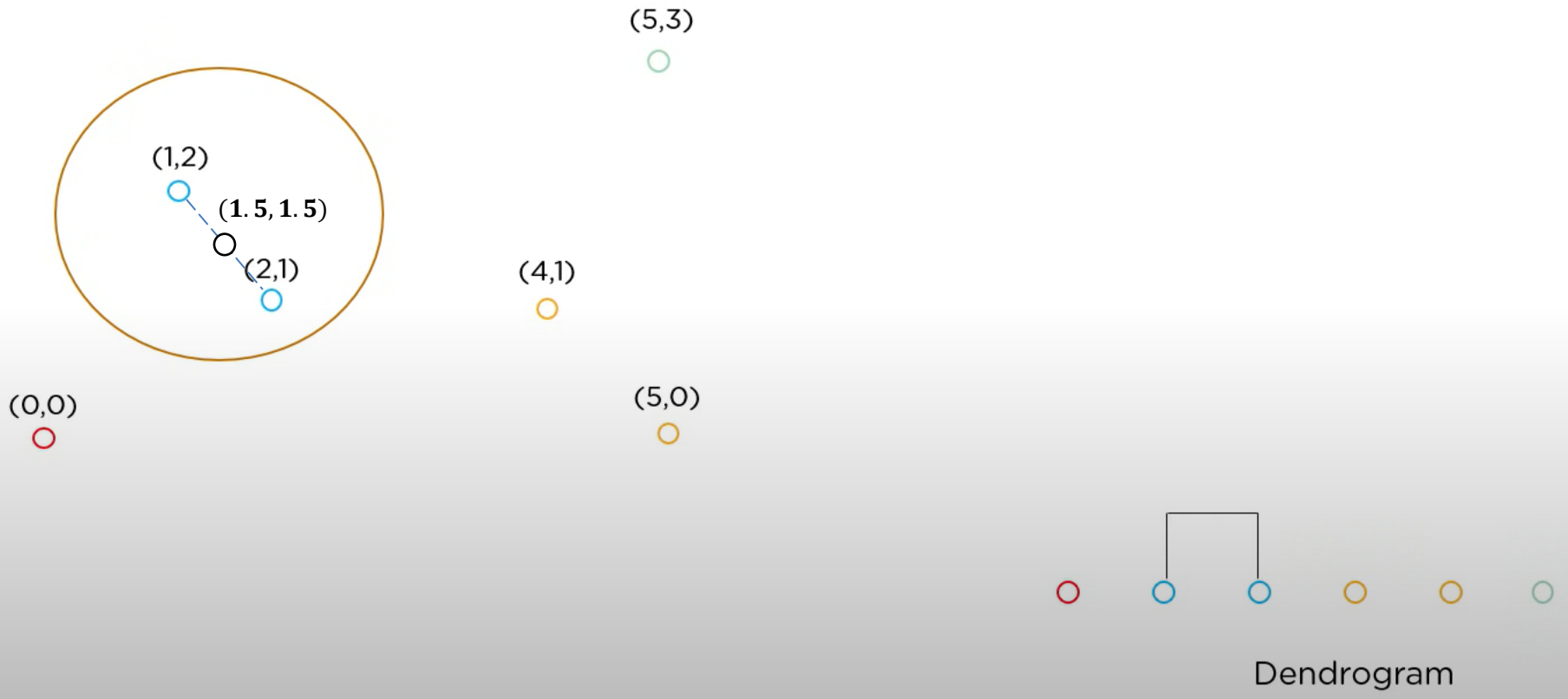
We make use of centroids which is the average of its points

How do we represent a cluster of more than one point?



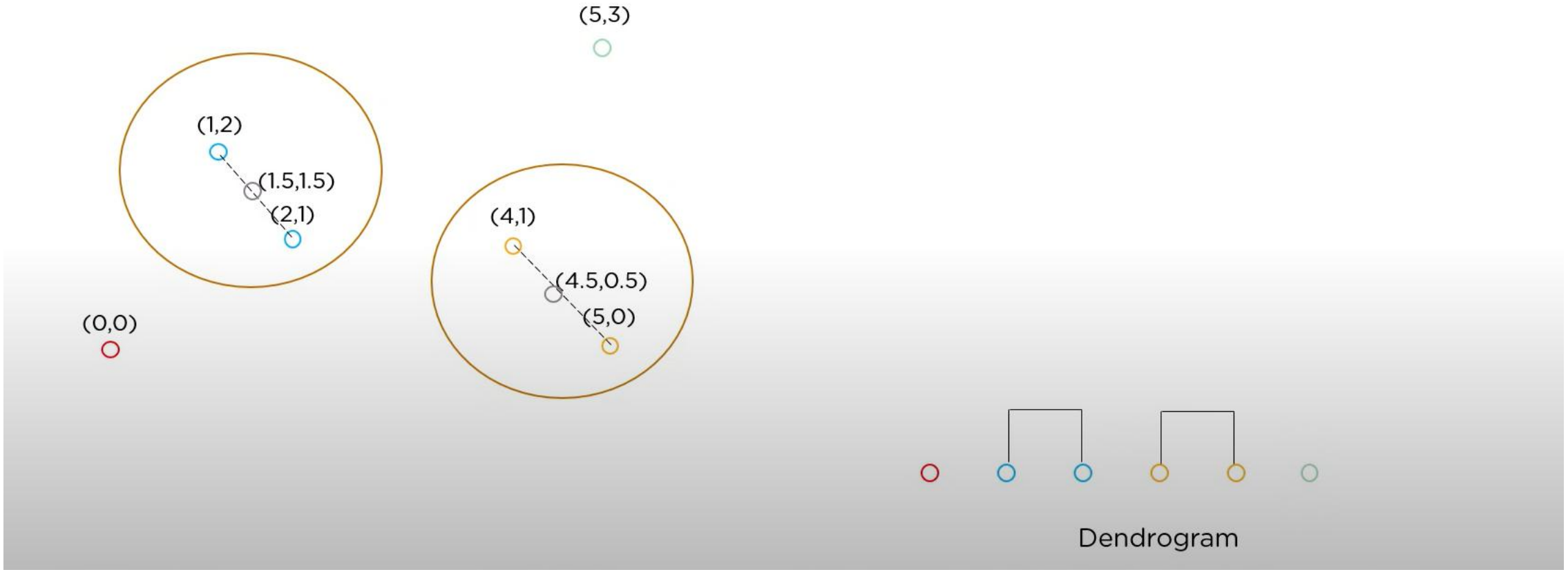
Example of Agglomerative Clustering

Let's assume that we have 6 data points in Euclidean space



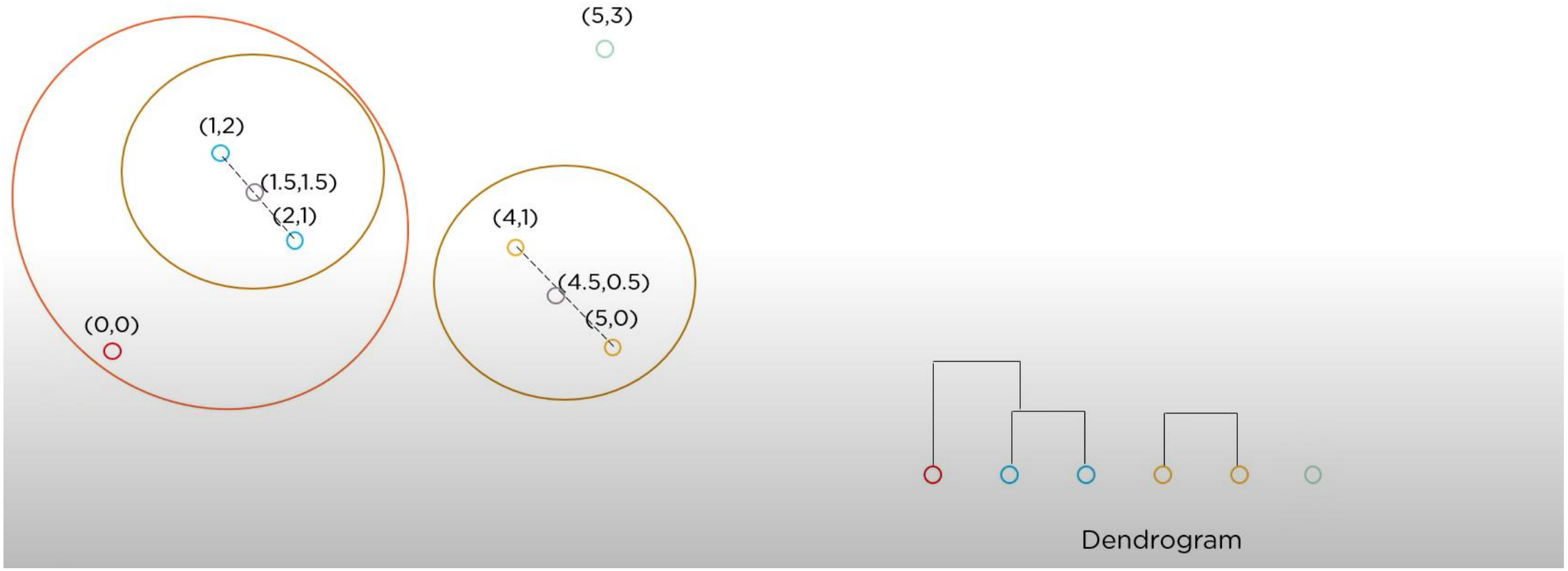
Example of Agglomerative Clustering

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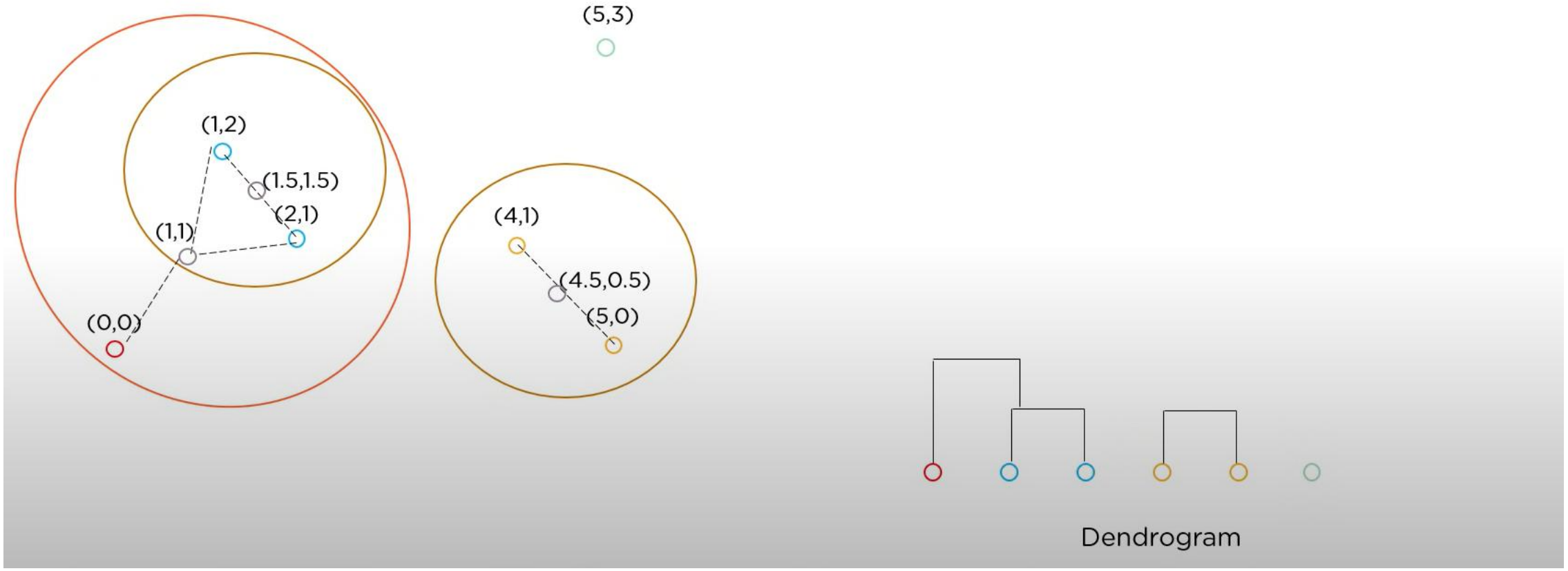
Example of Agglomerative Clustering

Let's assume that we have 6 data points in Euclidean space



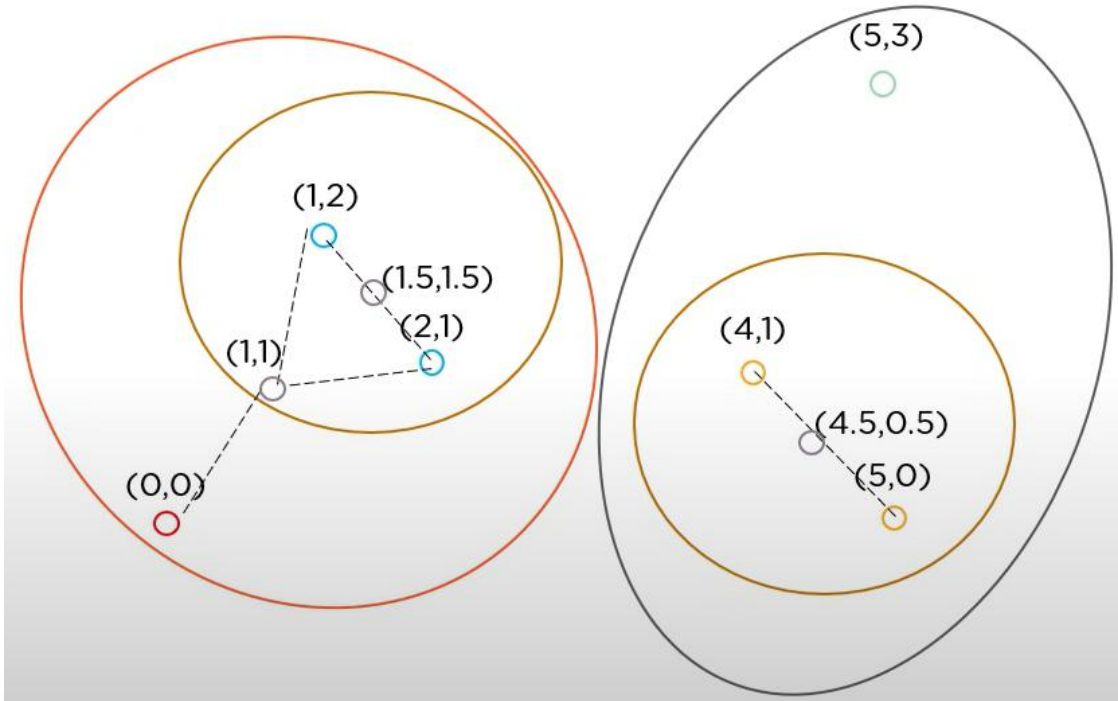
Example of Agglomerative Clustering

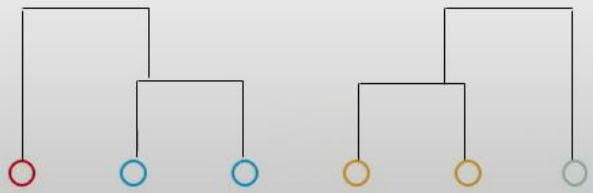
Let's assume that we have 6 data points in Euclidean space



Example of Agglomerative Clustering

Let's assume that we have 6 data points in Euclidean space

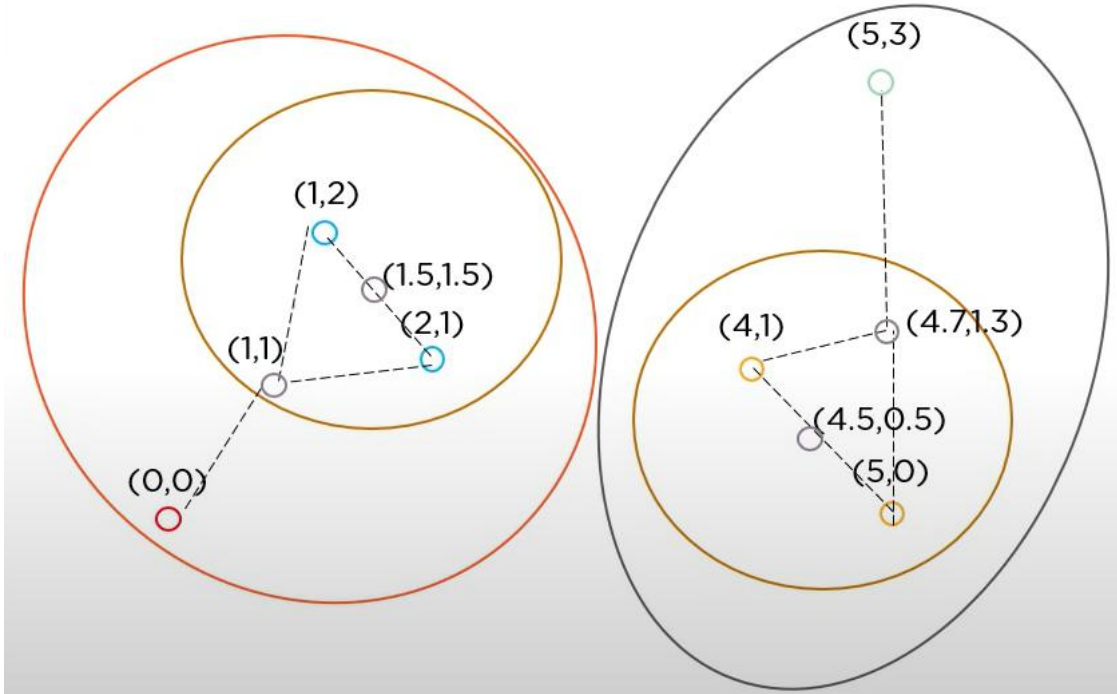


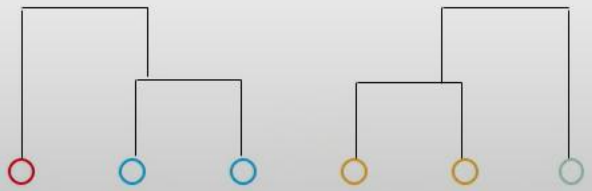


Dendrogram

Example of Agglomerative Clustering

Let's assume that we have 6 data points in Euclidean space





Dendrogram

What is Agglomerative Clustering

There are many approaches to it

Approach 1: Pick a number of clusters (k) upfront

We decide the number of clusters required in the beginning and we terminate when we reach the value (k)

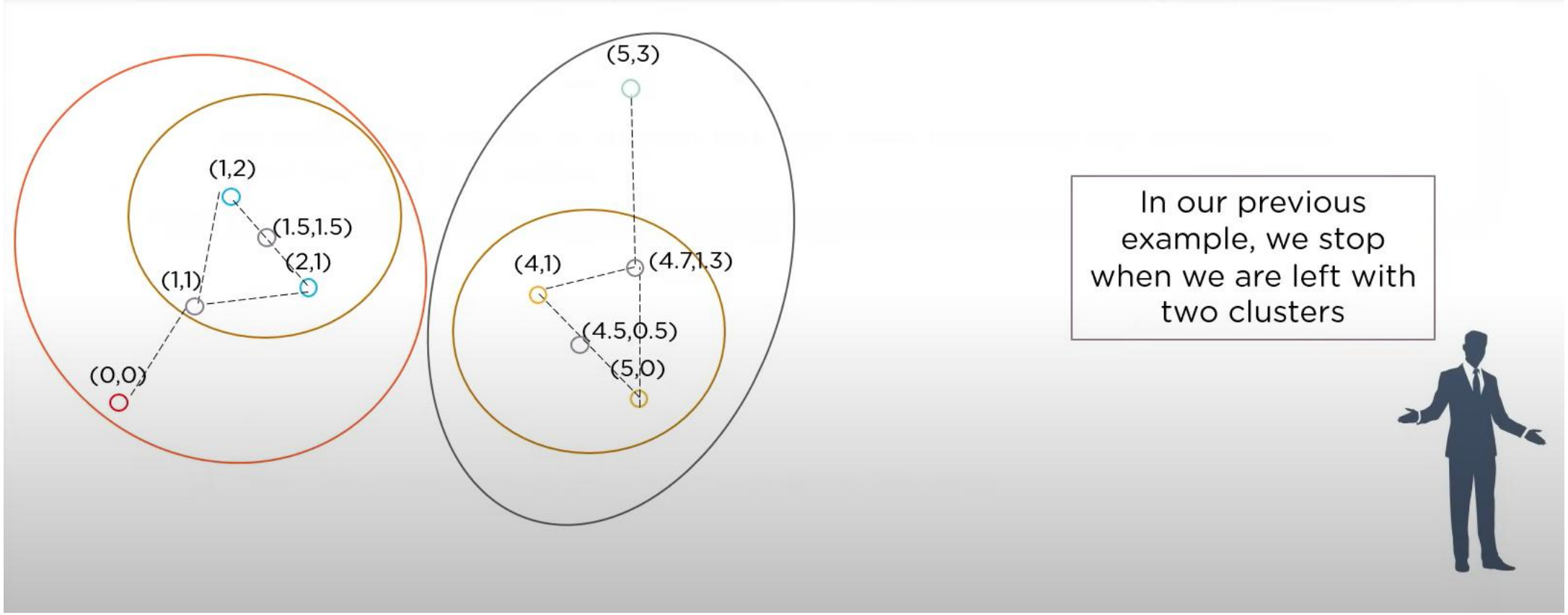
Possible Challenges

This only makes sense when we know the data well

Let’s consider the value of k to be 2

What is Agglomerative Clustering

There are many approaches to it



What is Agglomerative Clustering

There are many approaches to it

Approach 2: Stop when the next merge would create a cluster with low "cohesion"

We keep clustering the till the next merge of clusters creates a bad cluster/low cohesion.

Possible Challenges

This only makes sense when we know the data well.

What is Agglomerative Clustering

There are many approaches to it

Approach 3: Diameter of a cluster

Diameter is the maximum distance between any pair of points in a cluster.

We terminate when the diameter of a new cluster exceeds the threshold

What is Divisive Clustering

The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.

Step 1: Start with a single cluster composed of all the data points.

Step 2: Split it into different clusters. This can be done using the Monothetic divisive method

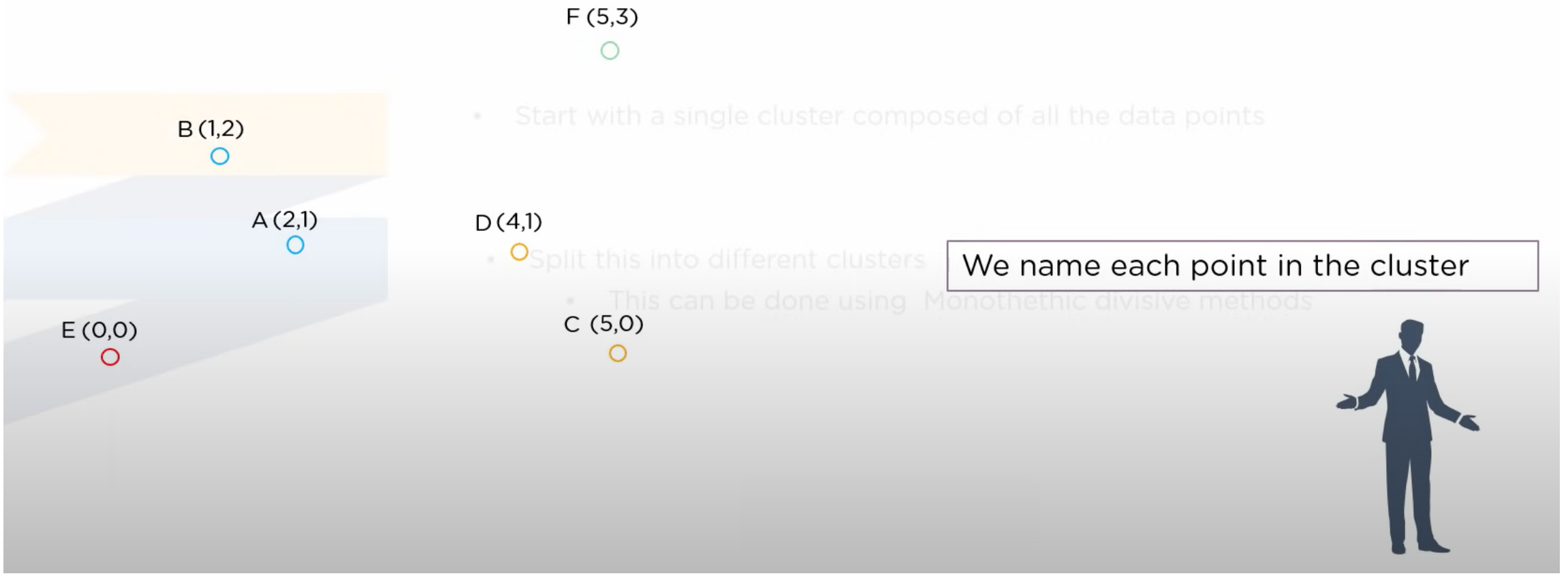
What is Divisive Clustering

The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.



What is Divisive Clustering

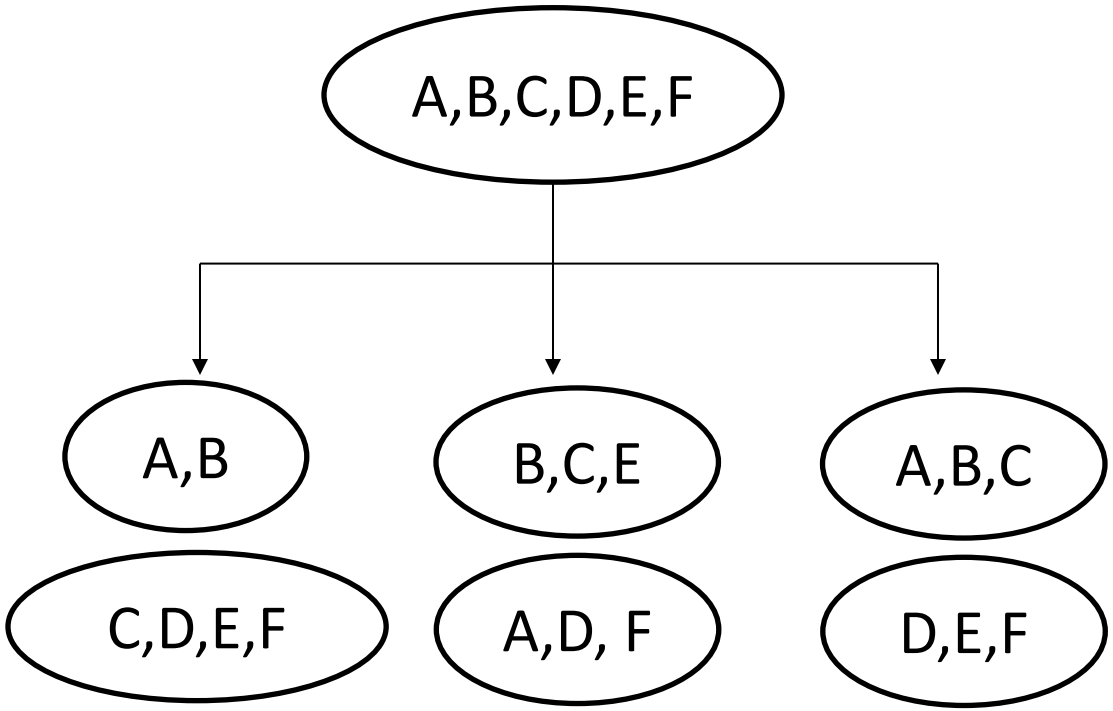
The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.



What is Divisive Clustering

The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.

- Obtain all possible splits into two clusters



Since we follow a top-down approach in Divisive Clustering

- For each split, compute cluster sum of squares

What is Divisive Clustering

The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.

- For each split, compute the cluster sum of squares

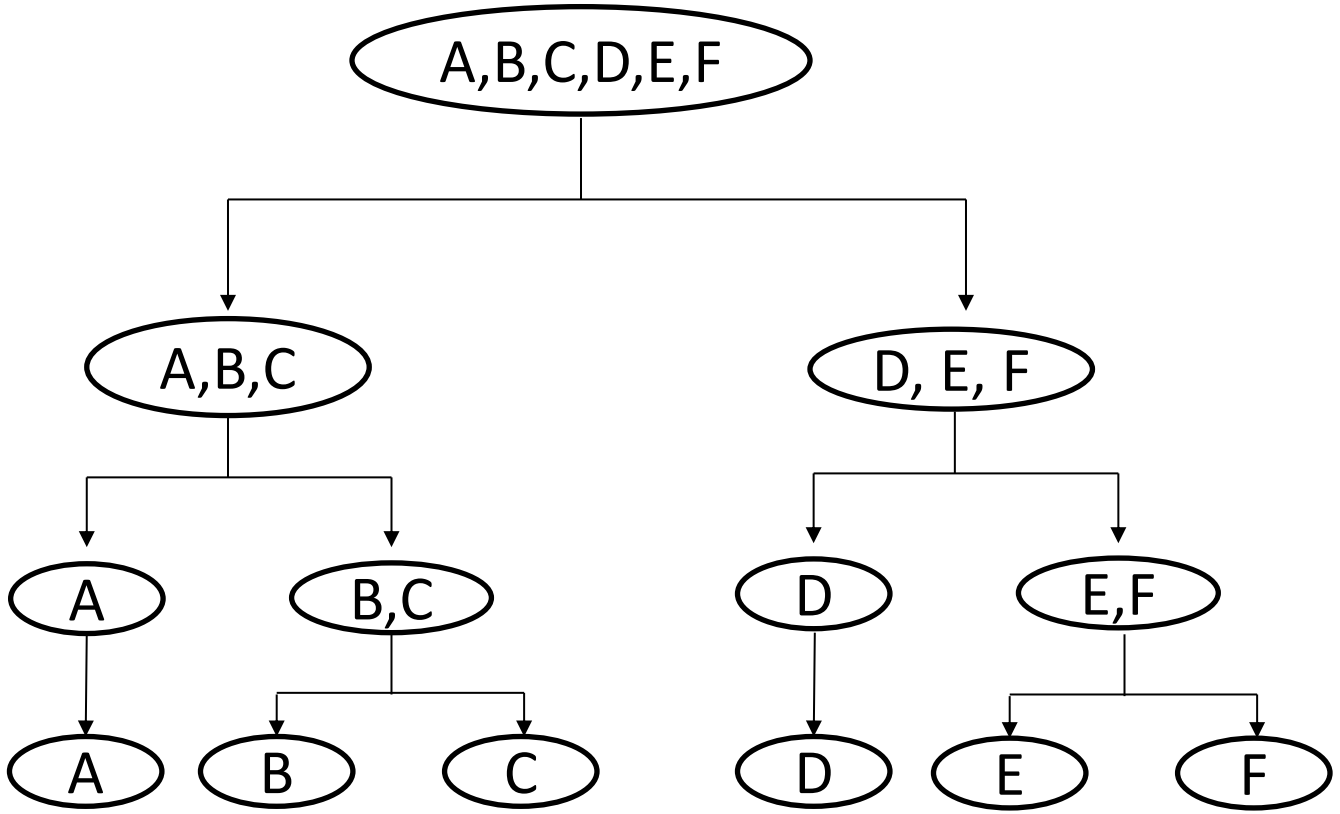
$$B_{j12} = n_1\left(\bar{x}_{j1} - \bar{x}_j\right)^2 + n_2\left(\bar{x}_{j2} - \bar{x}_j\right)^2$$

B_j	Between cluster 1 and 2	X_{j1}	Mean of the cluster
n	Number of members in cluster	X_j	Grand mean

- We select the cluster with largest sum of squares

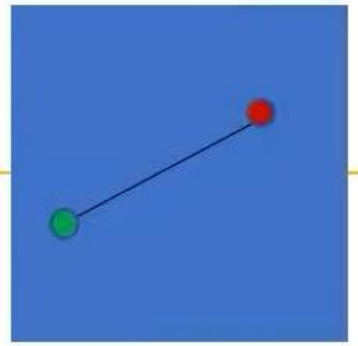
What is Divisive Clustering

The divisive Clustering approach begins with the whole set and proceeds to divide it into smaller clusters.

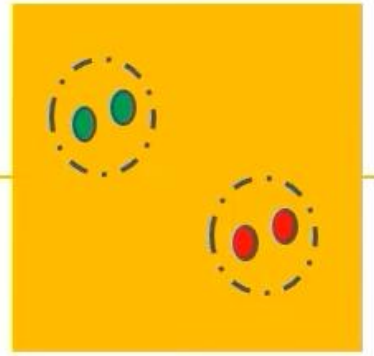


- Let's assume that the sum of squared distance is the largest for 3rd split
- We divide into two clusters
- We again find the sum of squared distance and split it into clusters.
- We terminate when every data point is its own cluster.

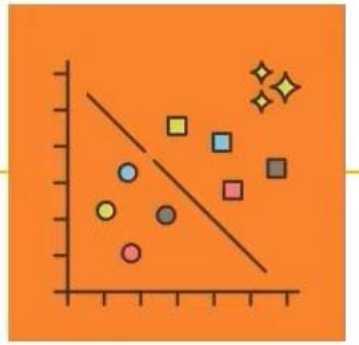
How to Decide the Group of Clusters



Based on the similarity, single linkage, complete linkage.

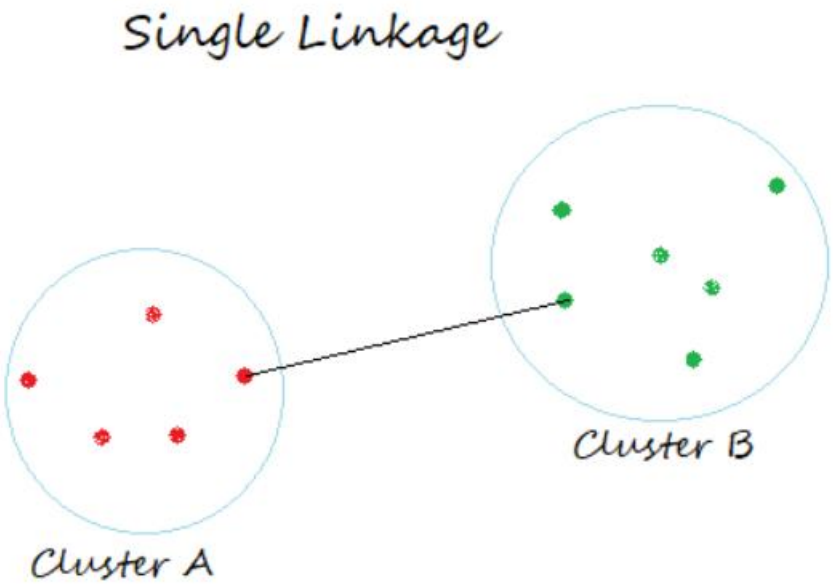


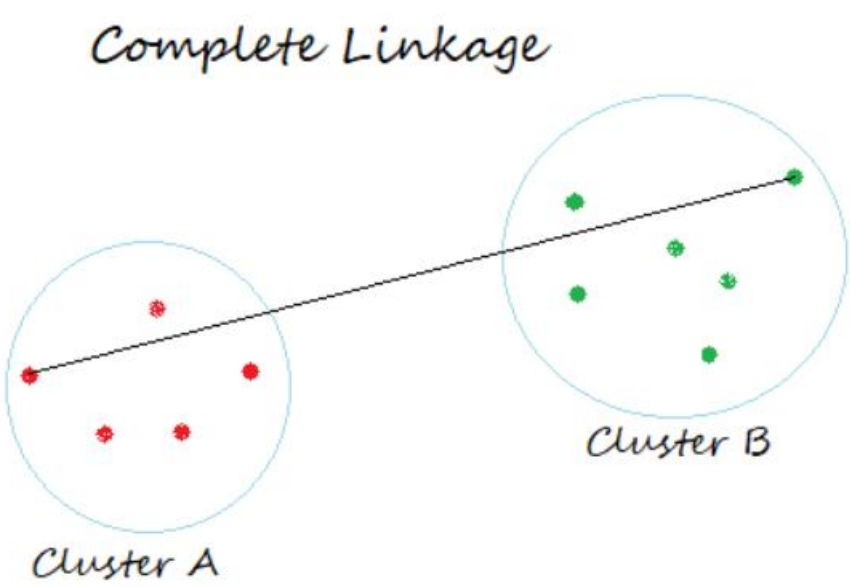
Similarity is obtained by calculating distance between clusters.

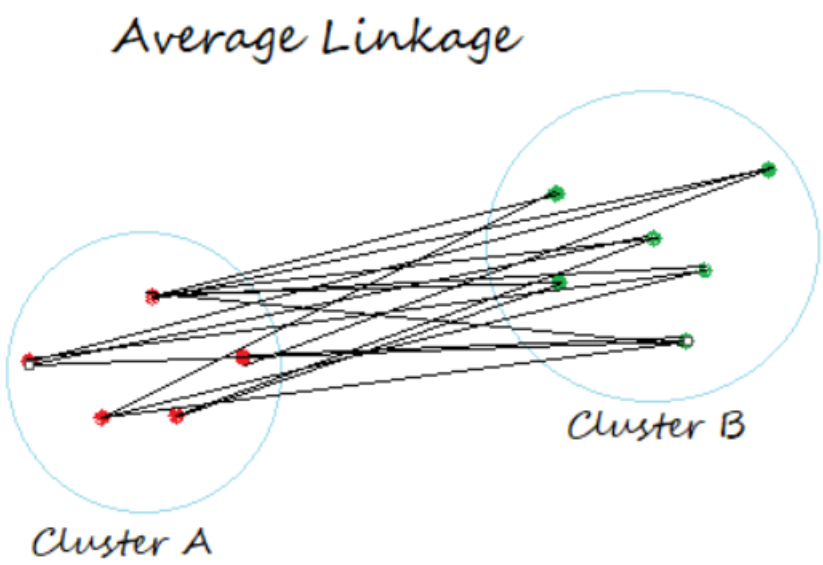


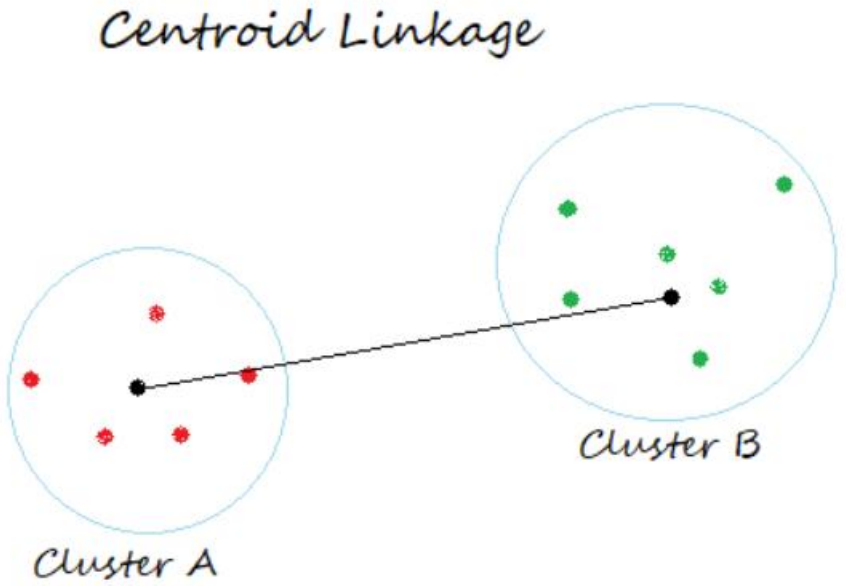
Distance metrics like Euclidian , Manhattan, Correlation distance

How to Calculate The Similarity Between Two Clusters







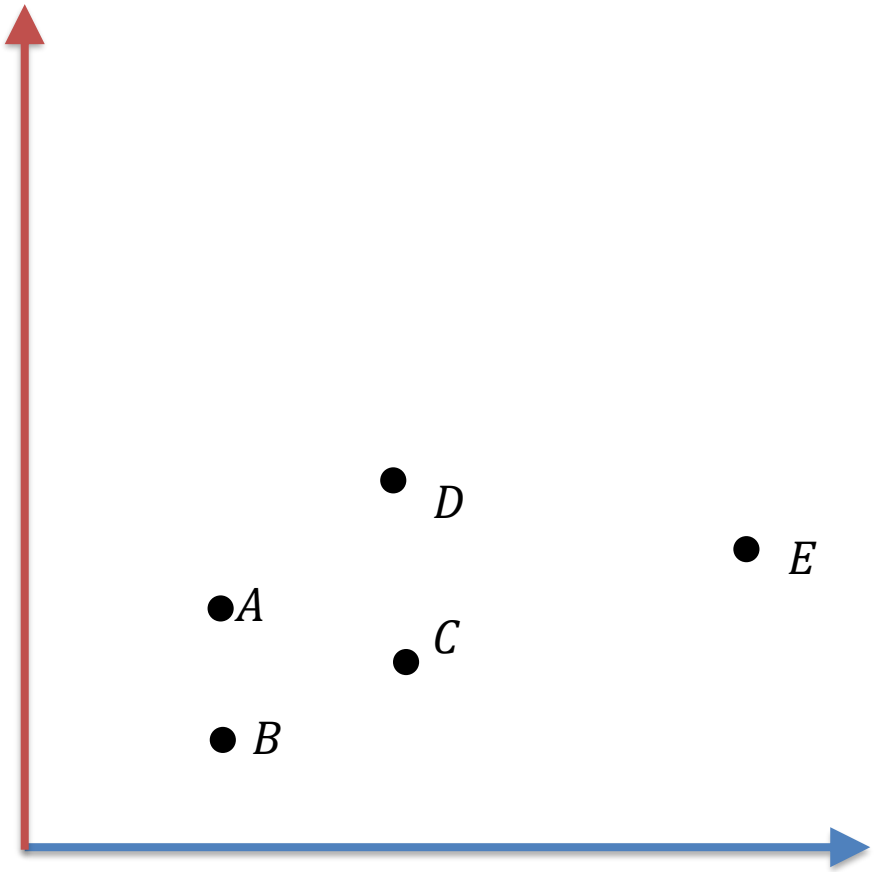


Class Exercise (Single Link approach)

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Single link approach.

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0



Class Exercise (Single Link approach)

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Single link approach.

Solution:

Step 1: Find the minimum distance in the distance matrix.

Since the minimum distance is '1' at object 'A' and 'B' therefore, object 'A' and 'B' will going to merge.

Now we have an updated-1 distance matrix in which we need to compute unknown distance using single link approach.

Updated-1 Distance Matrix

	A,B	C	D	E
A,B	0	2	2	3
C		0	3	6
D			0	6
E				0

d{(A,B), C}=min{d(A,C), d(B,C)} = min {2,2} = 2

d{(A,B), D}=min{d(A,D), d(B,D)} = min {2,5} = 2

d{(A,B), E}=min{d(A,E), d(B,E)} = min {3,4} = 3

d{C,D}=3 , d{C,E}=6, d{D,E}=6

Step 2: Find the minimum distance in the updated-1 distance matrix.

In the updated-1 distance matrix of step 1 "2" is the minimum value. Therefore, (A, B) will merge with "C" and "D"

Class Exercise (Single Link approach)

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Single link approach.

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Step 2: Find the minimum distance in the updated-1 distance matrix.

Now we have an updated-2 distance matrix in which we need to compute unknown distance using a single link approach.

$d\{(A,B,C,D), E\}=\min\{d(A,E), d(B,E),d(C,E),d(D,E)\} = \min\{3,4,6,6\} = 3$

Updated-2 Distance Matrix

	A,B,C,D	E
A,B,C,D	0	3
E		0

In the updated-2 distance matrix of step 2 “3” is the minimum value. Therefore, (A, B,C,D) will merge with “E” and will terminate here.

Class Exercise (Single Link approach)

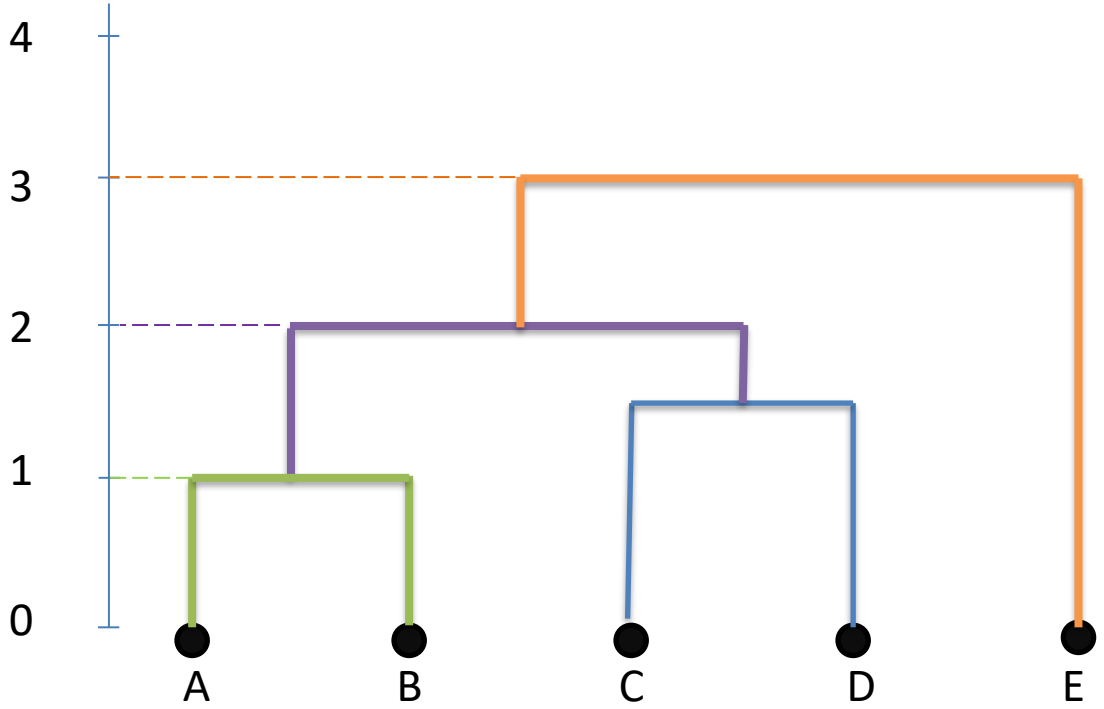
Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Single link approach.

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Plotting of dendrogram based on the updated distance matrix

Find the minimum distance at each updated distance matrix.



Updated-1 Distance Matrix

	A,B	C	D	E
A,B	0	2	2	3
C		0	3	6
D			0	6
E				0

Updated-2 Distance Matrix

	A,B,C,D	E
A,B,C,D	0	3
E		0

Class Exercise (Complete Link approach)

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Complete link approach.

Solution:

Step 1: Find the minimum distance in the distance matrix.

Since the minimum distance is '1' at object 'A' and 'B' therefore, object 'A' and 'B' will going to merge.

Now we have an updated-1 distance matrix in which we need to compute unknown distance using complete link approach.

Updated-1 Distance Matrix

	A,B	C	D	E
A,B	0	2	5	4
C		0	3	6
D			0	6
E				0

d{(A,B), C}=max{d(A,C), d(B,C)} = max {2,2} = 2

d{(A,B), D}=max{d(A,D), d(B,D)} = max {2,5} = 5

d{(A,B), E}=max{d(A,E), d(B,E)} = max {3,4} = 4

d{C,D}=3 , d{C,E}=6, d{D,E}=6

Step 2: Find the minimum distance in the updated-1 distance matrix.

In the updated-1 distance matrix of step 1 "2" is the minimum value. Therefore, (A, B) will merge with "C"

Class Exercise (Complete Link approach)

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the complete link approach.

Step 2: Find the minimum distance in the updated-1 distance matrix.

Now we have an updated-2 distance matrix in which we need to compute unknown distance using a complete link approach.

d{(A,B,C), D}=max{d(A,D), d(B,D),d(C,D)} = max {2,5,3} = 5

d{(A,B,C), E}=max{d(A,E), d(B,E),d(C,E)} = max {3,4,6} = 6

d{D,E}=6

Updated-2 Distance Matrix

	A,B,C	D	E
A,B,C	0	5	6
D		0	6
E			0

In the updated-2 distance matrix of step 2 “5” is the minimum value. Therefore, (A, B,C) will merge with “D”.

Class Exercise (Complete Link approach)

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the complete link approach.

Step 3: Find the minimum distance in the updated-2 distance matrix.

Now we have an updated-2 distance matrix in which we need to compute unknown distance using a complete link approach.

d{(A,B,C,D), E}=max{d(A,E), d(B,E),d(C,E),d(D,E)} = max {3,4,6,6} = 6

Updated-3 Distance Matrix

	A,B,C,D	E
A,B,C,D	0	6
E		0

In the updated-3 distance matrix of step 3 “6” is the minimum value. Therefore, (A, B,C, D) will merge with “E” and we will terminate it here.

Class Exercise (Complete Link approach)

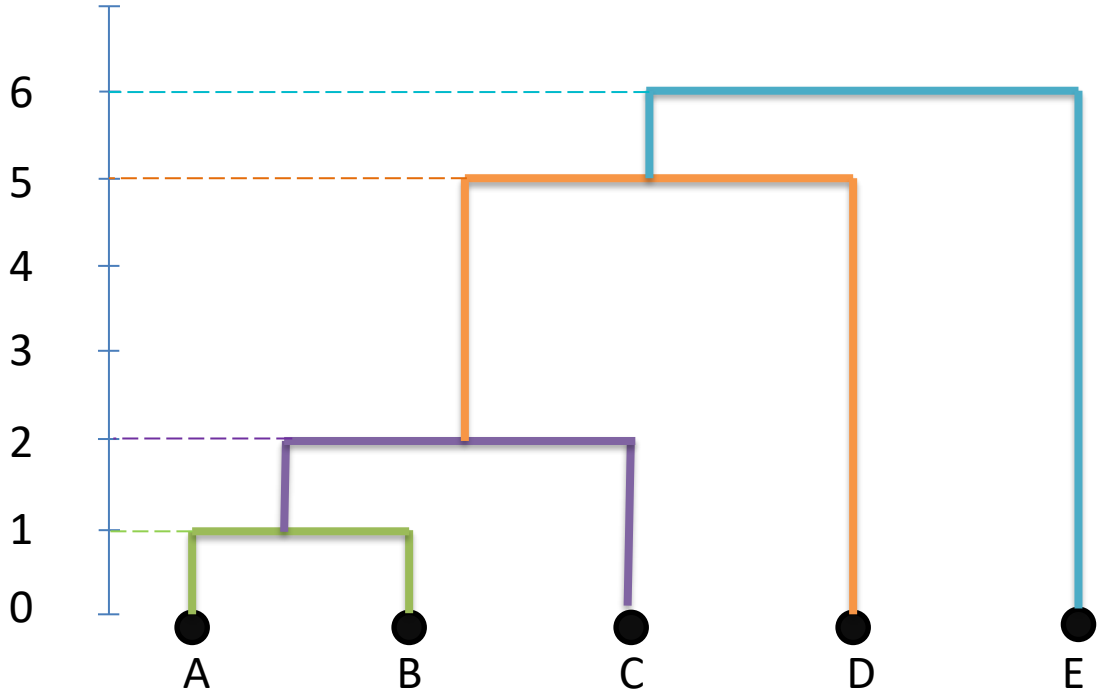
Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the Complete link approach.

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Plotting of dendrogram based on the updated distance matrix

Find the minimum distance at each updated distance matrix.



Updated-1 Distance Matrix

	A,B	C	D	E
A,B	0	2	5	4
C		0	3	6
D			0	6
E				0

Updated-2 Distance Matrix

	A,B,C	D	E
A,B,C	0	5	6
D		0	6
E			0

Updated-3 Distance Matrix

	A,B,C,D	E
A,B,C,D	0	6
E		0

Class Exercise (Average Link approach)

Q. For the given Distance Matrix apply hierarchical clustering using the agglomerative approach. Also, draw a dendrogram by using the average (centroid) link approach.

Distance Matrix

	A	B	C	D	E
A	0	1	2	2	3
B		0	2	5	4
C			0	3	6
D				0	6
E					0

Do it by yourself

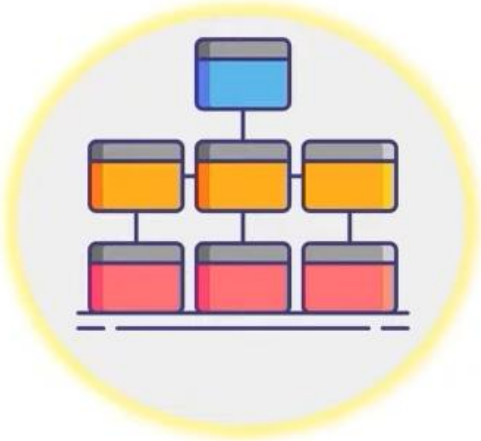
Hint d{(A,B), C}=average{d(A,C), d(B,C)} = average {2,2} = 2

Applications of Hierarchical Clustering



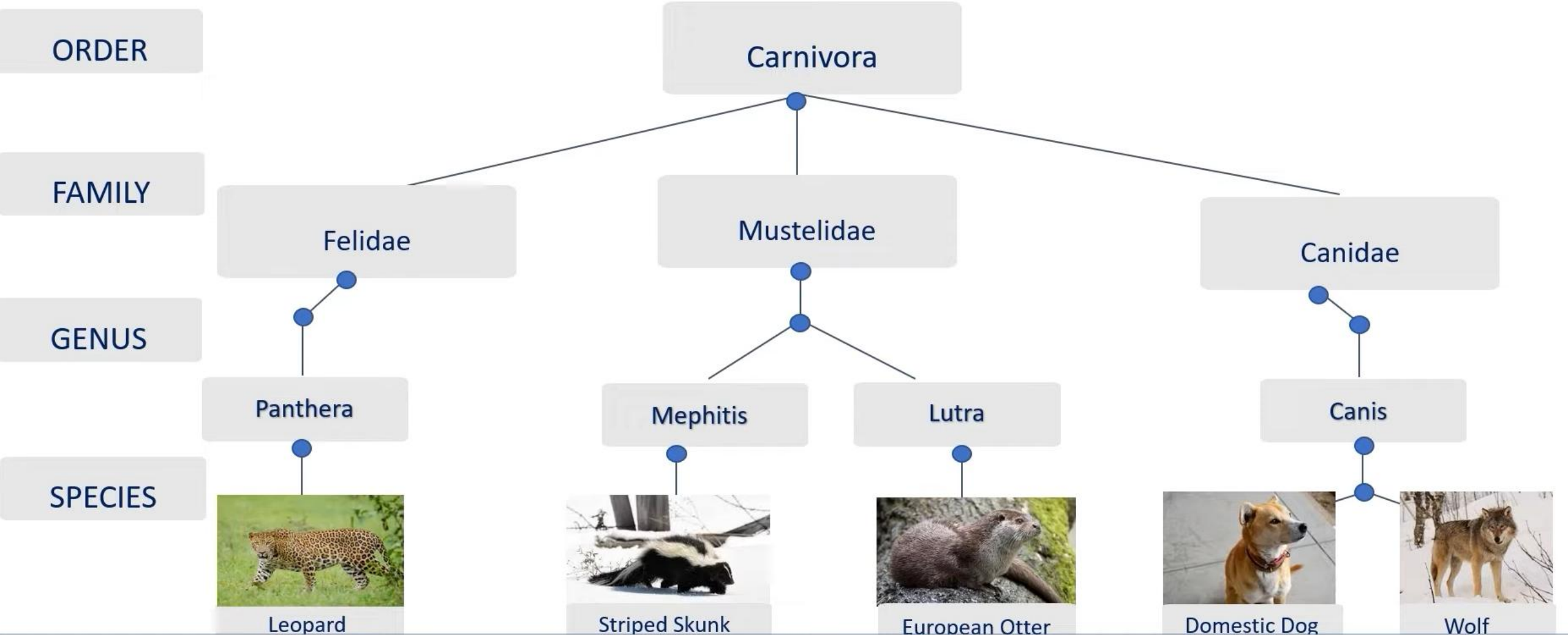
Used in Taxonomy, biological classification of animal or plant kingdom.







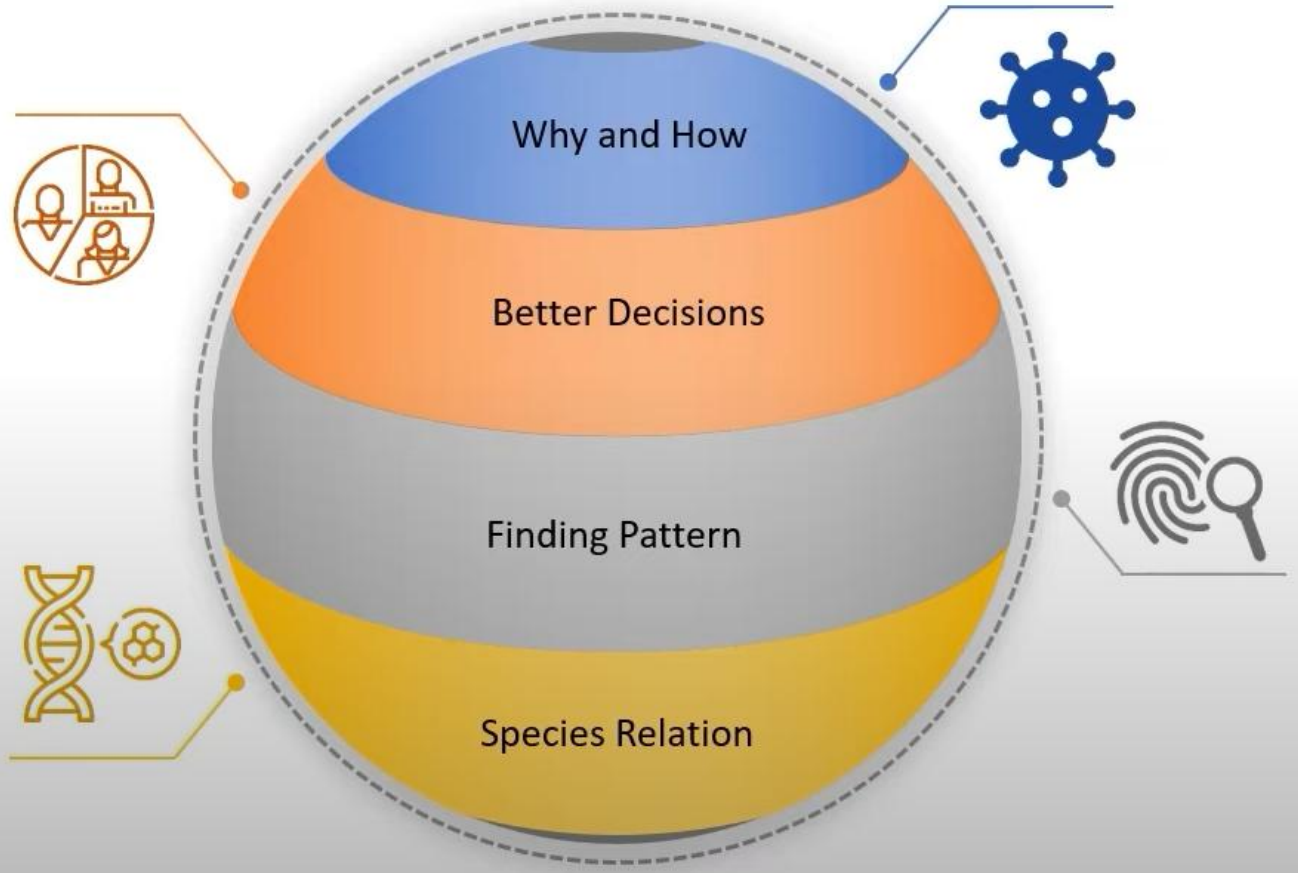
Applications of Hierarchical Clustering



Applications of Hierarchical Clustering

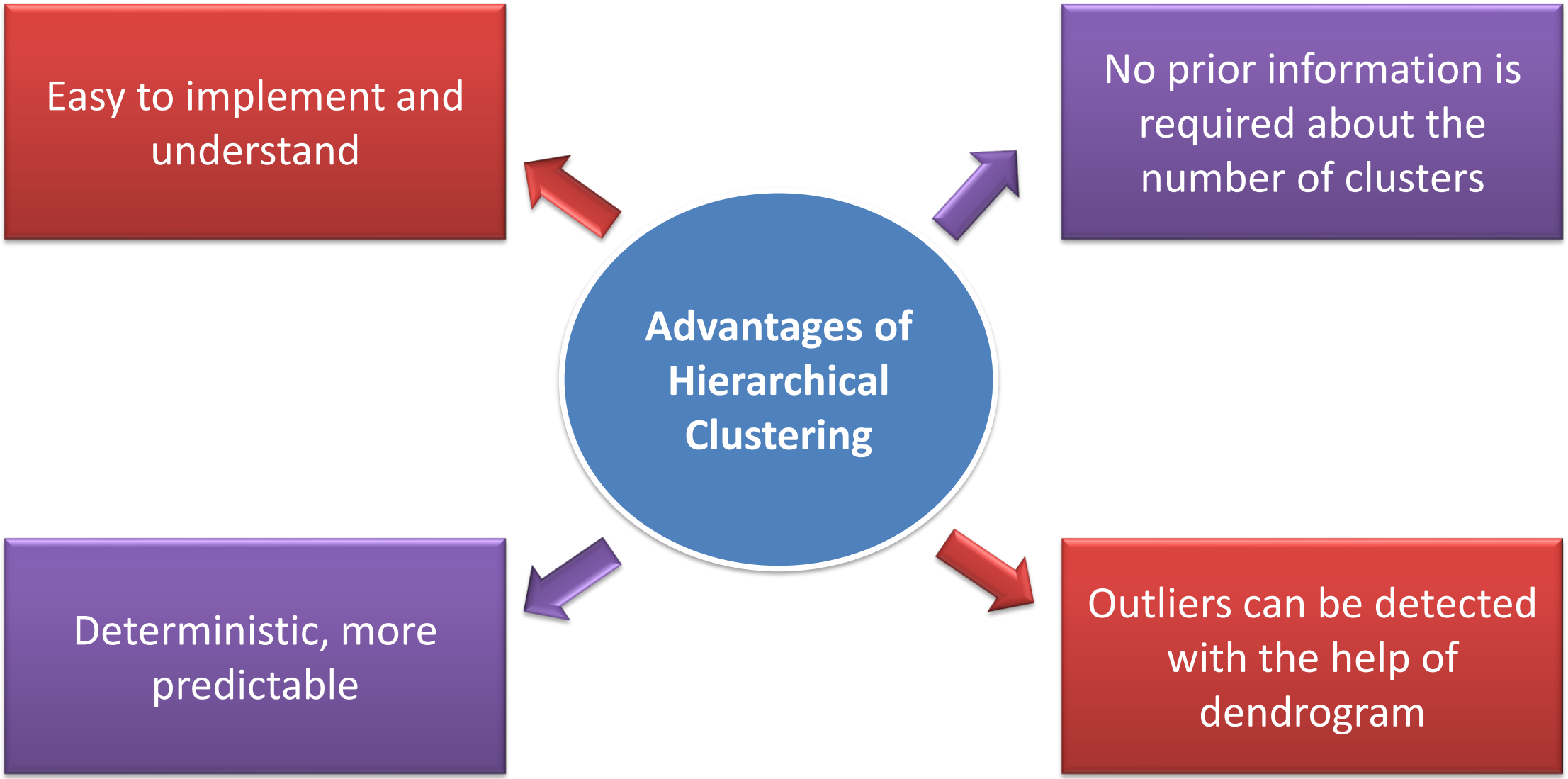
Segmenting the customers

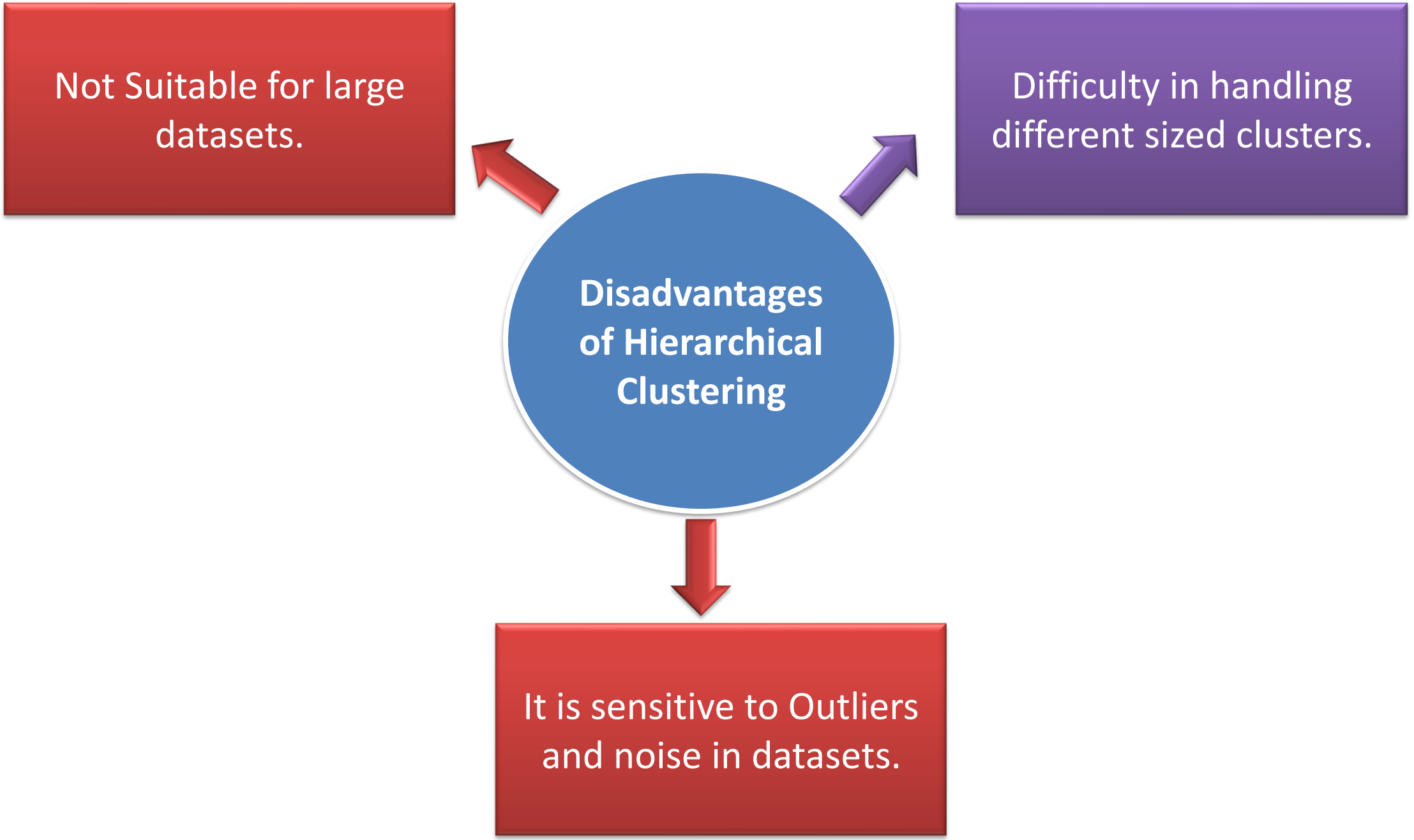
Tracking viral Outbreaks



Clustering Crimes sites in the city

Evolution through Phylogenetic trees


Easy to implement and understand
No prior information is required about the number of clusters
Advantages of Hierarchical Clustering
Deterministic, more predictable
Outliers can be detected with the help of dendrogram


Not Suitable for large datasets.
Difficulty in handling different sized clusters.
Disadvantages of Hierarchical Clustering
It is sensitive to Outliers and noise in datasets.

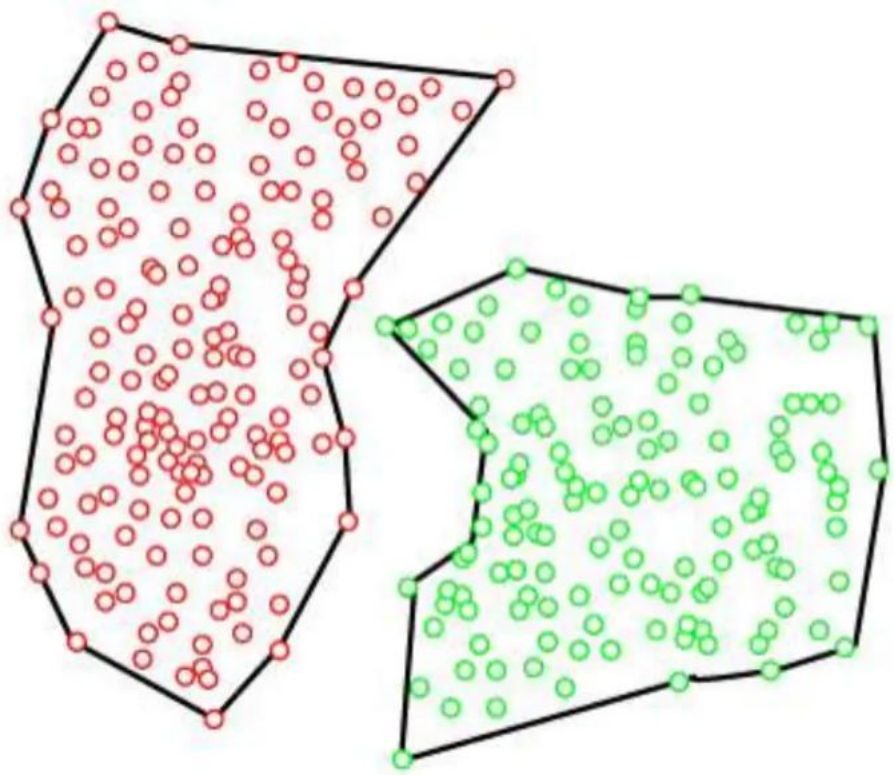
Density-based Clustering

- The density-based clustering method connects the highly dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected.

- This algorithm does it by identifying different clusters in the dataset and connecting the areas of high density into clusters.

- The dense areas in data space are divided from each other by sparse areas.

- These algorithms can face difficulty in clustering the data points if the dataset has varying densities and high dimensions.

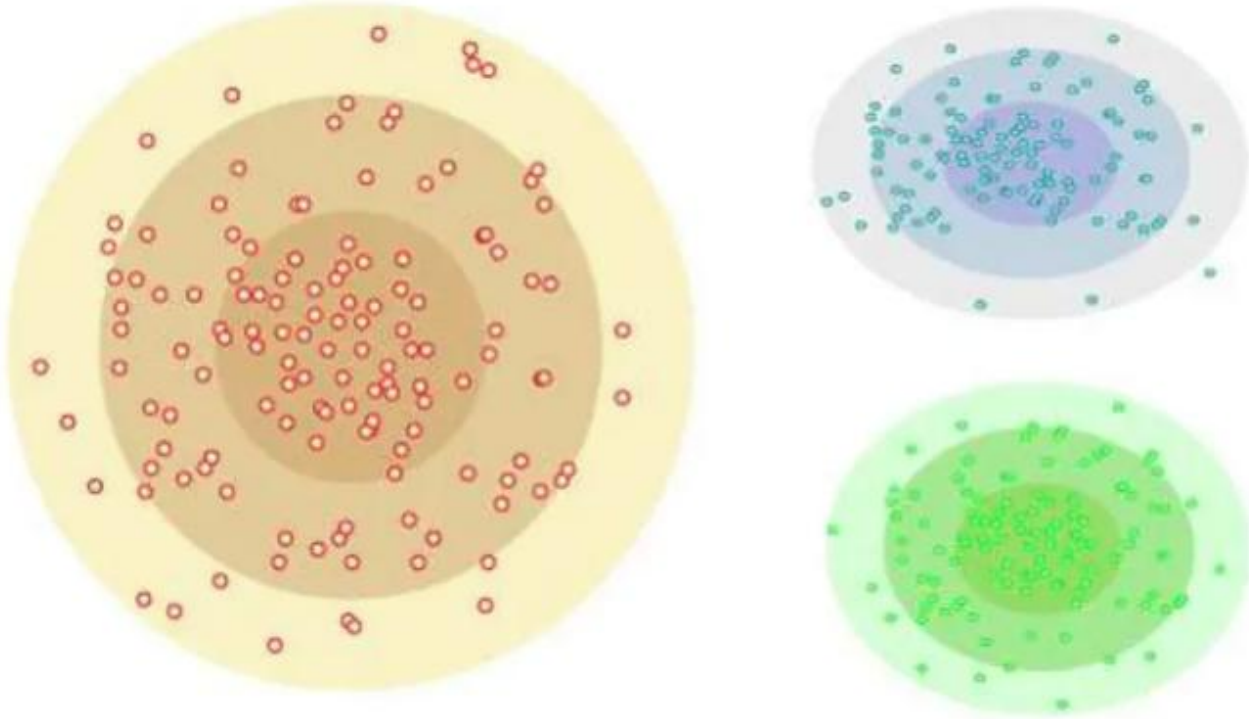


Distribution Model-Based Clustering

- In the distribution model-based clustering method, the data is divided based on the probability of how a dataset belongs to a particular distribution.

- The grouping is done by assuming some distributions, commonly Gaussian distributions.

- The example of this type is the Expectation-Maximization Clustering algorithm that uses Gaussian Mixture Models (GMM).



Association Rule Mining

Market Basket Analysis

Market basket analysis is one of the key techniques used by large retailers to uncover associations between items.





Bread and Jam





Laptop and Bag

Market Basket Analysis

Market basket analysis is one of the key techniques used by large retailers to uncover associations between items.





Bread and Butter

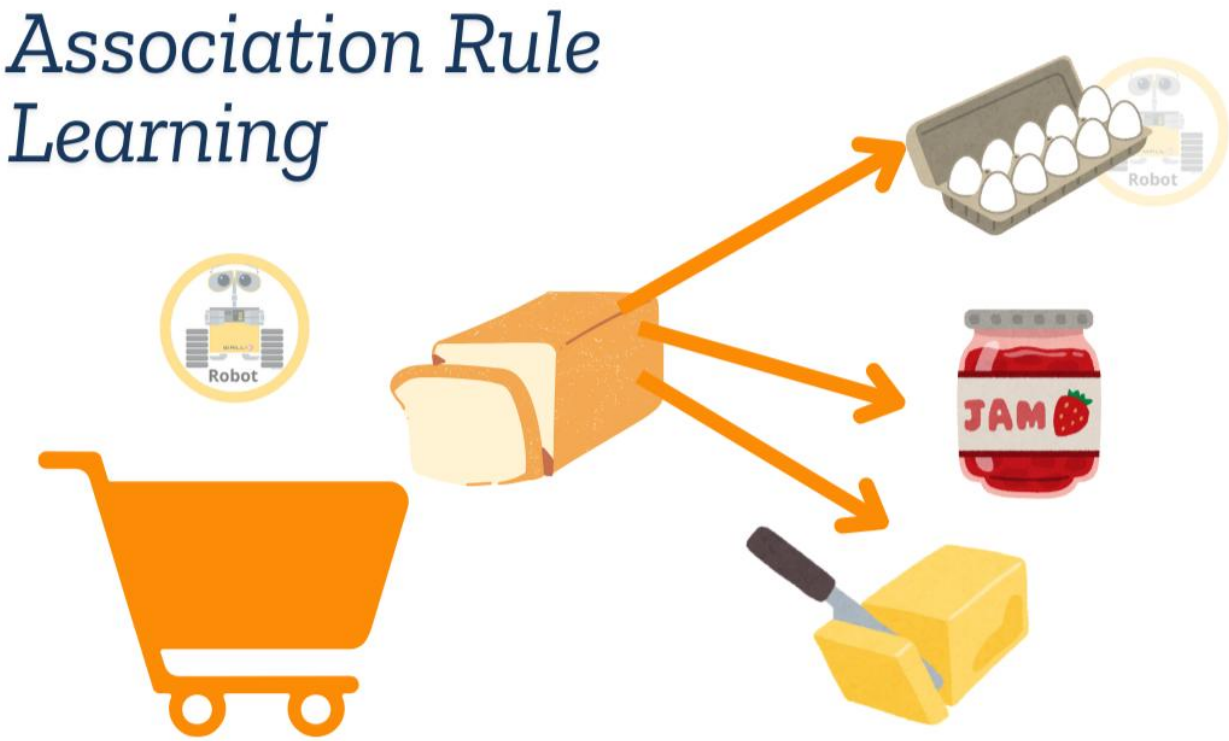




Association Rule Mining

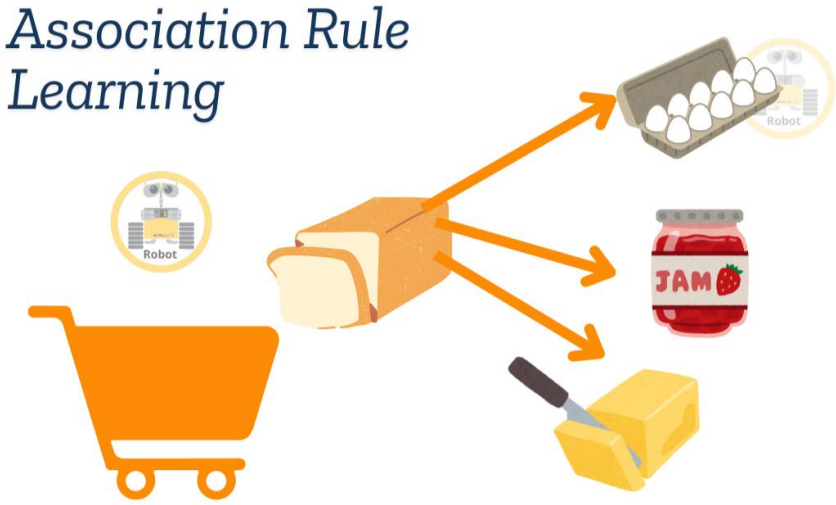
$$X \Rightarrow Y$$

IF THEN



What is Association

- An association rule is an unsupervised learning method that is used for finding the relationship between variables in a large database.
- It determines the set of items that occur together in the dataset.
- Association rules make marketing strategy more effective.
- For example, people who buy X(suppose bread) also tend to buy Y (Butter/Jam) items.
- A typical example of an Association rule is Market Basket Analysis.



Working of Association Rule Learning

- Here, the if element is called the **antecedent**, and then statement is called as **Consequent**.
- These types of relationships, where we can find out some association or relation between two items, is known as single cardinality.
- It is all about creating rules, and if the number of items increases, then cardinality also increases accordingly.
- So to measure the associations between thousands of data items, there are several metrics, which are:
    - Support
    - Confidence
    - Lift

Working of Association Rule Learning

Support:

- Support is the frequency of A or how frequently an item appears in the dataset. It is defined as the fraction of the transaction T that contains the itemset X. If there are X datasets, then for transaction T, it can be written as :

$$Supp(X) = \frac{Freq(X)}{T}$$

Working of Association Rule Learning

Confidence:

- Confidence indicates how often the rule is true. Or how often the items X and Y occur together in the dataset when the occurrence of X is already given.

- It is the ratio of the transaction that contains X and Y to the number of records that contain X.

$$Confiedence = \frac{Freq(X,Y)}{Freq(X)}$$

Working of Association Rule Learning

Lift:

$$Lift = \frac{Supp(X, Y)}{Supp(X) \times Supp(Y)}$$

- It is the ratio of the observed support measure and expected support if X and Y are independent of each other. It has three possible values:
  - **Lift =1:** The probability of occurrence of antecedent and consequent is independent of each other.
  - **Lift>1:** It determines the degree to which the two itemsets are dependent on each other.
  - **Lift<1:** It tells us that one item is a substitute for other items, which means one item has a negative effect on another.

Applications of Association Rule Learning

- It has various applications in machine learning and data mining. Below are some popular applications of association rule learning:
- **Market Basket Analysis:** It is one of the popular examples and applications of association rule mining. This technique is commonly used by big retailers to determine the association between items.
- **Medical Diagnosis:** With the help of association rules, patients can be cured easily, as it helps in identifying the probability of illness for a particular disease.
- **Protein Sequence:** The association rules help in determining the synthesis of artificial proteins.
- It is also used for the **Catalog Design** and **Loss-leader analysis** and many other applications.

Apriori Algorithm

- The Apriori algorithm is an algorithm that uses frequent itemsets to generate association rules, and it is designed to work on databases that contain transactions.
- With the help of these association rules, it determines how strongly or how weakly two objects are connected.
- This algorithm uses a **breadth-first search and a Hash Tree** to calculate the itemset association efficiently.
- It is an iterative process for finding the frequent itemsets from the large dataset.
- This algorithm was given by **R. Agrawal** and **Srikant** in the year **1994**.
- It is mainly used for market basket analysis and helps to find those products that can be bought together.

Apriori Algorithm

- It can also be used in the healthcare field to find drug reactions for patients.
- Frequent itemsets are those items whose support is greater than the threshold value or user-specified minimum support.
- It means if A&B are the frequent itemsets together, then individually A and B should also be the frequent itemsets.
- Suppose there are two transactions: A={1,2,3,4,5} and B={2,3,7}, in these two transactions, 2 and 3 are the frequent itemsets.

Steps for Apriori Algorithm

Below are the steps for the Apriori algorithm:

- **Step-1**: Determine the support of itemsets in the transactional database, and select the minimum support and confidence.
- **Step-2:** Take all supports in the transaction with a higher support value than the minimum or selected support value.
- **Step-3:** Find all the rules of these subsets that have a higher confidence value than the threshold or minimum confidence.
- **Step-4:** Sort the rules as decreasing order of lift.

Apriori Algorithm- Example 1

ARL- Movie Recommendation

User ID	Movies liked
46578	Movie1, Movie2, Movie3, Movie4
98989	Movie1, Movie2
71527	Movie1, Movie2, Movie4
78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:

Movie1 ⇒ Movie2

Movie2 ⇒ Movie4

Movie1 ⇒ Movie3

Apriori Algorithm- Example 1

ARL- Movie Recommendation

Movie Recommendation: $\text{support}(\boldsymbol{M}) = \dfrac{\#\text{ user watchlists containing } \boldsymbol{M}}{\#\text{ user watchlists}}$

User ID	Movies liked
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78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



user watchlists=100

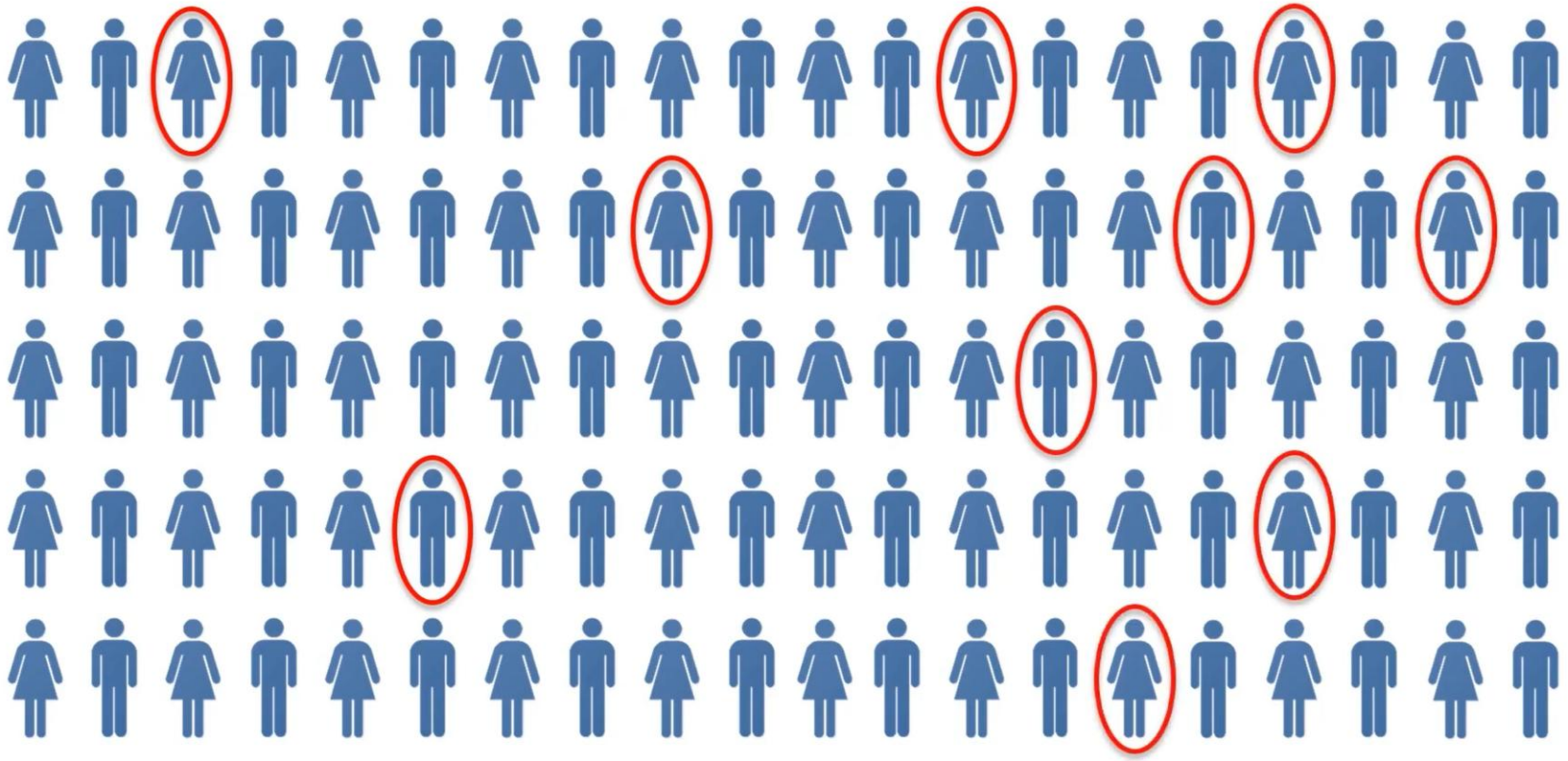
Apriori Algorithm- Example 1

ARL- Movie Recommendation

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61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



$$support(M) = \frac{10}{100} = 10\%$$

Apriori Algorithm- Example 1

ARL- Movie Recommendation

Movie Recommendation: $\text{confidence}(M_1 \rightarrow M_2) = \frac{\#\text{ user watchlists containing } M_1 \text{ and } M_2}{\#\text{ user watchlists containing } M_1}$

User ID	Movies liked
46578	Movie1, Movie2, Movie3, Movie4
98989	Movie1, Movie2
71527	Movie1, Movie2, Movie4
78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



Apriori Algorithm- Example 1

ARL- Movie Recommendation

Movie Recommendation: $\text{confidence}(M_1 \rightarrow M_2) = \frac{\#\text{ user watchlists containing } M_1 \text{ and } M_2}{\#\text{ user watchlists containing } M_1}$

User ID	Movies liked
46578	Movie1, Movie2, Movie3, Movie4
98989	Movie1, Movie2
71527	Movie1, Movie2, Movie4
78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



user watch M1 = 40

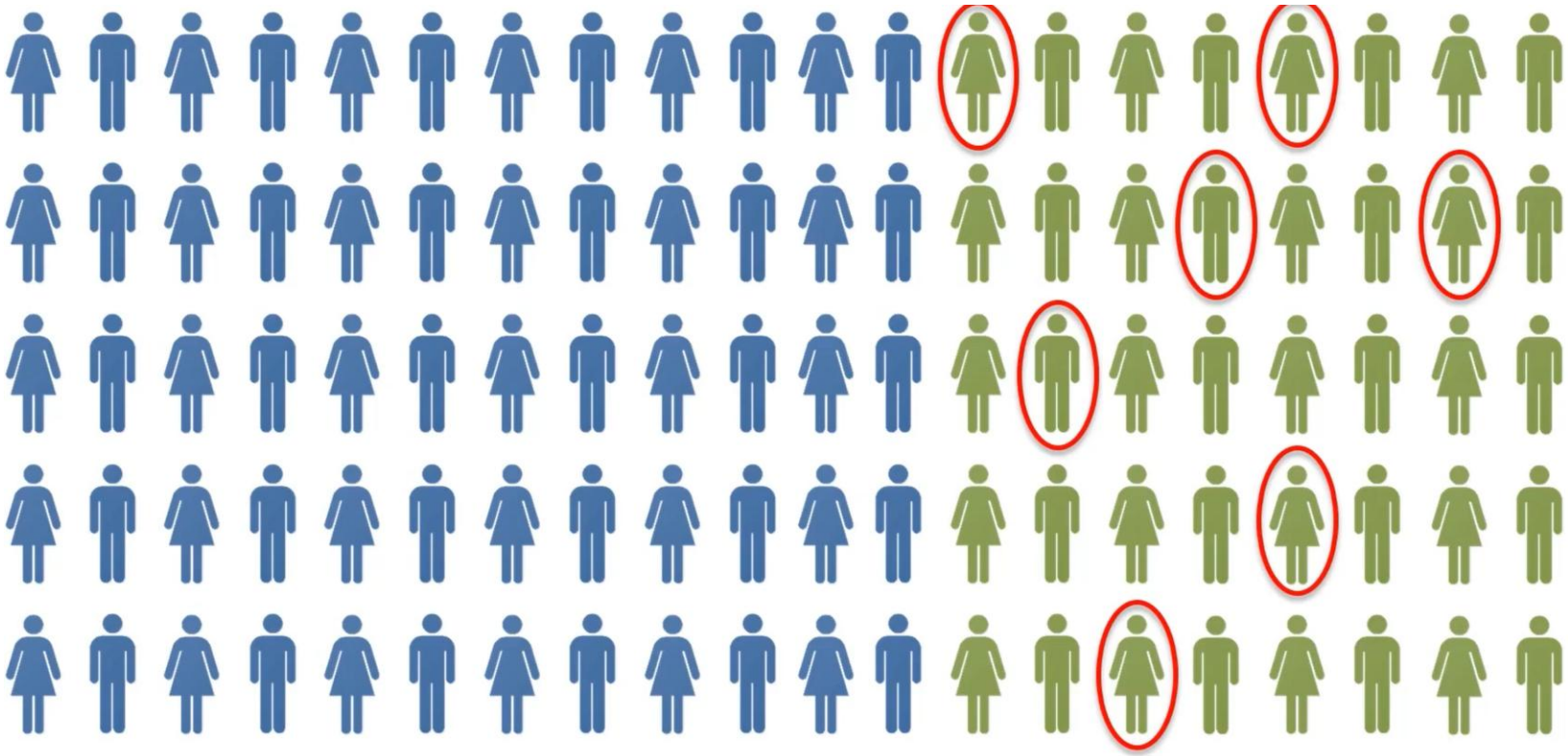
Apriori Algorithm- Example 1

ARL- Movie Recommendation

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61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



$$Confidence = \frac{7}{40} = 17.5\%$$

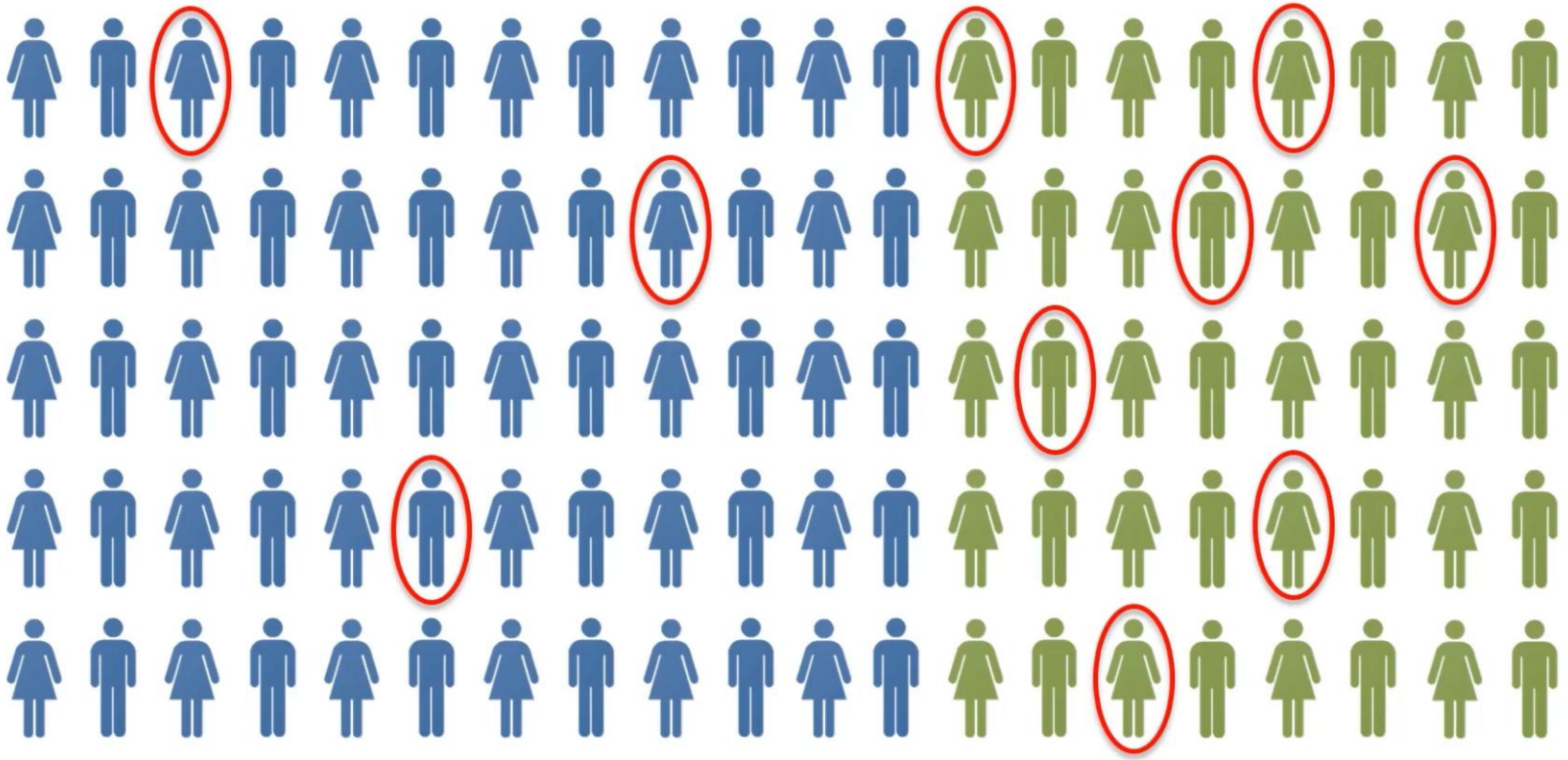
Apriori Algorithm- Example 1

ARL- Movie Recommendation

Movie Recommendation: $\text{lift}(M_1 \rightarrow M_2) = \frac{\text{confidence}(M_1 \rightarrow M_2)}{\text{support}(M_2)}$

User ID	Movies liked
46578	Movie1, Movie2, Movie3, Movie4
98989	Movie1, Movie2
71527	Movie1, Movie2, Movie4
78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:
- Movie1 ⇨ Movie2
- Movie2 ⇨ Movie4
- Movie1 ⇨ Movie3



$$Lift = \frac{17.5\%}{10\%} = 1.75$$

Apriori Algorithm- Example 2

Suppose we have the following dataset, which contains various transactions. From this dataset, we need to find the frequent itemsets and generate association rules using the Apriori algorithm. Minimum support = 2, Minimum Confidence = 50%

TID	ITEMSETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

Apriori Algorithm- Example 2

Solution: Minimum support = 2, Minimum Confidence = 50%

Step-1: Calculating C1 and L1:

- In the first step, we will create a table that contains the support count (the frequency of each itemset individually in the dataset) for each itemset in the given dataset. This is called the candidate set or C1.

1-freq Itemset	Count
A	6
B	7
C	5
D	2
E	1

C1

- Now, we will take out all the itemsets that have a greater support count than the Minimum Support(2). It will give us the table for the **frequent itemset L1**.

- Since all the itemsets have greater or equal support count than the minimum support, except the E, so E itemset will be removed.

1- Freq Itemset	Count
A	6
B	7
C	5
D	2

L1

Apriori Algorithm- Example 2

1-Freq Itemset	Count
A	6
B	7
C	5
D	2

L1

Solution: Minimum support = 2, Minimum Confidence = 50%

Step-2: Calculating C2 and L2:

- In the step, will generate C2 with the help of L1. In C2, we will create the pair of the itemset of L1 in the form of subsets.
- After creating the subsets, we will again find the support count from the main transaction table datasets, i.e., how many times these pairs have occurred together in the given dataset. So, we will get the table C2

2-Freq Itemset	Count
$\{A, B\}$	4
$\{A, C\}$	4
$\{A, D\}$	1
$\{B, C\}$	4
$\{B, D\}$	2
$\{C, E\}$	0

C2

- Again we need to compare the C2 support count with the minimum support count, and after comparing the itemset with less support count will be eliminated from the table C2. It will give us the Table L2.

2-Freq Itemset	Count
$\{A, B\}$	4
$\{A, C\}$	4
$\{B, C\}$	4
$\{B, D\}$	2

L2

Apriori Algorithm- Example 2

2-Freq Itemset	Count
$\{A, B\}$	4
$\{A, C\}$	4
$\{B, C\}$	4
$\{B, D\}$	2

L2

Solution: Minimum support = 2, Minimum Confidence = 50%

Step-3: Calculating C3 and L3:

- For C3, we will repeat the same two processes, but now we will form the C3 table with a subset of three itemsets together, and will calculate support count from the dataset.

3-Freq Itemset	Count
$\{A, B, C\}$	2
$\{B, C, D\}$	1
$\{A, C, D\}$	0
$\{A, B, D\}$	0

C3

- Now, we will create an L3 table. As we can see from the above C3 table, there is only one combination of itemsets that has a support count equal to the minimum count, so the L3 will have only one combination as shown below:

3-Freq Itemset	Count
$\{A, B, C\}$	2

L3

Apriori Algorithm- Example 2

1-Freq Itemset	Count
A	6
B	7
C	5
D	2

L1

2-Freq Itemset	Count
$\{A, B\}$	4
$\{A, C\}$	4
$\{B, C\}$	4
$\{B, D\}$	2

L2

3-Freq Itemset	Count
$\{A, B, C\}$	2

L3

Solution: **Minimum support = 2, Minimum Confidence = 50%**

- To generate the association rules, first we will create a new table with the possible rules from the occurred combination {A, B, C}. For all the rules, we will calculate the confidence and exclude the rules that have less confidence than the minimum confidence threshold (50%).

Rules	Confidence	Association
$\{A,B\} \longrightarrow \{C\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{A,B\}} = \frac{2}{4} = 0.50 = 50\% \geq \mathbf{50\%}$	Strong
$\{B,C\} \longrightarrow \{A\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{B,C\}} = \frac{2}{4} = 0.50 = 50\% \geq \mathbf{50\%}$	Strong
$\{A,C\} \longrightarrow \{B\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{A,C\}} = \frac{2}{4} = 0.50 = 50\% \geq \mathbf{50\%}$	Strong
$\{A\} \longrightarrow \{B,C\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{A\}} = \frac{2}{6} = 0.33 = 33\% < \mathbf{50\%}$	Not Strong
$\{B\} \longrightarrow \{A,C\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{B\}} = \frac{2}{7} = 0.28 = 28\% < \mathbf{50\%}$	Not Strong
$\{C\} \longrightarrow \{A,B\}$	$\text{Confidence} = \frac{support\{A,B,C\}}{support\{C\}} = \frac{2}{5} = 0.4 = 40\% < \mathbf{50\%}$	Not Strong

Apriori Algorithm- Example 3

For the given dataset, apply the Apriori algorithm to discover strong association rules among image tags. Assume that support threshold =40% , and min_confidence = 70%. Generate association rules from the frequent itemsets. Calculate the confidence of each rule and identify all strong association rules.

Transaction ID	List of items
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

Apriori Algorithm- Example 3

For the given dataset, apply the Apriori algorithm to discover strong association rules among image tags. Assume that support threshold =40% , and min_confidence = 70%. Generate association rules from the frequent itemsets. Calculate the confidence of each rule and identify all strong association rules.

Transaction ID	List of items
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

Solution:

Min sup = Support threshold * no of transactions

$$Min\ Sup = 0.4 \times 5 = 2$$

$$\boxed{Min\ Sup = 2}$$

$$\boxed{Confidence = 70\%}$$

Apriori Algorithm- Example 3

For the given dataset, apply the Apriori algorithm to discover strong association rules among image tags. Assume that support threshold =40% , and min_confidence = 70%. Generate association rules from the frequent itemsets. Calculate the confidence of each rule and identify all strong association rules.

Transaction ID	List of items
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2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

Solution:

Generate candidate itemsets (Ck) and qualified frequent itemsets (Lk) step by step until the largest frequent itemset is generated.

Apriori Algorithm- Example 3

Solution: Minimum support = 2, Minimum Confidence = 70%

Step-1: Calculating C1 and L1:

Transaction ID	List of items
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

1-Freq Itemset	Support_count
Bread	3
Butter	3
Milk	2
Beer	2
Cookies	1
Diapers	3

C1

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

L1

Apriori Algorithm- Example 3

Solution: Minimum support = 2, Minimum Confidence = 70%

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

L1

Step-2: Calculating C2 and L2:

Transaction ID	List of items
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

2-Itemset	Support_count
Bread, Butter	3
Bread, Diapers	1
Bread, Milk	2
Bread, Beer	0
Butter, Diapers	1
Butter, Milk	2
Butter, Beer	0
Diapers, Milk	1
Diapers, Beer	2
Milk, Beer	0

C2

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

L2

Apriori Algorithm- Example 3

Solution: Minimum support = 2, Minimum Confidence = 70%

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

L2

Step-2: Calculating C3 and L3:

Transaction ID	List of items
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Butter, Diapers}

Bread, Butter, Milk, Diapers, Beer

3-Itemset	Support_count
Bread, Butter, Milk	2
Bread, Butter, Diapers	1
Bread, Butter, Beer	0
Bread, Milk, Diapers	1
Bread, Milk, Beer	0
Bread, Diapers, Beer	0
Butter, Milk, Diapers	1
Butter, Milk, Beer	0
Butter, Diapers, Beer	0
Milk, Diapers, Beer	0

C3

3-Frequent Itemset	Support_count
Bread, Butter, Milk	2

L3

Apriori Algorithm- Example 3

Solution: Minimum support = 2, Minimum Confidence = 70%

L1

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

L2

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

L3

3-Frequent Itemset	Support_count
Bread, Butter, Milk	2

Rules	Confidence	Association
$\{Bread, Butter\} \longrightarrow \{Milk\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Bread, Butter\}} = \frac{2}{3} = 0.67 = 67\% < \mathbf{70\%}$	Not Strong
$\{Bread, Milk\} \longrightarrow \{Butter\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Bread, Milk\}} = \frac{2}{2} = 1 = 100\% \geq \mathbf{70\%}$	Strong
$\{Butter, Milk\} \longrightarrow \{Bread\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Butter, Milk\}} = \frac{2}{2} = 1 = 100\% \geq \mathbf{70\%}$	Strong
$\{Bread\} \longrightarrow \{Butter, Milk\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Bread\ \}} = \frac{2}{3} = 0.67 = 67\% < \mathbf{70\%}$	Not Strong
$\{Butter\} \longrightarrow \{Bread, Milk\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Butter\}} = \frac{2}{3} = 0.67 = 67\% < \mathbf{70\%}$	Not Strong
$\{Milk\} \longrightarrow \{Bread, Butter\}$	$\text{Confidence} = \frac{support\{Bread,Butter,Milk\}}{support\{Milk\}} = \frac{2}{2} = 1 = 100\% \geq \mathbf{70\%}$	Strong

Apriori Algorithm- Example 4

For the given dataset, apply the Apriori algorithm to discover strong association rules among a list of items. Assume that the support threshold = 50%, and the minimum confidence = 60%. Generate association rules from the frequent itemset.

DO IT BY YOURSELF

Transaction	List of items
T_1	I_1, I_2, I_3
T_2	I_2, I_3, I_4
T_3	I_4, I_5
T_4	I_1, I_2, I_4
T_5	I_1, I_2, I_3, I_5
T_6	I_1, I_2, I_3, I_4

Supervised Vs Unsupervised Learnings

- Unsupervised learning is used for more complex tasks as compared to supervised learning.

- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

- Unsupervised Learning is intrinsically more difficult than supervised learning.

- The result of an unsupervised learning algorithm might be less accurate as input data is not labelled, and algorithms do not know the exact output in advance.

References

1. Alpaydin, E. (2020). Introduction to machine learning. MIT press.

Resources: Datasets

- UCI Repository: http://www.ics.uci.edu/~mlearn/MLRepository.html
- UCI KDD Archive: http://kdd.ics.uci.edu/summary.data.application.html
- Statlib: http://lib.stat.cmu.edu/
- Delve: http://www.cs.utoronto.ca/~delve/